

VZ class project April 17 2026

```
knitr::opts_chunk$set(warning = FALSE, message = FALSE)

library(rethinking)
library(dplyr)
library(lubridate)

project_dir <- "C:/Users/vezuc/OneDrive - Florida International University/PhD/Classes/Bayesian Statistics/2026 Spring/Week 10/Project 10"
```

Load prepared data

```
track_centers_env <- readRDS(file.path(project_dir, "track_centers_env.rds"))
gps_env <- readRDS(file.path(project_dir, "gps_env.rds"))
```

Bayesian Models

```
### Bayesian track-level model of movement

## set factor levels
month_levels <- if (is.factor(track_centers_env$month)) {
  levels(track_centers_env$month)
} else {
  unique(as.character(track_centers_env$month))
}

year_levels <- sort(unique(c(
  as.character(track_centers_env$year),
  as.character(gps_env$year)
```

```

)))

## Track-level Gaussian models

track_bayes <- track_centers_env %>%
  filter(
    !is.na(center_y),
    !is.na(center_x),
    !is.na(month),
    !is.na(year),
    !is.na(tide_height_m),
    !is.na(moon_illumination)
  ) %>%
  mutate(
    month = factor(as.character(month), levels = month_levels),
    year = factor(as.character(year), levels = year_levels),
    month_id = as.integer(month),
    year_id = as.integer(year),
    tide_s = as.numeric(scale(tide_height_m)),
    moon_s = as.numeric(scale(moon_illumination)),
    center_y_s = as.numeric(scale(center_y)),
    center_x_s = as.numeric(scale(center_x))
  )

d_track_y_scaled <- list(
  N = nrow(track_bayes),
  Y = track_bayes$center_y_s,
  month = track_bayes$month_id,
  tide = track_bayes$tide_s,
  moon = track_bayes$moon_s,
  year = track_bayes$year_id,
  n_month = nlevels(track_bayes$month),
  n_year = nlevels(track_bayes$year)
)

d_track_x_scaled <- list(
  N = nrow(track_bayes),
  Y = track_bayes$center_x_s,
  month = track_bayes$month_id,
  tide = track_bayes$tide_s,
  moon = track_bayes$moon_s,

```

```

year = track_bayes$year_id,
n_month = nlevels(track_bayes$month),
n_year = nlevels(track_bayes$year)
)

m_track_y <- ulam(
  alist(
    Y ~ dnorm(mu, sigma),
    mu <- a + a_month[month] + b_tide * tide + b_moon * moon + a_year[year],

    a ~ dnorm(0, 1),
    a_month[month] ~ dnorm(0, sigma_month),
    a_year[year] ~ dnorm(0, sigma_year),

    b_tide ~ dnorm(0, 1),
    b_moon ~ dnorm(0, 1),

    sigma ~ dexp(1),
    sigma_month ~ dexp(1),
    sigma_year ~ dexp(1)
  ),
  data = d_track_y_scaled,
  chains = 4, cores = 4, iter = 2000
)

```

Running MCMC with 4 parallel chains, with 1 thread(s) per chain...

```

Chain 1 Iteration:    1 / 2000 [  0%] (Warmup)
Chain 2 Iteration:    1 / 2000 [  0%] (Warmup)
Chain 3 Iteration:    1 / 2000 [  0%] (Warmup)

Chain 4 Iteration:    1 / 2000 [  0%] (Warmup)
Chain 1 Iteration:   100 / 2000 [  5%] (Warmup)
Chain 2 Iteration:   100 / 2000 [  5%] (Warmup)
Chain 3 Iteration:   100 / 2000 [  5%] (Warmup)
Chain 4 Iteration:   100 / 2000 [  5%] (Warmup)
Chain 1 Iteration:   200 / 2000 [ 10%] (Warmup)
Chain 2 Iteration:   200 / 2000 [ 10%] (Warmup)
Chain 1 Iteration:   300 / 2000 [ 15%] (Warmup)
Chain 3 Iteration:   200 / 2000 [ 10%] (Warmup)
Chain 1 Iteration:   400 / 2000 [ 20%] (Warmup)

```

Chain 4 Iteration: 200 / 2000 [10%] (Warmup)
 Chain 2 Iteration: 300 / 2000 [15%] (Warmup)
 Chain 3 Iteration: 300 / 2000 [15%] (Warmup)
 Chain 1 Iteration: 500 / 2000 [25%] (Warmup)
 Chain 2 Iteration: 400 / 2000 [20%] (Warmup)
 Chain 4 Iteration: 300 / 2000 [15%] (Warmup)
 Chain 3 Iteration: 400 / 2000 [20%] (Warmup)
 Chain 1 Iteration: 600 / 2000 [30%] (Warmup)
 Chain 4 Iteration: 400 / 2000 [20%] (Warmup)
 Chain 2 Iteration: 500 / 2000 [25%] (Warmup)
 Chain 3 Iteration: 500 / 2000 [25%] (Warmup)
 Chain 1 Iteration: 700 / 2000 [35%] (Warmup)
 Chain 2 Iteration: 600 / 2000 [30%] (Warmup)
 Chain 4 Iteration: 500 / 2000 [25%] (Warmup)
 Chain 1 Iteration: 800 / 2000 [40%] (Warmup)
 Chain 3 Iteration: 600 / 2000 [30%] (Warmup)
 Chain 2 Iteration: 700 / 2000 [35%] (Warmup)
 Chain 4 Iteration: 600 / 2000 [30%] (Warmup)
 Chain 1 Iteration: 900 / 2000 [45%] (Warmup)
 Chain 2 Iteration: 800 / 2000 [40%] (Warmup)
 Chain 3 Iteration: 700 / 2000 [35%] (Warmup)
 Chain 4 Iteration: 700 / 2000 [35%] (Warmup)
 Chain 1 Iteration: 1000 / 2000 [50%] (Warmup)
 Chain 3 Iteration: 800 / 2000 [40%] (Warmup)
 Chain 4 Iteration: 800 / 2000 [40%] (Warmup)
 Chain 1 Iteration: 1001 / 2000 [50%] (Sampling)
 Chain 2 Iteration: 900 / 2000 [45%] (Warmup)
 Chain 3 Iteration: 900 / 2000 [45%] (Warmup)
 Chain 1 Iteration: 1100 / 2000 [55%] (Sampling)
 Chain 4 Iteration: 900 / 2000 [45%] (Warmup)
 Chain 2 Iteration: 1000 / 2000 [50%] (Warmup)
 Chain 1 Iteration: 1200 / 2000 [60%] (Sampling)
 Chain 2 Iteration: 1001 / 2000 [50%] (Sampling)
 Chain 3 Iteration: 1000 / 2000 [50%] (Warmup)
 Chain 4 Iteration: 1000 / 2000 [50%] (Warmup)
 Chain 2 Iteration: 1100 / 2000 [55%] (Sampling)
 Chain 3 Iteration: 1001 / 2000 [50%] (Sampling)
 Chain 4 Iteration: 1001 / 2000 [50%] (Sampling)
 Chain 1 Iteration: 1300 / 2000 [65%] (Sampling)
 Chain 3 Iteration: 1100 / 2000 [55%] (Sampling)
 Chain 4 Iteration: 1100 / 2000 [55%] (Sampling)
 Chain 2 Iteration: 1200 / 2000 [60%] (Sampling)
 Chain 4 Iteration: 1200 / 2000 [60%] (Sampling)

```

Chain 1 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 2 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 3 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 4 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 1 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 2 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 3 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 4 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 1 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 2 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 3 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 4 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 3 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 1 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 2 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 4 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 2 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 3 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 4 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 1 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 2 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 3 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 4 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 1 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 3 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 4 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 1 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 2 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 4 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 1 finished in 4.3 seconds.
Chain 4 finished in 4.3 seconds.
Chain 3 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 2 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 3 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 2 finished in 4.4 seconds.
Chain 3 finished in 4.5 seconds.

```

```

All 4 chains finished successfully.
Mean chain execution time: 4.4 seconds.
Total execution time: 4.7 seconds.

```

```

m_track_x <- ulam(
  alist(
    Y ~ dnorm(mu, sigma),
    mu <- a + a_month[month] + b_tide * tide + b_moon * moon + a_year[year],

    a ~ dnorm(0, 1),
    a_month[month] ~ dnorm(0, sigma_month),
    a_year[year] ~ dnorm(0, sigma_year),

    b_tide ~ dnorm(0, 1),
    b_moon ~ dnorm(0, 1),

    sigma ~ dexp(1),
    sigma_month ~ dexp(1),
    sigma_year ~ dexp(1)
  ),
  data = d_track_x_scaled,
  chains = 4, cores = 4, iter = 2000
)

```

Running MCMC with 4 parallel chains, with 1 thread(s) per chain...

```

Chain 1 Iteration:    1 / 2000 [  0%] (Warmup)
Chain 2 Iteration:    1 / 2000 [  0%] (Warmup)
Chain 3 Iteration:    1 / 2000 [  0%] (Warmup)

Chain 4 Iteration:    1 / 2000 [  0%] (Warmup)
Chain 1 Iteration:   100 / 2000 [  5%] (Warmup)
Chain 2 Iteration:   100 / 2000 [  5%] (Warmup)
Chain 3 Iteration:   100 / 2000 [  5%] (Warmup)
Chain 3 Iteration:   200 / 2000 [ 10%] (Warmup)
Chain 4 Iteration:   100 / 2000 [  5%] (Warmup)
Chain 1 Iteration:   200 / 2000 [ 10%] (Warmup)
Chain 3 Iteration:   300 / 2000 [ 15%] (Warmup)
Chain 1 Iteration:   300 / 2000 [ 15%] (Warmup)
Chain 3 Iteration:   400 / 2000 [ 20%] (Warmup)
Chain 4 Iteration:   200 / 2000 [ 10%] (Warmup)
Chain 4 Iteration:   300 / 2000 [ 15%] (Warmup)
Chain 1 Iteration:   400 / 2000 [ 20%] (Warmup)
Chain 3 Iteration:   500 / 2000 [ 25%] (Warmup)
Chain 4 Iteration:   400 / 2000 [ 20%] (Warmup)

```

Chain 1 Iteration: 500 / 2000 [25%] (Warmup)
 Chain 4 Iteration: 500 / 2000 [25%] (Warmup)
 Chain 3 Iteration: 600 / 2000 [30%] (Warmup)
 Chain 1 Iteration: 600 / 2000 [30%] (Warmup)
 Chain 3 Iteration: 700 / 2000 [35%] (Warmup)
 Chain 4 Iteration: 600 / 2000 [30%] (Warmup)
 Chain 1 Iteration: 700 / 2000 [35%] (Warmup)
 Chain 3 Iteration: 800 / 2000 [40%] (Warmup)
 Chain 4 Iteration: 700 / 2000 [35%] (Warmup)
 Chain 1 Iteration: 800 / 2000 [40%] (Warmup)
 Chain 2 Iteration: 200 / 2000 [10%] (Warmup)
 Chain 3 Iteration: 900 / 2000 [45%] (Warmup)
 Chain 4 Iteration: 800 / 2000 [40%] (Warmup)
 Chain 1 Iteration: 900 / 2000 [45%] (Warmup)
 Chain 3 Iteration: 1000 / 2000 [50%] (Warmup)
 Chain 4 Iteration: 900 / 2000 [45%] (Warmup)
 Chain 3 Iteration: 1001 / 2000 [50%] (Sampling)
 Chain 1 Iteration: 1000 / 2000 [50%] (Warmup)
 Chain 2 Iteration: 300 / 2000 [15%] (Warmup)
 Chain 3 Iteration: 1100 / 2000 [55%] (Sampling)
 Chain 4 Iteration: 1000 / 2000 [50%] (Warmup)
 Chain 1 Iteration: 1001 / 2000 [50%] (Sampling)
 Chain 1 Iteration: 1100 / 2000 [55%] (Sampling)
 Chain 4 Iteration: 1001 / 2000 [50%] (Sampling)
 Chain 2 Iteration: 400 / 2000 [20%] (Warmup)
 Chain 3 Iteration: 1200 / 2000 [60%] (Sampling)
 Chain 4 Iteration: 1100 / 2000 [55%] (Sampling)
 Chain 1 Iteration: 1200 / 2000 [60%] (Sampling)
 Chain 1 Iteration: 1300 / 2000 [65%] (Sampling)
 Chain 3 Iteration: 1300 / 2000 [65%] (Sampling)
 Chain 4 Iteration: 1200 / 2000 [60%] (Sampling)
 Chain 2 Iteration: 500 / 2000 [25%] (Warmup)
 Chain 1 Iteration: 1400 / 2000 [70%] (Sampling)
 Chain 3 Iteration: 1400 / 2000 [70%] (Sampling)
 Chain 1 Iteration: 1500 / 2000 [75%] (Sampling)
 Chain 2 Iteration: 600 / 2000 [30%] (Warmup)
 Chain 4 Iteration: 1300 / 2000 [65%] (Sampling)
 Chain 3 Iteration: 1500 / 2000 [75%] (Sampling)
 Chain 1 Iteration: 1600 / 2000 [80%] (Sampling)
 Chain 2 Iteration: 700 / 2000 [35%] (Warmup)
 Chain 4 Iteration: 1400 / 2000 [70%] (Sampling)
 Chain 1 Iteration: 1700 / 2000 [85%] (Sampling)
 Chain 2 Iteration: 800 / 2000 [40%] (Warmup)

```

Chain 3 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 2 Iteration:  900 / 2000 [ 45%] (Warmup)
Chain 1 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 3 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 4 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 2 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 3 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 1 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 2 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 1 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 2 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 4 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 1 finished in 3.7 seconds.
Chain 2 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 3 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 4 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 2 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 3 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 3 finished in 3.9 seconds.
Chain 2 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 4 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 2 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 4 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 2 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 4 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 4 finished in 4.4 seconds.
Chain 2 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 2 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 2 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 2 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 2 finished in 4.9 seconds.

```

All 4 chains finished successfully.
 Mean chain execution time: 4.2 seconds.
 Total execution time: 4.9 seconds.

```
### 2-second GPS cell-based model
```

```

cell_size <- 25

gps_use <- gps_env %>%
  filter(

```

```

!is.na(file_name),
!is.na(x_true_m),
!is.na(y_true_m),
!is.na(month),
!is.na(year),
!is.na(tide_height_m),
!is.na(moon_illumination)
) %>%
mutate(
  month = factor(as.character(month), levels = month_levels),
  year = factor(as.character(year), levels = year_levels),
  x_cell = floor(x_true_m / cell_size),
  y_cell = floor(y_true_m / cell_size)
)

# Track effort = total number of 2-sec fixes in each track
track_effort <- gps_use %>%
  count(file_name, name = "n_fixes_track") %>%
  mutate(log_effort = log(n_fixes_track))

# Aggregate number of 2-sec fixes per track x cell
gps_cells <- gps_use %>%
  group_by(
    file_name, year, month,
    tide_height_m, moon_illumination,
    x_cell, y_cell
  ) %>%
  summarise(
    n_fixes_cell = n(),
    .groups = "drop"
  ) %>%
  left_join(track_effort, by = "file_name") %>%
  mutate(
    seconds_in_cell = n_fixes_cell * 2,
    cell_id = as.integer(factor(paste(x_cell, y_cell, sep = "_"))),
    track_id = as.integer(factor(file_name)),
    month_id = as.integer(month),
    year_id = as.integer(year),
    tide_s = as.numeric(scale(tide_height_m)),
    moon_s = as.numeric(scale(moon_illumination))
  )

```

```

d_cells_nb <- list(
  N = nrow(gps_cells),
  count = gps_cells$n_fixes_cell,
  month = gps_cells$month_id,
  year = gps_cells$year_id,
  track = gps_cells$track_id,
  cell = gps_cells$cell_id,
  tide = gps_cells$tide_s,
  moon = gps_cells$moon_s,
  log_effort = gps_cells$log_effort,
  n_month = nlevels(gps_use$month),
  n_year = nlevels(gps_use$year),
  n_track = length(unique(gps_cells$track_id)),
  n_cell = length(unique(gps_cells$cell_id))
)

# Negative binomial (gamma-Poisson) with effort offset and cell effect
m_space_use <- ulam(
  alist(
    count ~ dgamma_pois(lambda, phi),
    log(lambda) <- log_effort +
      a +
      a_month[month] +
      a_year[year] +
      a_track[track] +
      a_cell[cell] +
      b_tide * tide +
      b_moon * moon,

    a ~ dnorm(0, 1),

    a_month[month] ~ dnorm(0, sigma_month),
    a_year[year] ~ dnorm(0, sigma_year),
    a_track[track] ~ dnorm(0, sigma_track),
    a_cell[cell] ~ dnorm(0, sigma_cell),

    b_tide ~ dnorm(0, 1),
    b_moon ~ dnorm(0, 1),

    sigma_month ~ dexp(1),
    sigma_year ~ dexp(1),
    sigma_track ~ dexp(1),

```

```

    sigma_cell ~ dexp(1),

    phi ~ dexp(1)
  ),
  data = d_cells_nb,
  chains = 4, cores = 4, iter = 2000
)

```

Running MCMC with 4 parallel chains, with 1 thread(s) per chain...

Chain 1 Iteration: 1 / 2000 [0%] (Warmup)

Chain 2 Iteration: 1 / 2000 [0%] (Warmup)

Chain 3 Iteration: 1 / 2000 [0%] (Warmup)

Chain 4 Iteration: 1 / 2000 [0%] (Warmup)

Chain 2 Iteration: 100 / 2000 [5%] (Warmup)

Chain 3 Iteration: 100 / 2000 [5%] (Warmup)

Chain 4 Iteration: 100 / 2000 [5%] (Warmup)

Chain 1 Iteration: 100 / 2000 [5%] (Warmup)

Chain 1 Iteration: 200 / 2000 [10%] (Warmup)

Chain 4 Iteration: 200 / 2000 [10%] (Warmup)

Chain 3 Iteration: 200 / 2000 [10%] (Warmup)

Chain 2 Iteration: 200 / 2000 [10%] (Warmup)

Chain 1 Iteration: 300 / 2000 [15%] (Warmup)

Chain 4 Iteration: 300 / 2000 [15%] (Warmup)

Chain 3 Iteration: 300 / 2000 [15%] (Warmup)

Chain 1 Iteration: 400 / 2000 [20%] (Warmup)

Chain 2 Iteration: 300 / 2000 [15%] (Warmup)

Chain 4 Iteration: 400 / 2000 [20%] (Warmup)

Chain 1 Iteration: 500 / 2000 [25%] (Warmup)

Chain 2 Iteration: 400 / 2000 [20%] (Warmup)

Chain 3 Iteration: 400 / 2000 [20%] (Warmup)

Chain 4 Iteration: 500 / 2000 [25%] (Warmup)

Chain 1 Iteration: 600 / 2000 [30%] (Warmup)

Chain 2 Iteration: 500 / 2000 [25%] (Warmup)

Chain 4 Iteration: 600 / 2000 [30%] (Warmup)

Chain 3 Iteration: 500 / 2000 [25%] (Warmup)

Chain 1 Iteration: 700 / 2000 [35%] (Warmup)
Chain 2 Iteration: 600 / 2000 [30%] (Warmup)
Chain 4 Iteration: 700 / 2000 [35%] (Warmup)
Chain 3 Iteration: 600 / 2000 [30%] (Warmup)
Chain 1 Iteration: 800 / 2000 [40%] (Warmup)
Chain 2 Iteration: 700 / 2000 [35%] (Warmup)
Chain 4 Iteration: 800 / 2000 [40%] (Warmup)
Chain 1 Iteration: 900 / 2000 [45%] (Warmup)
Chain 3 Iteration: 700 / 2000 [35%] (Warmup)
Chain 2 Iteration: 800 / 2000 [40%] (Warmup)
Chain 4 Iteration: 900 / 2000 [45%] (Warmup)
Chain 3 Iteration: 800 / 2000 [40%] (Warmup)
Chain 1 Iteration: 1000 / 2000 [50%] (Warmup)
Chain 1 Iteration: 1001 / 2000 [50%] (Sampling)
Chain 2 Iteration: 900 / 2000 [45%] (Warmup)
Chain 3 Iteration: 900 / 2000 [45%] (Warmup)
Chain 1 Iteration: 1100 / 2000 [55%] (Sampling)
Chain 4 Iteration: 1000 / 2000 [50%] (Warmup)
Chain 4 Iteration: 1001 / 2000 [50%] (Sampling)
Chain 1 Iteration: 1200 / 2000 [60%] (Sampling)
Chain 2 Iteration: 1000 / 2000 [50%] (Warmup)
Chain 2 Iteration: 1001 / 2000 [50%] (Sampling)
Chain 4 Iteration: 1100 / 2000 [55%] (Sampling)
Chain 3 Iteration: 1000 / 2000 [50%] (Warmup)
Chain 3 Iteration: 1001 / 2000 [50%] (Sampling)
Chain 1 Iteration: 1300 / 2000 [65%] (Sampling)
Chain 2 Iteration: 1100 / 2000 [55%] (Sampling)
Chain 3 Iteration: 1100 / 2000 [55%] (Sampling)
Chain 4 Iteration: 1200 / 2000 [60%] (Sampling)
Chain 2 Iteration: 1200 / 2000 [60%] (Sampling)
Chain 1 Iteration: 1400 / 2000 [70%] (Sampling)
Chain 3 Iteration: 1200 / 2000 [60%] (Sampling)
Chain 4 Iteration: 1300 / 2000 [65%] (Sampling)
Chain 2 Iteration: 1300 / 2000 [65%] (Sampling)
Chain 1 Iteration: 1500 / 2000 [75%] (Sampling)
Chain 3 Iteration: 1300 / 2000 [65%] (Sampling)
Chain 4 Iteration: 1400 / 2000 [70%] (Sampling)
Chain 2 Iteration: 1400 / 2000 [70%] (Sampling)
Chain 1 Iteration: 1600 / 2000 [80%] (Sampling)
Chain 3 Iteration: 1400 / 2000 [70%] (Sampling)
Chain 4 Iteration: 1500 / 2000 [75%] (Sampling)
Chain 1 Iteration: 1700 / 2000 [85%] (Sampling)
Chain 2 Iteration: 1500 / 2000 [75%] (Sampling)

```

Chain 3 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 4 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 1 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 3 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 4 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 2 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 1 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 3 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 4 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 2 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 1 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 1 finished in 453.5 seconds.
Chain 3 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 4 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 2 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 4 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 4 finished in 481.5 seconds.
Chain 3 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 2 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 3 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 3 finished in 497.9 seconds.
Chain 2 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 2 finished in 501.5 seconds.

```

All 4 chains finished successfully.
 Mean chain execution time: 483.6 seconds.
 Total execution time: 501.7 seconds.

```

#### Diagnostics and posterior summaries for track-level models
#| cache: true

## posterior summaries
precis(m_track_y, depth = 2)

```

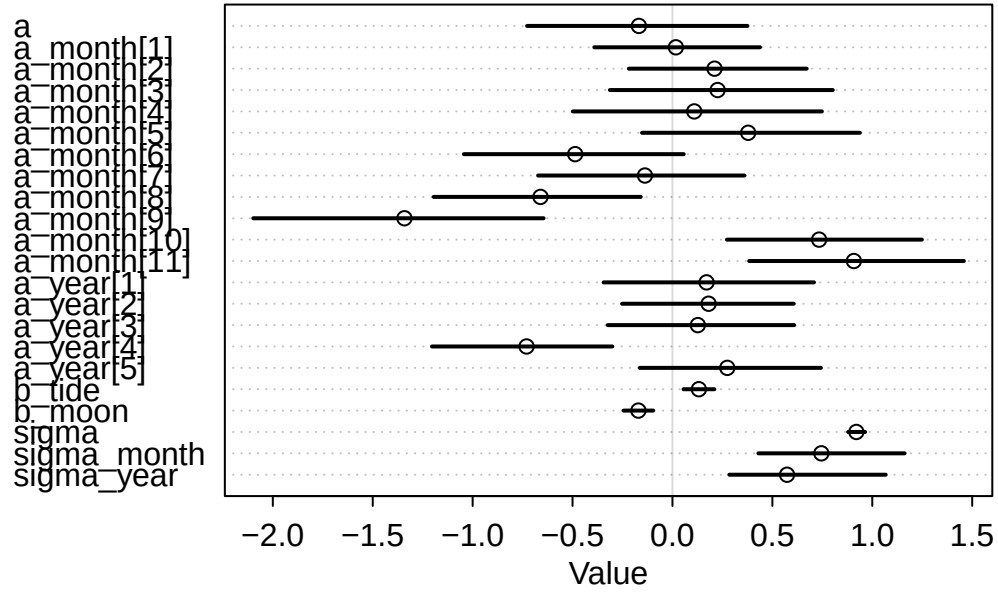
	mean	sd	5.5%	94.5%	rhat	ess_bulk
a	-0.16806048	0.35420475	-0.72751941	0.37551909	1.0044495	972.5753
a_month[1]	0.01714644	0.26504619	-0.39131779	0.43831151	1.0050477	1017.2969
a_month[2]	0.21045712	0.28325423	-0.21846651	0.67187760	1.0044609	1045.5305
a_month[3]	0.22586333	0.35656002	-0.31302720	0.80249020	1.0026021	1390.3381
a_month[4]	0.10900988	0.39302687	-0.49866553	0.74788392	1.0034663	1704.0212
a_month[5]	0.37888923	0.34935174	-0.15234872	0.93806118	1.0035996	1574.3630
a_month[6]	-0.48686452	0.35449900	-1.04390579	0.05639083	1.0034696	1694.3630

a_month[7]	-0.13766443	0.32802937	-0.67242583	0.36043323	1.0018276	1381.5086
a_month[8]	-0.66045610	0.33435064	-1.19524477	-0.15849345	1.0029310	1392.6975
a_month[9]	-1.34136019	0.46048672	-2.09781625	-0.64493008	1.0010826	1978.2607
a_month[10]	0.73383656	0.30800210	0.27245955	1.24825471	1.0039254	1035.7039
a_month[11]	0.90785569	0.34163519	0.38394213	1.45837059	1.0030220	1077.7033
a_year[1]	0.17067456	0.33214269	-0.34482687	0.70843291	1.0003644	1707.5827
a_year[2]	0.18083507	0.27281483	-0.25226957	0.60692487	1.0012221	1453.3071
a_year[3]	0.12686512	0.29859535	-0.32403118	0.60951706	1.0005982	1576.1488
a_year[4]	-0.73023776	0.28970160	-1.20375428	-0.30084596	1.0002079	1510.2293
a_year[5]	0.27433965	0.28945233	-0.16379920	0.74289781	0.9999534	1644.6313
b_tide	0.13210998	0.04810162	0.05551629	0.20979015	1.0003705	3548.9395
b_moon	-0.17057506	0.04603779	-0.24480418	-0.09648046	1.0007674	2895.4295
sigma	0.92022884	0.02733641	0.87899096	0.96460614	1.0025642	3660.5869
sigma_month	0.74512021	0.24285267	0.43001325	1.16193264	1.0006370	1834.8233
sigma_year	0.57337554	0.27101492	0.28445140	1.06728852	1.0000916	2196.5021

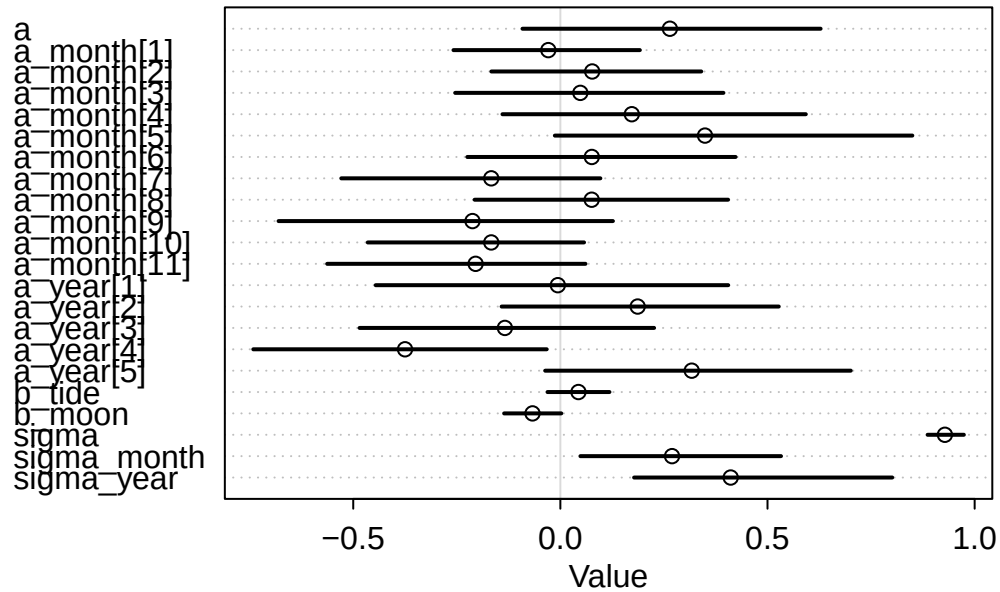
```
precis(m_track_x, depth = 2)
```

	mean	sd	5.5%	94.5%	rhat	ess_bulk
a	0.264343096	0.23712637	-0.09146921	0.62819351	1.005584	750.4324
a_month[1]	-0.029105867	0.14263830	-0.25807013	0.19190748	1.002998	2083.6694
a_month[2]	0.076863415	0.16022432	-0.16653099	0.34011461	1.001863	1905.4858
a_month[3]	0.048162661	0.20254751	-0.25397161	0.39367439	1.001628	2760.3416
a_month[4]	0.172430929	0.23769029	-0.13981374	0.59252724	1.001341	1749.7383
a_month[5]	0.349114055	0.28218681	-0.01332281	0.84962909	1.003035	662.9863
a_month[6]	0.075845362	0.20392258	-0.22466655	0.42299882	1.000771	2440.5421
a_month[7]	-0.166798150	0.20204639	-0.52890435	0.09658176	1.003274	1210.7935
a_month[8]	0.075631684	0.19131245	-0.20699477	0.40483829	1.003176	2629.7732
a_month[9]	-0.212121153	0.26271029	-0.68026399	0.12686027	1.001809	1446.7150
a_month[10]	-0.166814753	0.17056512	-0.46492465	0.05721285	1.003702	829.8530
a_month[11]	-0.204605982	0.20098204	-0.56290564	0.06064223	1.003376	800.5066
a_year[1]	-0.006148838	0.27729699	-0.44584785	0.40516647	1.003372	1304.2875
a_year[2]	0.186579865	0.22712182	-0.14184305	0.52725339	1.004388	842.2309
a_year[3]	-0.133934299	0.23692565	-0.48473279	0.22632900	1.005103	837.2132
a_year[4]	-0.374985253	0.24006487	-0.74127136	-0.03290805	1.003518	938.2006
a_year[5]	0.317083635	0.24407601	-0.03673131	0.70101899	1.003829	923.4143
b_tide	0.043652410	0.04689474	-0.03111666	0.11852939	1.001011	3728.9031
b_moon	-0.067200005	0.04331704	-0.13576255	0.00230185	1.000438	3579.9532
sigma	0.928208838	0.02730536	0.88640084	0.97401881	1.003157	3639.3331
sigma_month	0.269572695	0.15024561	0.04835321	0.53262800	1.007859	341.9074
sigma_year	0.411166507	0.22445685	0.17795492	0.80110723	1.001372	1675.7532

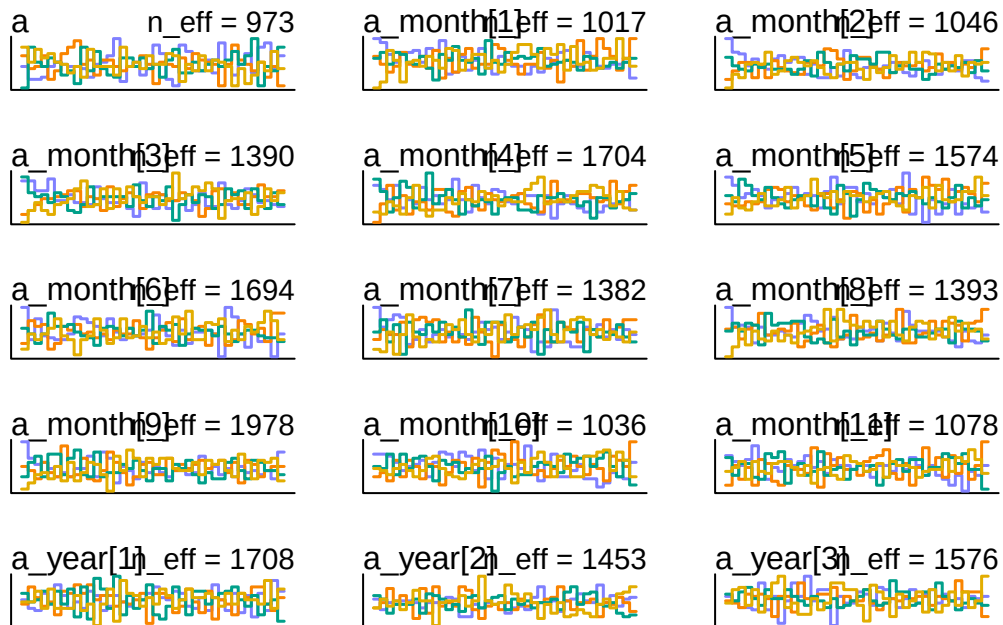
```
plot(precis(m_track_y, depth = 2))
```



```
plot(precis(m_track_x, depth = 2))
```



```
## trace/rank plots
# Track Y
par(mfrow = c(2,2))
trankplot(m_track_y, ask = FALSE)
```



```
par(mfrow = c(1,1))
```

a_year[4]_eff = 1510



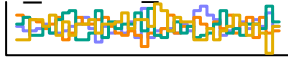
a_year[5]_eff = 1645



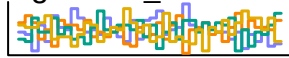
b_tide n_eff = 3549



b_moon n_eff = 2895



sigma n_eff = 3661



sigma_month n_eff = 1835

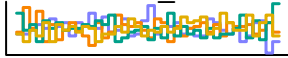


sigma_year n_eff = 2197



```
# Track X  
par(mfrow = c(2,2))  
trankplot(m_track_x, ask = FALSE)
```

a n_eff = 750



a_month[1]_eff = 2084



a_month[2]_eff = 1905



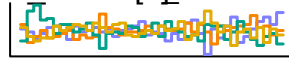
a_month[3]_eff = 2760



a_month[4]_eff = 1750



a_month[5]_eff = 663



a_month[6]_eff = 2441



a_month[7]_eff = 1211



a_month[8]_eff = 2630



a_month[9]_eff = 1447



a_month[10]_eff = 830



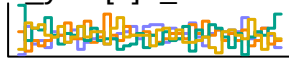
a_month[11]_eff = 801



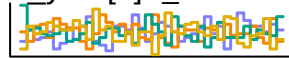
a_year[1]_eff = 1304



a_year[2] n_eff = 842



a_year[3] n_eff = 837

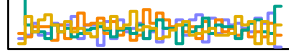


```
par(mfrow = c(1,1))
```

a_year[4] n_eff = 938



a_year[5] n_eff = 923



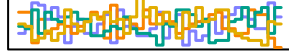
b_tide n_eff = 3729



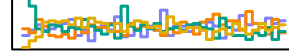
b_moon n_eff = 3580



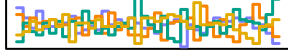
sigma n_eff = 3639



sigma_month n_eff = 342



sigma_year n_eff = 1676



```
## convergence check
summary(m_track_y)
```

	mean	sd	5.5%	94.5%	rhat	ess_bulk
a	-0.16806048	0.35420475	-0.72751941	0.37551909	1.0044495	972.5753
a_month[1]	0.01714644	0.26504619	-0.39131779	0.43831151	1.0050477	1017.2969
a_month[2]	0.21045712	0.28325423	-0.21846651	0.67187760	1.0044609	1045.5305
a_month[3]	0.22586333	0.35656002	-0.31302720	0.80249020	1.0026021	1390.3381
a_month[4]	0.10900988	0.39302687	-0.49866553	0.74788392	1.0034663	1704.0212
a_month[5]	0.37888923	0.34935174	-0.15234872	0.93806118	1.0035996	1574.3630
a_month[6]	-0.48686452	0.35449900	-1.04390579	0.05639083	1.0034696	1694.3630
a_month[7]	-0.13766443	0.32802937	-0.67242583	0.36043323	1.0018276	1381.5086
a_month[8]	-0.66045610	0.33435064	-1.19524477	-0.15849345	1.0029310	1392.6975
a_month[9]	-1.34136019	0.46048672	-2.09781625	-0.64493008	1.0010826	1978.2607
a_month[10]	0.73383656	0.30800210	0.27245955	1.24825471	1.0039254	1035.7039
a_month[11]	0.90785569	0.34163519	0.38394213	1.45837059	1.0030220	1077.7033
a_year[1]	0.17067456	0.33214269	-0.34482687	0.70843291	1.0003644	1707.5827
a_year[2]	0.18083507	0.27281483	-0.25226957	0.60692487	1.0012221	1453.3071
a_year[3]	0.12686512	0.29859535	-0.32403118	0.60951706	1.0005982	1576.1488
a_year[4]	-0.73023776	0.28970160	-1.20375428	-0.30084596	1.0002079	1510.2293

```

a_year[5]      0.27433965 0.28945233 -0.16379920  0.74289781 0.9999534 1644.6313
b_tide         0.13210998 0.04810162  0.05551629  0.20979015 1.0003705 3548.9395
b_moon        -0.17057506 0.04603779 -0.24480418 -0.09648046 1.0007674 2895.4295
sigma          0.92022884 0.02733641  0.87899096  0.96460614 1.0025642 3660.5869
sigma_month    0.74512021 0.24285267  0.43001325  1.16193264 1.0006370 1834.8233
sigma_year     0.57337554 0.27101492  0.28445140  1.06728852 1.0000916 2196.5021

```

```
summary(m_track_x)
```

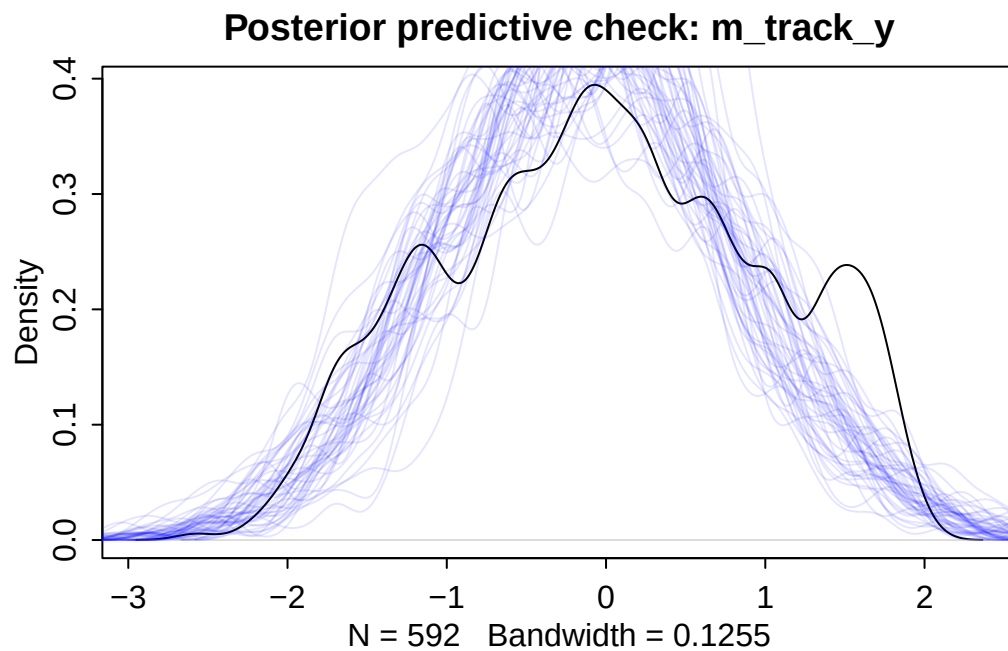
	mean	sd	5.5%	94.5%	rhat	ess_bulk
a	0.264343096	0.23712637	-0.09146921	0.62819351	1.005584	750.4324
a_month[1]	-0.029105867	0.14263830	-0.25807013	0.19190748	1.002998	2083.6694
a_month[2]	0.076863415	0.16022432	-0.16653099	0.34011461	1.001863	1905.4858
a_month[3]	0.048162661	0.20254751	-0.25397161	0.39367439	1.001628	2760.3416
a_month[4]	0.172430929	0.23769029	-0.13981374	0.59252724	1.001341	1749.7383
a_month[5]	0.349114055	0.28218681	-0.01332281	0.84962909	1.003035	662.9863
a_month[6]	0.075845362	0.20392258	-0.22466655	0.42299882	1.000771	2440.5421
a_month[7]	-0.166798150	0.20204639	-0.52890435	0.09658176	1.003274	1210.7935
a_month[8]	0.075631684	0.19131245	-0.20699477	0.40483829	1.003176	2629.7732
a_month[9]	-0.212121153	0.26271029	-0.68026399	0.12686027	1.001809	1446.7150
a_month[10]	-0.166814753	0.17056512	-0.46492465	0.05721285	1.003702	829.8530
a_month[11]	-0.204605982	0.20098204	-0.56290564	0.06064223	1.003376	800.5066
a_year[1]	-0.006148838	0.27729699	-0.44584785	0.40516647	1.003372	1304.2875
a_year[2]	0.186579865	0.22712182	-0.14184305	0.52725339	1.004388	842.2309
a_year[3]	-0.133934299	0.23692565	-0.48473279	0.22632900	1.005103	837.2132
a_year[4]	-0.374985253	0.24006487	-0.74127136	-0.03290805	1.003518	938.2006
a_year[5]	0.317083635	0.24407601	-0.03673131	0.70101899	1.003829	923.4143
b_tide	0.043652410	0.04689474	-0.03111666	0.11852939	1.001011	3728.9031
b_moon	-0.067200005	0.04331704	-0.13576255	0.00230185	1.000438	3579.9532
sigma	0.928208838	0.02730536	0.88640084	0.97401881	1.003157	3639.3331
sigma_month	0.269572695	0.15024561	0.04835321	0.53262800	1.007859	341.9074
sigma_year	0.411166507	0.22445685	0.17795492	0.80110723	1.001372	1675.7532

```

## posterior predictive check
y_sim <- sim(m_track_y, n = 1000)

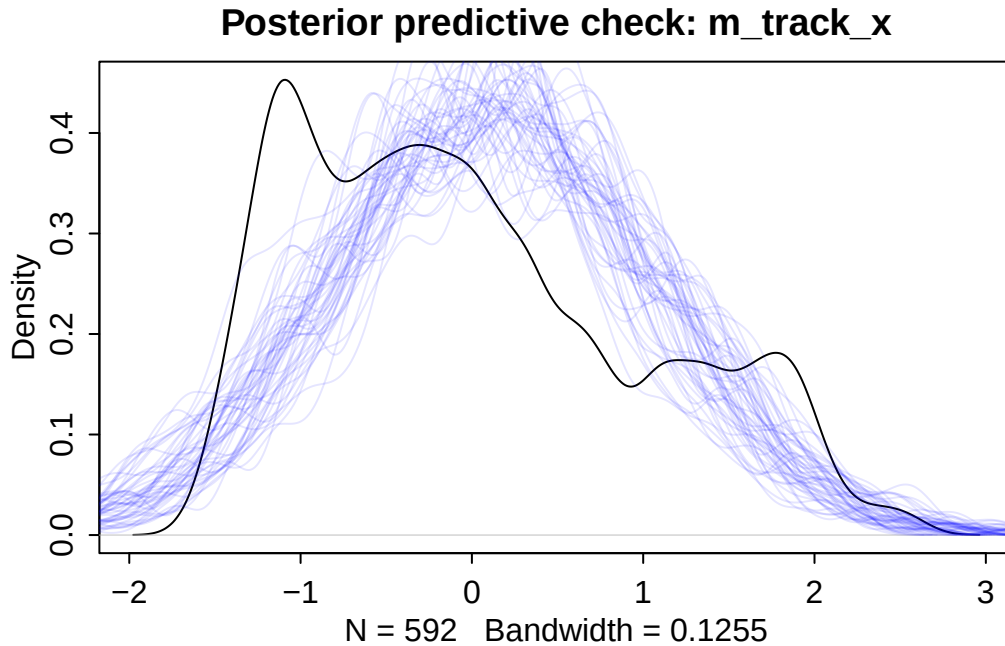
dens(d_track_y_scaled$Y)
for(i in 1:50) dens(y_sim[i,], add = TRUE, col = col.alpha("blue", 0.1))
title("Posterior predictive check: m_track_y")

```



```
x_sim <- sim(m_track_x, n = 1000)

dens(d_track_x_scaled$Y)
for(i in 1:50) dens(x_sim[i,], add = TRUE, col = col.alpha("blue", 0.1))
title("Posterior predictive check: m_track_x")
```



```
#### Diagnostics and posterior summaries for space-use model
```

```
## posterior summaries
```

```
precis(m_space_use, depth = 2)
```

	mean	sd	5.5%	94.5%	rhat
a	-1.445107e+00	0.31631276	-1.900561296	-0.9155428765	1.0014583
a_month[1]	-1.450397e-01	0.14408829	-0.375085352	0.0713376433	1.0018410
a_month[2]	-3.903910e-01	0.15795403	-0.653530632	-0.1619841807	1.0021315
a_month[3]	-5.546982e-01	0.20456904	-0.897087022	-0.2498652388	1.0007412
a_month[4]	-2.798650e-01	0.22095437	-0.640181237	0.0527249246	1.0008298
a_month[5]	1.191828e-01	0.18908293	-0.184192797	0.4179400641	1.0008336
a_month[6]	-4.314699e-02	0.19116577	-0.349128444	0.2533594574	1.0028893
a_month[7]	3.205005e-01	0.17812703	0.046532757	0.5991325840	1.0014763
a_month[8]	3.267421e-01	0.18186829	0.048351504	0.6152736918	1.0017435
a_month[9]	5.164152e-01	0.24186211	0.145352136	0.9132331801	1.0016556
a_month[10]	-6.738535e-02	0.16620927	-0.328675068	0.1704865476	1.0017021
a_month[11]	-5.010591e-02	0.18407372	-0.352275596	0.2199075737	1.0008470
a_year[1]	-6.317773e-02	0.31522494	-0.580959059	0.3985824100	1.0007860
a_year[2]	-5.180224e-01	0.29611075	-1.018284925	-0.0868758523	1.0009025
a_year[3]	-5.134087e-01	0.30264673	-1.035817570	-0.0789229332	1.0009667
a_year[4]	6.136916e-01	0.29540679	0.119406844	1.0420436260	1.0008166

a_year[5]	-1.010020e-01	0.30080328	-0.616880550	0.3404886892	1.0010135
a_track[1]	-3.774271e-01	0.21463855	-0.715267461	-0.0361673556	1.0004256
a_track[2]	-1.513528e-01	0.22874739	-0.517373768	0.2203682608	1.0003257
a_track[3]	-1.519733e-01	0.22350905	-0.510056014	0.2169170725	1.0008599
a_track[4]	-1.560206e-01	0.23975716	-0.532252780	0.2356238555	1.0024700
a_track[5]	2.187780e-01	0.24817429	-0.174083680	0.6106338374	0.9997928
a_track[6]	5.997229e-04	0.29615740	-0.472377454	0.4693242881	1.0021253
a_track[7]	-2.993517e-02	0.21354067	-0.365755222	0.3139148400	1.0015800
a_track[8]	-2.930947e-02	0.22058278	-0.381812629	0.3285467349	1.0010143
a_track[9]	4.924901e-02	0.22887747	-0.308373965	0.4282147746	1.0008254
a_track[10]	-3.095314e-01	0.20226536	-0.632637269	0.0198071653	1.0006322
a_track[11]	-2.626772e-02	0.22058400	-0.379814310	0.3328285692	1.0016180
a_track[12]	-1.112962e-01	0.21145261	-0.438745543	0.2329777120	1.0007688
a_track[13]	-1.068260e-01	0.20588692	-0.430412519	0.2270574750	1.0001234
a_track[14]	-2.416467e-02	0.21833020	-0.369412183	0.3272963581	1.0024537
a_track[15]	4.462505e-02	0.22737978	-0.318552429	0.4084420554	1.0005367
a_track[16]	1.364397e-01	0.22437160	-0.211134726	0.4954251475	1.0005266
a_track[17]	1.344523e-01	0.23773527	-0.239146214	0.5161973113	1.0012231
a_track[18]	-9.718829e-02	0.21095085	-0.430826523	0.2478937895	1.0024682
a_track[19]	-1.097754e-01	0.20457734	-0.431276509	0.2227252344	1.0023201
a_track[20]	-3.134196e-01	0.20381789	-0.638807626	0.0113686458	0.9998787
a_track[21]	-2.544562e-01	0.20281725	-0.574682139	0.0784966580	1.0018813
a_track[22]	-1.784411e-01	0.20970216	-0.506752731	0.1535231680	1.0022010
a_track[23]	-1.780253e-01	0.20291461	-0.490903304	0.1428172924	1.0001018
a_track[24]	9.594414e-02	0.27474753	-0.344659260	0.5278794709	0.9997055
a_track[25]	-1.119906e-02	0.22466655	-0.370036999	0.3575149989	1.0006620
a_track[26]	-1.586679e-01	0.21571064	-0.496146355	0.1894792325	0.9998200
a_track[27]	3.302092e-01	0.28095297	-0.118299708	0.7782228128	1.0017717
a_track[28]	4.020375e-01	0.32906702	-0.125282060	0.9291091514	1.0026442
a_track[29]	3.806747e-01	0.26916143	-0.040720697	0.8199914913	1.0031243
a_track[30]	1.919295e-01	0.23724194	-0.181165323	0.5802338413	1.0022129
a_track[31]	1.025104e-01	0.22845368	-0.257703496	0.4741303092	0.9998107
a_track[32]	-1.531372e-01	0.20880159	-0.472631110	0.1870050772	1.0023912
a_track[33]	6.114854e-03	0.21964827	-0.343605634	0.3540807539	1.0009908
a_track[34]	9.316732e-03	0.21672908	-0.336785698	0.3575737409	1.0004305
a_track[35]	-7.175232e-02	0.21486852	-0.409800713	0.2740337440	1.0013324
a_track[36]	9.577877e-03	0.21804449	-0.342277825	0.3628314771	1.0004801
a_track[37]	-2.900415e-01	0.20835038	-0.628053769	0.0416242425	1.0016262
a_track[38]	-3.004489e-01	0.20745226	-0.619863927	0.0400266414	1.0009531
a_track[39]	-4.273982e-01	0.19430079	-0.742736722	-0.1143658036	1.0012302
a_track[40]	-5.389111e-01	0.18709765	-0.836368544	-0.2442831712	1.0019361
a_track[41]	-8.017437e-02	0.21808598	-0.427555565	0.2775385915	0.9997728
a_track[42]	4.112451e-01	0.29434672	-0.056656042	0.8865955464	1.0012939

a_track[43]	-2.169618e-01	0.21374691	-0.553372749	0.1276494903	1.0020116
a_track[44]	-1.428218e-01	0.22282677	-0.496672909	0.2214939170	1.0014126
a_track[45]	1.908462e-01	0.24867705	-0.198097758	0.5970768479	0.9997872
a_track[46]	-5.048056e-03	0.22398291	-0.364439851	0.3420170322	0.9997040
a_track[47]	7.426245e-02	0.22816408	-0.289724509	0.4444869775	1.0025439
a_track[48]	1.480431e-01	0.23378217	-0.221148327	0.5266830129	1.0012657
a_track[49]	2.306244e-01	0.23791349	-0.153156751	0.6129891219	1.0002112
a_track[50]	2.351405e-02	0.25291693	-0.370987984	0.4312343387	1.0009874
a_track[51]	-4.227226e-01	0.22521639	-0.784385405	-0.0591706299	1.0004801
a_track[52]	1.230974e-01	0.26940721	-0.310358558	0.5537006030	1.0017198
a_track[53]	3.123883e-01	0.29592723	-0.156076552	0.7822956366	1.0008125
a_track[54]	-7.129497e-02	0.25696808	-0.484844116	0.3431122381	1.0007508
a_track[55]	5.677176e-01	0.30688848	0.083836379	1.0523500900	1.0028559
a_track[56]	-4.937936e-01	0.23067083	-0.854675251	-0.1292452959	1.0021594
a_track[57]	1.253291e-01	0.25690402	-0.283145209	0.5460303700	1.0000049
a_track[58]	-2.538452e-01	0.24364794	-0.638667112	0.1293854806	1.0000596
a_track[59]	-2.477426e-01	0.23727655	-0.623345609	0.1287593844	1.0011344
a_track[60]	4.497221e-01	0.29666094	-0.009776617	0.9409100676	1.0002893
a_track[61]	-6.912851e-02	0.23683692	-0.444975031	0.3105336325	1.0002332
a_track[62]	6.488030e-01	0.31124886	0.159411264	1.1451094520	1.0007917
a_track[63]	1.155743e-01	0.24818475	-0.273281853	0.5143066604	1.0007083
a_track[64]	-3.718035e-01	0.22741953	-0.729571273	-0.0046385047	1.0015041
a_track[65]	2.868191e-01	0.36514110	-0.301627420	0.9004310400	1.0001767
a_track[66]	-4.426311e-01	0.22697079	-0.805790995	-0.0780767073	1.0013850
a_track[67]	2.700169e-01	0.30226620	-0.205084114	0.7714086953	1.0042992
a_track[68]	-5.936497e-02	0.24198468	-0.442629443	0.3357419757	0.9997807
a_track[69]	-1.403057e-01	0.23232155	-0.504811240	0.2388121522	1.0017241
a_track[70]	-1.415362e-01	0.23266234	-0.516247922	0.2337659566	1.0009577
a_track[71]	-1.351790e-01	0.23630318	-0.503519547	0.2433145495	1.0012388
a_track[72]	-8.815708e-03	0.26772118	-0.431280188	0.4203308669	1.0003713
a_track[73]	2.306715e-01	0.28802183	-0.220081998	0.6823459670	0.9994940
a_track[74]	-2.921764e-01	0.23913785	-0.674462709	0.0896851840	1.0000765
a_track[75]	2.341845e-01	0.28954965	-0.213873835	0.7042186906	1.0005717
a_track[76]	-1.922246e-01	0.25249853	-0.580333559	0.2252204232	0.9998529
a_track[77]	-1.917814e-01	0.24105404	-0.576691532	0.2012058964	1.0014913
a_track[78]	1.488677e-01	0.26053343	-0.272837138	0.5692621372	1.0000796
a_track[79]	-1.911708e-01	0.24750609	-0.576140762	0.2073857005	1.0010885
a_track[80]	-2.915053e-01	0.23850552	-0.665400936	0.0958023317	1.0003016
a_track[81]	-4.920715e-03	0.26190168	-0.414404482	0.4090182660	1.0004475
a_track[82]	1.231300e-01	0.26567644	-0.288885464	0.5531985593	1.0029535
a_track[83]	-1.071756e-01	0.26622132	-0.536100810	0.3122705975	1.0015951
a_track[84]	5.314563e-01	0.30278556	0.056593635	1.0293253760	1.0022679
a_track[85]	-2.114112e-01	0.24904409	-0.615218866	0.1925977986	1.0017234

a_track[86]	-2.034533e-01	0.25291003	-0.603626653	0.1966761363	1.0010002
a_track[87]	-2.068497e-01	0.24344081	-0.602250394	0.1898252891	1.0025712
a_track[88]	3.542227e-01	0.29828887	-0.127037858	0.8392225843	1.0007652
a_track[89]	4.806669e-01	0.31181265	-0.005083657	0.9837989127	1.0021701
a_track[90]	2.390858e-01	0.28709166	-0.212481797	0.6903771254	1.0000809
a_track[91]	-1.225279e-01	0.24830447	-0.505039088	0.2832421179	1.0002490
a_track[92]	-4.246604e-01	0.23681085	-0.798629304	-0.0427812166	0.9998186
a_track[93]	-1.170474e-01	0.25467360	-0.526479919	0.2988130077	0.9999316
a_track[94]	3.376584e-01	0.31136277	-0.163466739	0.8406961291	1.0004806
a_track[95]	1.324405e-01	0.28064583	-0.309270364	0.5866103495	1.0004330
a_track[96]	-9.330665e-02	0.25480299	-0.502965692	0.3171970372	1.0013936
a_track[97]	-3.873428e-01	0.22898262	-0.741207913	-0.0196287252	1.0014343
a_track[98]	3.436386e-01	0.30772387	-0.148642170	0.8419288619	0.9999154
a_track[99]	2.342123e-02	0.26602847	-0.404605259	0.4478530454	1.0006456
a_track[100]	3.844023e-01	0.30116361	-0.083993339	0.8773763825	1.0012665
a_track[101]	2.367787e-01	0.28639474	-0.218597095	0.6914700587	1.0016810
a_track[102]	2.375828e-01	0.37090255	-0.358282615	0.8384218863	1.0017411
a_track[103]	1.328627e-01	0.27413213	-0.310797171	0.5792091195	1.0015171
a_track[104]	-8.851360e-02	0.24999826	-0.487927883	0.3178087548	1.0003235
a_track[105]	-1.968267e-01	0.24304752	-0.579430752	0.2048540721	1.0053463
a_track[106]	-1.984749e-01	0.24058053	-0.570019482	0.1854869610	1.0004842
a_track[107]	2.618159e-01	0.28033860	-0.181777433	0.7150286876	1.0002080
a_track[108]	-3.759422e-01	0.19750530	-0.688115509	-0.0731262642	1.0032986
a_track[109]	-1.053813e-01	0.20754610	-0.437170212	0.2262127456	1.0033952
a_track[110]	-2.527546e-01	0.21004596	-0.580173130	0.0809917355	1.0019776
a_track[111]	5.048634e-01	0.33874293	-0.050591041	1.0428100300	1.0007851
a_track[112]	-3.731965e-01	0.20782085	-0.706298360	-0.0373367774	1.0008232
a_track[113]	3.433166e-01	0.35837174	-0.224350955	0.9204572551	1.0011766
a_track[114]	2.844899e-01	0.32728953	-0.231988095	0.8126214996	1.0008308
a_track[115]	3.358970e-01	0.31474833	-0.166168898	0.8235449992	0.9995173
a_track[116]	3.037918e-01	0.32894874	-0.235729290	0.8386284340	1.0028984
a_track[117]	3.038085e-01	0.27583047	-0.125522771	0.7444708161	1.0052585
a_track[118]	1.949831e-01	0.27163511	-0.229501226	0.6328818458	1.0017060
a_track[119]	3.853224e-01	0.33980805	-0.163853023	0.9185865649	1.0002009
a_track[120]	1.410209e-01	0.28097441	-0.301670774	0.6056856093	1.0005855
a_track[121]	3.802135e-01	0.30510226	-0.096486822	0.8783850549	1.0013615
a_track[122]	2.422461e-01	0.28855173	-0.216758666	0.6980815340	1.0003491
a_track[123]	-1.925150e-01	0.24072473	-0.573321813	0.1920937358	0.9996342
a_track[124]	3.105122e-01	0.33241118	-0.228111732	0.8409512482	1.0034097
a_track[125]	2.634421e-01	0.35147948	-0.298535213	0.8273118843	1.0018892
a_track[126]	-4.465999e-01	0.22121516	-0.803083932	-0.0856936622	1.0011855
a_track[127]	-1.079626e-01	0.24474589	-0.503169899	0.2899684516	1.0019543
a_track[128]	-4.448501e-01	0.23021709	-0.810939472	-0.0751191781	1.0013516

a_track[129]	-3.451728e-01	0.22241791	-0.703208612	0.0170563610	1.0010343
a_track[130]	2.938505e-01	0.32056946	-0.214246134	0.8126887793	1.0031007
a_track[131]	1.591713e-01	0.26543094	-0.253444332	0.5867374099	1.0009322
a_track[132]	-6.075406e-01	0.21515944	-0.947725540	-0.2540510815	1.0005020
a_track[133]	1.200733e-01	0.28129020	-0.318742271	0.5690957895	1.0035744
a_track[134]	-5.342078e-01	0.21524990	-0.876343635	-0.1944445974	1.0031817
a_track[135]	3.197379e-01	0.30833302	-0.175324190	0.8144294615	1.0029224
a_track[136]	4.073590e-01	0.34526257	-0.148872112	0.9663797715	1.0005436
a_track[137]	3.566438e-02	0.26051085	-0.375641144	0.4451788581	0.9996317
a_track[138]	-6.117008e-01	0.21004306	-0.959135061	-0.2690592615	1.0013615
a_track[139]	1.514289e-01	0.26481305	-0.268931283	0.5761860634	1.0027244
a_track[140]	2.456867e-01	0.28669223	-0.212149482	0.6890293162	1.0032193
a_track[141]	4.453740e-01	0.32104004	-0.072850782	0.9738726955	0.9993499
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a_track[143]	-2.851791e-01	0.22812886	-0.649434895	0.0928070958	1.0023847
a_track[144]	1.877961e-01	0.24670323	-0.200675442	0.5880681649	1.0001254
a_track[145]	2.884062e-01	0.26619579	-0.134935020	0.7184398780	1.0020208
a_track[146]	-5.009761e-01	0.21138825	-0.839912723	-0.1593419772	1.0013786
a_track[147]	-1.160742e-01	0.23014617	-0.478645503	0.2522685637	1.0002196
a_track[148]	1.834497e-01	0.25040407	-0.209268683	0.5957603545	1.0007449
a_track[149]	7.546228e-02	0.23884196	-0.299640779	0.4629692549	1.0008622
a_track[150]	-1.512076e-02	0.23291262	-0.378665952	0.3651090734	1.0033037
a_track[151]	2.945528e-01	0.26426416	-0.120090307	0.7278583774	1.0014736
a_track[152]	5.338221e-01	0.28453660	0.089565187	0.9885570799	1.0003997
a_track[153]	-7.154486e-01	0.19107805	-1.015031286	-0.4063169634	1.0001367
a_track[154]	-4.847507e-01	0.20416311	-0.807858705	-0.1557823792	1.0009319
a_track[155]	-4.157603e-01	0.20934097	-0.746063266	-0.0754613872	0.9998356
a_track[156]	-2.683910e-01	0.21040435	-0.599988617	0.0677263797	1.0009561
a_track[157]	9.066124e-02	0.24161329	-0.288400287	0.4741869103	1.0002069
a_track[158]	8.675946e-02	0.24331732	-0.304233079	0.4827793430	1.0022249
a_track[159]	-2.073020e-01	0.22432405	-0.559817739	0.1574147580	1.0016595
a_track[160]	2.962613e-02	0.24093362	-0.363588061	0.4125779720	1.0007935
a_track[161]	-4.906849e-02	0.23225985	-0.408292912	0.3240498400	1.0017523
a_track[162]	-3.356579e-01	0.22000615	-0.684672115	-0.0001332224	1.0014741
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a_track[164]	-5.107847e-01	0.21233039	-0.838785695	-0.1716638357	1.0016935
a_track[165]	1.926901e-01	0.26038370	-0.216095396	0.6078385027	1.0011455
a_track[166]	-7.425812e-01	0.19986299	-1.064901226	-0.4215529767	1.0023561
a_track[167]	-4.280868e-01	0.21055552	-0.764846083	-0.0962391795	1.0003947
a_track[168]	3.871651e-01	0.27951786	-0.052701224	0.8419507253	1.0003783
a_track[169]	1.607775e-01	0.26904287	-0.261984657	0.5911838601	1.0010142
a_track[170]	1.705504e-02	0.25352070	-0.377834310	0.4231478370	1.0011630
a_track[171]	5.425362e-01	0.30387823	0.061353914	1.0387688570	1.0012335

a_track[172]	-8.616327e-02	0.24642476	-0.476684881	0.3122072824	1.0000257
a_track[173]	2.181789e-01	0.30882350	-0.277632855	0.7066100288	1.0011458
a_track[174]	-8.419796e-02	0.25280914	-0.485739405	0.3248132975	1.0005145
a_track[175]	1.384838e-01	0.27634988	-0.301488418	0.5875259976	1.0000610
a_track[176]	2.810691e-01	0.32165910	-0.236187097	0.7915689199	1.0019913
a_track[177]	5.491660e-01	0.30827200	0.058630242	1.0643115970	1.0008448
a_track[178]	3.114053e-02	0.24997393	-0.357158774	0.4295632264	1.0000857
a_track[179]	3.041407e-01	0.27944417	-0.133995973	0.7649764630	1.0010463
a_track[180]	-8.514032e-02	0.25025227	-0.473898020	0.3289323514	1.0014203
a_track[181]	-5.245887e-01	0.22984424	-0.895775182	-0.1558571764	1.0037061
a_track[182]	-8.836678e-02	0.29543961	-0.551448374	0.4033758416	1.0020234
a_track[183]	-1.101615e+00	0.22386931	-1.454473102	-0.7368466840	1.0017400
a_track[184]	-7.483703e-01	0.25394466	-1.147279756	-0.3420126621	1.0012644
a_track[185]	-3.677949e-01	0.27231032	-0.800778978	0.0718038013	1.0019795
a_track[186]	-6.343022e-01	0.26110698	-1.046788088	-0.2131653413	1.0004282
a_track[187]	-2.284165e-01	0.28380681	-0.679028557	0.2327172970	1.0009176
a_track[188]	-3.527278e-01	0.28173857	-0.795262706	0.0979554911	1.0002215
a_track[189]	-6.896811e-02	0.31835554	-0.573113750	0.4483075056	1.0034052
a_track[190]	2.191974e-01	0.33351529	-0.306470119	0.7492997779	0.9996769
a_track[191]	-6.220291e-01	0.25697267	-1.026138712	-0.1912604683	1.0017328
a_track[192]	7.470245e-02	0.31414266	-0.423046322	0.5810888927	1.0015710
a_track[193]	5.206673e-02	0.32660992	-0.471470987	0.5692085317	0.9998656
a_track[194]	2.665128e-01	0.32698474	-0.244384520	0.8006141798	1.0048916
a_track[195]	2.746172e-01	0.31205889	-0.210330204	0.7782600919	1.0005269
a_track[196]	2.156155e-01	0.33016189	-0.307321204	0.7375074862	1.0015691
a_track[197]	2.775060e-01	0.32731798	-0.246199478	0.8033931999	1.0028436
a_track[198]	-5.920532e-01	0.26076303	-1.008514995	-0.1704915876	0.9999784
a_track[199]	2.611476e-01	0.32999136	-0.252490619	0.7800546645	0.9994130
a_track[200]	-7.392220e-01	0.24533984	-1.123355926	-0.3350711696	0.9997864
a_track[201]	2.629273e-01	0.31766335	-0.226685337	0.7740267332	1.0011952
a_track[202]	-1.303050e+00	0.22738603	-1.662152483	-0.9428097769	1.0003064
a_track[203]	4.497475e-02	0.32128934	-0.468988746	0.5657183294	1.0023832
a_track[204]	9.912178e-02	0.31026800	-0.398192653	0.6131187077	1.0002238
a_track[205]	-2.242045e-01	0.27634223	-0.658247145	0.2215830236	1.0008552
a_track[206]	-6.689106e-02	0.29605165	-0.534173476	0.4086559479	1.0002217
a_track[207]	1.637317e-01	0.34128102	-0.383025294	0.7128238703	1.0001886
a_track[208]	8.595768e-02	0.32886428	-0.437081611	0.6193770423	1.0048419
a_track[209]	-4.331492e-02	0.29802284	-0.511487637	0.4372442805	1.0034886
a_track[210]	-3.240569e-01	0.28108349	-0.761064308	0.1274166884	0.9998530
a_track[211]	2.719671e-01	0.32426221	-0.243604937	0.7792068161	1.0024933
a_track[212]	-3.180733e-01	0.27153643	-0.751232178	0.1219518813	1.0014231
a_track[213]	1.752433e-01	0.34837193	-0.372724381	0.7258068294	1.0010704
a_track[214]	-3.798025e-02	0.30103872	-0.527348321	0.4357644695	1.0043998

a_track[215]	-3.570053e-02	0.30365675	-0.514912630	0.4493578159	0.9996715
a_track[216]	9.216831e-02	0.32273597	-0.421925832	0.6146227182	1.0002794
a_track[217]	8.811196e-02	0.32117292	-0.418066211	0.6058680800	1.0000696
a_track[218]	-4.125983e-02	0.29828773	-0.523316608	0.4465799475	1.0000566
a_track[219]	1.531177e-01	0.35753005	-0.419656874	0.7199519066	1.0016571
a_track[220]	-3.603601e-01	0.26668372	-0.791114300	0.0836960302	1.0005758
a_track[221]	-3.929911e-01	0.27705366	-0.826287164	0.0458871533	1.0021212
a_track[222]	8.423261e-02	0.33412705	-0.437867990	0.6257863275	1.0006763
a_track[223]	1.450726e-01	0.36778310	-0.441109976	0.7441768834	1.0005564
a_track[224]	1.078744e-01	0.31011582	-0.383153467	0.6037004078	1.0007980
a_track[225]	9.272313e-02	0.31730225	-0.411285154	0.5951507899	1.0007497
a_track[226]	9.892135e-02	0.31609710	-0.398039830	0.6096556533	1.0024700
a_track[227]	1.152921e-01	0.30308220	-0.366863394	0.6031584481	1.0011166
a_track[228]	2.291765e-01	0.33145686	-0.284789216	0.7629908754	1.0002599
a_track[229]	-4.202943e-02	0.30663442	-0.527277997	0.4477046044	1.0023004
a_track[230]	2.291261e-01	0.33462529	-0.303582821	0.7627870170	1.0010808
a_track[231]	2.690357e-01	0.31885119	-0.244470175	0.7739738568	1.0015793
a_track[232]	2.350761e-01	0.32663835	-0.287670437	0.7659258400	1.0028392
a_track[233]	-4.123227e-02	0.30134805	-0.528435245	0.4354537695	0.9998755
a_track[234]	8.715843e-02	0.32259600	-0.429888053	0.5995766494	1.0002759
a_track[235]	-1.827528e-01	0.28449110	-0.631235774	0.2764264212	1.0045534
a_track[236]	1.121684e-01	0.32505893	-0.404668041	0.6471847081	1.0012407
a_track[237]	2.269938e-01	0.32848485	-0.301330469	0.7533650619	1.0004765
a_track[238]	-1.132290e+00	0.23215630	-1.505539292	-0.7556497924	1.0009027
a_track[239]	-4.327042e-01	0.30368913	-0.917631938	0.0566156027	1.0032547
a_track[240]	-5.544832e-01	0.27138179	-0.989796347	-0.1235588046	1.0002789
a_track[241]	-7.270493e-02	0.29477917	-0.529499079	0.4006742738	1.0012385
a_track[242]	-5.706568e-01	0.26976189	-0.995149990	-0.1326111573	1.0003013
a_track[243]	-2.080050e-01	0.29290720	-0.683599938	0.2658461417	1.0010143
a_track[244]	-7.003271e-02	0.31129122	-0.570639116	0.4320794816	1.0019213
a_track[245]	7.818432e-02	0.31109840	-0.418942498	0.5771518879	1.0014200
a_track[246]	2.269843e-01	0.32884225	-0.295839322	0.7468697893	1.0030441
a_track[247]	-8.754816e-02	0.30371101	-0.568631963	0.4086464952	1.0026020
a_track[248]	2.315303e-01	0.33440648	-0.299150803	0.7544173078	0.9997601
a_track[249]	1.795023e-01	0.34796085	-0.375807868	0.7457859092	1.0010889
a_track[250]	2.198770e-01	0.31975333	-0.295900445	0.7334007961	1.0014817
a_track[251]	1.666213e-01	0.34408058	-0.379083721	0.7149670918	1.0007453
a_track[252]	-8.665042e-02	0.29899760	-0.557500014	0.3996780988	1.0022960
a_track[253]	6.772868e-02	0.31333461	-0.431873410	0.5749380240	0.9996903
a_track[254]	5.669697e-02	0.31906409	-0.456917553	0.5733259840	1.0016183
a_track[255]	-2.260158e-01	0.29943069	-0.706410728	0.2697708339	1.0010668
a_track[256]	-7.295109e-02	0.30258577	-0.547730587	0.4088769575	1.0012540
a_track[257]	2.106026e-01	0.34035703	-0.330908920	0.7658275264	1.0008926

a_track[258]	1.943603e-01	0.33791239	-0.342032038	0.7480061166	1.0016388
a_track[259]	1.978019e-01	0.34235719	-0.349674081	0.7581682810	1.0014349
a_track[260]	-8.069575e-02	0.29536583	-0.543319856	0.3938209701	1.0021203
a_track[261]	7.119403e-02	0.30988194	-0.413799235	0.5578972664	1.0020009
a_track[262]	7.757542e-02	0.31592789	-0.422027742	0.5853775328	1.0000620
a_track[263]	7.432501e-02	0.29988178	-0.394015624	0.5599624304	1.0009740
a_track[264]	7.126556e-02	0.31571269	-0.435108205	0.5806316601	1.0021453
a_track[265]	2.033048e-01	0.32966608	-0.329583564	0.7223972207	0.9999284
a_track[266]	7.655783e-02	0.31687139	-0.417655459	0.5930322976	1.0026126
a_track[267]	-7.786086e-02	0.29199171	-0.533122134	0.3965942926	1.0023844
a_track[268]	6.276110e-02	0.33084410	-0.471277258	0.5932503970	1.0020374
a_track[269]	-7.117934e-02	0.31342218	-0.573766186	0.4355419460	1.0008620
a_track[270]	2.219104e-01	0.32915827	-0.303823278	0.7527290765	1.0028936
a_track[271]	-7.262881e-02	0.30318001	-0.541818724	0.4090444736	1.0005891
a_track[272]	6.725047e-02	0.31490060	-0.432625848	0.5573744305	1.0026133
a_track[273]	-8.555718e-02	0.29098110	-0.544017647	0.3847554344	1.0015076
a_track[274]	1.787640e-01	0.33787400	-0.356830319	0.7254185329	1.0036596
a_track[275]	-8.411011e-02	0.30560128	-0.580883094	0.4033957469	1.0005945
a_track[276]	6.930097e-02	0.32542237	-0.441012519	0.5923567344	1.0019121
a_track[277]	6.879227e-02	0.31831401	-0.433631887	0.5782904391	0.9998037
a_track[278]	-8.388832e-02	0.29190418	-0.547020351	0.3918795154	1.0010298
a_track[279]	5.051909e-02	0.32961709	-0.476790987	0.5792717280	0.9998217
a_track[280]	1.784305e-01	0.35037728	-0.370796863	0.7435281747	1.0030036
a_track[281]	1.424230e-01	0.35095362	-0.428338707	0.6960744366	0.9996320
a_track[282]	2.020524e-01	0.33546500	-0.327127593	0.7410312498	1.0011685
a_track[283]	-2.202013e-01	0.29115641	-0.684020794	0.2511160390	1.0010012
a_track[284]	2.473850e-01	0.32109065	-0.265725282	0.7678753674	1.0003536
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a_track[286]	7.351039e-02	0.31498912	-0.426076133	0.5809771306	1.0007811
a_track[287]	1.874655e-01	0.34765174	-0.367921058	0.7362979063	1.0023833
a_track[288]	1.689052e-01	0.35199871	-0.402525626	0.7311362950	1.0040363
a_track[289]	1.791161e-01	0.34454161	-0.377853469	0.7378976359	1.0013989
a_track[290]	-1.992189e-01	0.30113912	-0.680906224	0.2800966812	1.0031133
a_track[291]	4.107129e-02	0.34661635	-0.512618349	0.5979877099	1.0008142
a_track[292]	-6.204660e-01	0.25737091	-1.035865896	-0.2095617977	1.0024562
a_track[293]	5.720915e-02	0.33629621	-0.483500053	0.5870150598	1.0023260
a_track[294]	-2.034539e-01	0.29378600	-0.669345243	0.2668379403	1.0022747
a_track[295]	-7.113177e-02	0.30001107	-0.543719630	0.4054016222	1.0013374
a_track[296]	5.826284e-02	0.33520816	-0.472549502	0.5991968217	1.0018227
a_track[297]	1.984812e-01	0.34200739	-0.345299945	0.7428136559	1.0012150
a_track[298]	2.519928e-01	0.32182898	-0.260782433	0.7832141126	1.0017887
a_track[299]	7.517503e-02	0.32746273	-0.444342262	0.6059787866	1.0022346
a_track[300]	2.349336e-01	0.31918856	-0.276411774	0.7505239070	1.0003349

a_track[301]	2.405911e-01	0.33919684	-0.299268733	0.7896488395	1.0002348
a_track[302]	8.316717e-02	0.31363058	-0.414965233	0.5803573915	1.0008272
a_track[303]	-6.591270e-02	0.30330534	-0.541973878	0.4350395676	1.0024872
a_track[304]	7.789556e-02	0.31600467	-0.424374912	0.5942840847	1.0046958
a_track[305]	-5.473677e-02	0.30693693	-0.528315038	0.4417048680	1.0014679
a_track[306]	1.715677e-01	0.34830281	-0.389615002	0.7174478312	1.0014079
a_track[307]	2.369827e-01	0.32478641	-0.274127884	0.7567372592	1.0012269
a_track[308]	8.364765e-02	0.32034654	-0.415863130	0.5941696420	1.0009148
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a_track[310]	8.304764e-02	0.31329014	-0.417011742	0.5865206851	1.0000204
a_track[311]	1.862152e-01	0.34776513	-0.361305424	0.7475154699	1.0017695
a_track[312]	-2.097510e-01	0.29551609	-0.683584745	0.2680979257	1.0006372
a_track[313]	-6.249281e-02	0.29420450	-0.532867804	0.4266327434	1.0017490
a_track[314]	8.984745e-02	0.31647884	-0.411915542	0.6006706822	1.0016268
a_track[315]	-2.072342e-01	0.28180783	-0.659722824	0.2503938937	1.0052462
a_track[316]	-6.264516e-02	0.29783778	-0.535623513	0.4192665139	1.0011105
a_track[317]	1.015539e-01	0.30771906	-0.375030018	0.6044014041	1.0027495
a_track[318]	7.708761e-02	0.31146793	-0.407167803	0.5790243044	1.0002578
a_track[319]	-1.641164e-01	0.31385902	-0.657809394	0.3278527029	1.0035009
a_track[320]	-5.586085e-02	0.31301039	-0.549837525	0.4513610163	1.0002640
a_track[321]	2.172829e-01	0.34067050	-0.323585182	0.7566961860	1.0023489
a_track[322]	-1.996098e-01	0.28438686	-0.647405735	0.2638121024	1.0025211
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a_track[324]	8.056331e-02	0.32715432	-0.437894528	0.6162971286	1.0014484
a_track[325]	1.501991e-01	0.36605143	-0.453753558	0.7334427651	1.0008811
a_track[326]	-6.579679e-02	0.29723364	-0.542848266	0.4200014413	1.0002550
a_track[327]	7.552784e-02	0.31767484	-0.414052123	0.5954433632	1.0025429
a_track[328]	2.717304e-01	0.31835224	-0.224981478	0.7775794207	1.0011567
a_track[329]	2.481538e-01	0.33860284	-0.286820372	0.7989492033	1.0005818
a_track[330]	7.843672e-02	0.31433336	-0.419080084	0.5820098908	0.9996266
a_track[331]	2.428648e-01	0.32083711	-0.257711652	0.7538656584	1.0019269
a_track[332]	7.745293e-02	0.32313938	-0.442621129	0.5844362902	0.9993720
a_track[333]	-2.020406e-01	0.29212618	-0.655298874	0.2773110259	1.0008349
a_track[334]	9.127310e-02	0.31277497	-0.397494155	0.5801881979	1.0002989
a_track[335]	9.566226e-02	0.30458258	-0.378960942	0.5888686183	1.0024404
a_track[336]	7.805474e-02	0.32387243	-0.435566736	0.5898254020	1.0010671
a_track[337]	2.051444e-01	0.32704729	-0.311929216	0.7430914701	1.0029534
a_track[338]	2.117124e-01	0.33738882	-0.319698155	0.7444326640	1.0023494
a_track[339]	-5.829138e-02	0.29637521	-0.522511640	0.4193877328	1.0009997
a_track[340]	-2.082381e-01	0.28389536	-0.649893375	0.2540611796	1.0056866
a_track[341]	-6.830406e-02	0.29291841	-0.531997976	0.4080981691	1.0029104
a_track[342]	-3.525561e-01	0.27770743	-0.784296020	0.0932668123	1.0008060
a_track[343]	8.328056e-02	0.31584326	-0.415313034	0.5934849399	1.0015327

a_track[344]	-2.251294e-01	0.27838766	-0.665335216	0.2193665179	1.0003722
a_track[345]	2.213247e-01	0.32039411	-0.289841870	0.7382763959	1.0039849
a_track[346]	2.277580e-01	0.32654594	-0.292895433	0.7579448695	1.0011514
a_track[347]	-7.634480e-02	0.29610158	-0.554409870	0.3916434994	0.9999687
a_track[348]	6.790608e-02	0.33410147	-0.459427217	0.5964770515	1.0011092
a_track[349]	5.562585e-02	0.33744558	-0.481693469	0.5814365895	1.0010310
a_track[350]	2.148344e-01	0.32397755	-0.304143141	0.7308394281	1.0010408
a_track[351]	-2.406361e-01	0.28993975	-0.698399220	0.2247071889	1.0015615
a_track[352]	1.750020e-01	0.35212603	-0.376088088	0.7500373985	1.0011128
a_track[353]	7.594166e-02	0.30589570	-0.403530611	0.5702150988	1.0014235
a_track[354]	6.782043e-02	0.32218039	-0.436087999	0.5821809215	1.0025759
a_track[355]	-8.833760e-02	0.29433084	-0.562327127	0.3842779308	1.0020833
a_track[356]	-2.216985e-01	0.28998650	-0.681017616	0.2529493877	1.0013387
a_track[357]	4.905938e-02	0.33414316	-0.494314138	0.5925606255	1.0008984
a_track[358]	1.334263e-01	0.34445535	-0.411452675	0.6816662178	1.0013433
a_track[359]	4.841182e-02	0.33837609	-0.491580568	0.5827897266	1.0054337
a_track[360]	6.774407e-02	0.32575027	-0.457920686	0.5979570907	1.0006109
a_track[361]	1.882474e-01	0.34235860	-0.350764609	0.7455302234	1.0001532
a_track[362]	5.682773e-02	0.31664873	-0.451501501	0.5627092331	1.0010598
a_track[363]	-8.660365e-02	0.28710320	-0.527653257	0.3680823315	1.0059985
a_track[364]	-8.610437e-02	0.30034411	-0.562811781	0.3966854223	0.9999213
a_track[365]	-8.549036e-02	0.30164430	-0.559126727	0.4036500694	1.0002147
a_track[366]	3.573958e-02	0.35373036	-0.513978965	0.5983047995	1.0005209
a_track[367]	4.871982e-02	0.33314608	-0.497281782	0.5779720770	1.0011676
a_track[368]	1.572376e-01	0.34180199	-0.385623600	0.7124101653	1.0008106
a_track[369]	5.260883e-02	0.33436071	-0.475242711	0.5928505739	1.0008011
a_track[370]	6.471904e-02	0.32482838	-0.462383298	0.5967531602	1.0009585
a_track[371]	-4.898975e-01	0.27657476	-0.917612085	-0.0530642174	1.0008043
a_track[372]	-8.611840e-02	0.31285298	-0.592098619	0.4334767440	1.0027939
a_track[373]	-2.221513e-01	0.29057870	-0.677905148	0.2483448192	1.0019487
a_track[374]	5.212133e-02	0.32555093	-0.459387270	0.5803250072	1.0021774
a_track[375]	1.334823e-01	0.37041693	-0.436846728	0.7311519356	1.0030547
a_track[376]	-6.803274e-02	0.32892892	-0.591709929	0.4679887556	0.9998352
a_track[377]	2.131855e-01	0.32664567	-0.294703660	0.7304761253	1.0016493
a_track[378]	2.275840e-01	0.32266414	-0.289531715	0.7452255447	1.0016671
a_track[379]	1.867267e-01	0.34180808	-0.344694255	0.7306944664	0.9993370
a_track[380]	6.457890e-02	0.32519188	-0.457218847	0.5853935565	1.0015783
a_track[381]	8.069429e-02	0.31370859	-0.417627064	0.5863315306	1.0033007
a_track[382]	6.896739e-02	0.32765620	-0.452961260	0.5903940396	1.0008124
a_track[383]	2.730148e-01	0.32584112	-0.238831037	0.7995859647	1.0009062
a_track[384]	8.408141e-02	0.32051852	-0.433881189	0.6011953489	1.0013216
a_track[385]	-4.967922e-02	0.30085763	-0.529680349	0.4360408638	1.0021821
a_track[386]	-5.217783e-02	0.30990448	-0.546432643	0.4516529626	0.9999703

a_track[387]	2.209286e-01	0.34778629	-0.325062947	0.7797961919	1.0004103
a_track[388]	-5.297573e-02	0.29164361	-0.505963110	0.4217065524	1.0016172
a_track[389]	9.728841e-02	0.30965186	-0.390935456	0.5802125294	1.0019550
a_track[390]	-5.619617e-02	0.29389437	-0.524135343	0.4182931208	0.9997701
a_track[391]	-6.067755e-02	0.29921478	-0.540702656	0.4311168301	1.0007491
a_track[392]	-6.330116e-02	0.29086406	-0.523533365	0.4080564664	1.0028878
a_track[393]	2.119476e-01	0.33347153	-0.310791874	0.7566501774	1.0003481
a_track[394]	-2.170092e-01	0.27345265	-0.651860293	0.2261503847	1.0009531
a_track[395]	-5.921157e-02	0.29433093	-0.533166704	0.4182064793	1.0012072
a_track[396]	-6.361509e-02	0.29037653	-0.516899653	0.4126297375	1.0007208
a_track[397]	7.778720e-02	0.33494977	-0.460474395	0.6112189987	1.0031245
a_track[398]	8.653928e-02	0.30626969	-0.404539611	0.6007611207	1.0035412
a_track[399]	7.501911e-02	0.30898186	-0.412608064	0.5597517704	1.0033878
a_track[400]	9.221719e-02	0.30587308	-0.383304957	0.5888594068	1.0021867
a_track[401]	-5.515933e-02	0.30799706	-0.547018822	0.4363199225	1.0001511
a_track[402]	8.767499e-02	0.32506826	-0.431046742	0.6091864319	1.0013578
a_track[403]	-5.248129e-02	0.30665324	-0.541984103	0.4439994542	1.0010227
a_track[404]	-5.612388e-02	0.30522085	-0.547953524	0.4405837083	1.0007339
a_track[405]	7.843567e-02	0.32634440	-0.429511281	0.5925635139	1.0024159
a_track[406]	1.718752e-01	0.34535610	-0.379983478	0.7137357536	0.9998679
a_track[407]	-5.228440e-02	0.30185157	-0.520344764	0.4276032772	1.0024445
a_track[408]	-5.703783e-02	0.29647663	-0.536040397	0.4075789181	1.0005641
a_track[409]	-5.246237e-02	0.32176208	-0.573509722	0.4596086752	1.0008535
a_track[410]	-3.426099e-01	0.27882499	-0.782305995	0.1001700578	1.0010093
a_track[411]	4.453664e-02	0.34886905	-0.511782340	0.5968796686	1.0015370
a_track[412]	4.705963e-02	0.34570263	-0.516232720	0.5955449912	1.0012529
a_track[413]	-3.328219e-01	0.28469192	-0.786047515	0.1275214141	1.0073048
a_track[414]	1.911068e-01	0.33913933	-0.344675394	0.7309456298	1.0013370
a_track[415]	-5.460379e-02	0.28738790	-0.504812790	0.4008913219	1.0023895
a_track[416]	6.916975e-02	0.31740182	-0.431718050	0.5729960539	1.0009960
a_track[417]	2.205108e-01	0.33526914	-0.314538370	0.7467869780	1.0016656
a_track[418]	-6.173625e-02	0.29669133	-0.519987714	0.4064476304	1.0021176
a_track[419]	-6.345432e-02	0.30038586	-0.537257572	0.4221174765	1.0015520
a_track[420]	-5.984316e-02	0.30015292	-0.543559922	0.4221900008	1.0009579
a_track[421]	-6.133728e-02	0.29412401	-0.533886572	0.4060475762	1.0018302
a_track[422]	1.711072e-01	0.34976806	-0.377779115	0.7143120494	1.0008059
a_track[423]	8.789204e-02	0.30113099	-0.394377740	0.5825526614	1.0005914
a_track[424]	1.773666e-01	0.34259169	-0.371382397	0.7284603487	1.0014583
a_track[425]	-5.815834e-02	0.30316186	-0.527117585	0.4208138365	1.0022313
a_track[426]	9.102838e-02	0.29989178	-0.377744961	0.5626787466	1.0015532
a_track[427]	-2.166795e-01	0.29460917	-0.677111932	0.2632050568	1.0006855
a_track[428]	7.629742e-02	0.31915907	-0.427955529	0.5781403564	1.0002931
a_track[429]	-6.492831e-02	0.31200757	-0.561755904	0.4230827560	1.0006877

a_track[430]	-7.311353e-02	0.30802578	-0.554371428	0.4294303170	1.0023315
a_track[431]	7.348518e-02	0.30588502	-0.413077739	0.5866907805	1.0014332
a_track[432]	2.116447e-01	0.33156885	-0.320667669	0.7586925406	1.0035607
a_track[433]	7.128705e-02	0.32103058	-0.439864299	0.5700480907	1.0005687
a_track[434]	2.248842e-01	0.32928942	-0.292398178	0.7480018544	1.0000392
a_track[435]	-3.528867e-01	0.28329425	-0.790008951	0.0909338224	1.0016752
a_track[436]	1.920913e-01	0.35396633	-0.374138733	0.7494312136	1.0007557
a_track[437]	7.252952e-02	0.32697101	-0.449514666	0.6058798668	1.0018796
a_track[438]	-6.658935e-02	0.29823315	-0.534390457	0.4121417451	1.0019096
a_track[439]	8.448298e-02	0.30988334	-0.409985437	0.5856619013	1.0005784
a_track[440]	-6.910298e-02	0.30229123	-0.547028427	0.4124262889	1.0007666
a_track[441]	-7.701863e-02	0.29797745	-0.554204218	0.3938856703	1.0002946
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a_track[443]	-2.206394e-01	0.29151056	-0.686230965	0.2416167517	0.9997590
a_track[444]	-3.592695e-01	0.27805431	-0.799344578	0.0916865509	1.0016391
a_track[445]	-6.225550e-02	0.31409971	-0.561495172	0.4321725475	0.9997249
a_track[446]	7.257796e-02	0.32913344	-0.442297840	0.5947168639	1.0018698
a_track[447]	1.881845e-01	0.34595247	-0.359403415	0.7330919666	1.0012131
a_track[448]	6.283079e-02	0.32204287	-0.437371153	0.5789023309	1.0018456
a_track[449]	8.615460e-02	0.30146405	-0.390626576	0.5690897149	1.0021751
a_track[450]	-6.667713e-01	0.24937552	-1.057502076	-0.2614085334	0.9999268
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a_track[452]	-7.123785e-02	0.29573228	-0.534034809	0.4127599399	1.0021119
a_track[453]	-4.386690e-01	0.28601898	-0.895269967	0.0126120090	1.0014747
a_track[454]	4.598893e-02	0.34273889	-0.502176122	0.6031851039	1.0025197
a_track[455]	6.947185e-02	0.30281029	-0.412866091	0.5555537261	1.0020216
a_track[456]	9.676796e-02	0.30259768	-0.389882627	0.5875649339	1.0023329
a_track[457]	9.358108e-02	0.32769286	-0.420162017	0.6153393120	1.0044831
a_track[458]	-6.641774e-02	0.30086351	-0.546863671	0.4135156892	1.0044728
a_track[459]	8.528952e-02	0.31095380	-0.421679600	0.5761446953	1.0020428
a_track[460]	9.334251e-02	0.31611460	-0.403520040	0.6033261754	1.0024303
a_track[461]	1.842006e-01	0.34148075	-0.354332361	0.7275029187	1.0033890
a_track[462]	7.623601e-02	0.30720960	-0.424610222	0.5750064347	1.0015158
a_track[463]	-6.988524e-02	0.31292121	-0.561704952	0.4362809042	1.0014331
a_track[464]	-5.911151e-02	0.30819832	-0.545643215	0.4325315672	0.9997614
a_track[465]	7.013861e-02	0.32412164	-0.447479427	0.5804439590	1.0023968
a_track[466]	1.775218e-01	0.33816754	-0.373760974	0.7170319437	1.0012407
a_track[467]	-5.510199e-02	0.29675456	-0.522238647	0.4234210814	1.0013891
a_track[468]	1.661700e-01	0.35260633	-0.396419498	0.7306395922	1.0022114
a_track[469]	-6.300052e-02	0.29935700	-0.532988889	0.4189337690	1.0028670
a_track[470]	-6.331428e-02	0.32132302	-0.577103888	0.4544471873	1.0033423
a_track[471]	-2.015212e-01	0.28321694	-0.647765989	0.2518204492	1.0003313
a_track[472]	2.272927e-01	0.33489806	-0.306994424	0.7679323207	1.0006939

a_track[473]	8.527795e-02	0.30111203	-0.391508312	0.5610932980	1.0021335
a_track[474]	5.912602e-02	0.32493396	-0.464737783	0.5761953489	1.0010155
a_track[475]	1.025835e-01	0.30643910	-0.377050437	0.5863971852	1.0006343
a_track[476]	-5.667018e-02	0.30702373	-0.550093175	0.4319324428	1.0048375
a_track[477]	1.899829e-01	0.34432168	-0.341301303	0.7400636235	1.0030161
a_track[478]	7.242911e-02	0.31045403	-0.420756626	0.5773964696	1.0005681
a_track[479]	6.649104e-02	0.33506390	-0.462811764	0.6011847933	1.0000686
a_track[480]	2.061618e-01	0.34818056	-0.357729493	0.7711116221	1.0006749
a_track[481]	-5.905888e-02	0.30350523	-0.539500682	0.4258567462	1.0020256
a_track[482]	8.654067e-02	0.31814620	-0.418424010	0.6028446400	1.0004421
a_track[483]	8.700859e-02	0.31504061	-0.419944564	0.6036069705	1.0013781
a_track[484]	-1.910468e-01	0.29592739	-0.664987975	0.2901188520	1.0029422
a_track[485]	9.214872e-02	0.30826615	-0.403738865	0.5948629900	1.0051006
a_track[486]	-1.892167e-01	0.28555807	-0.642608069	0.2763014653	1.0027651
a_track[487]	-5.942398e-02	0.30365589	-0.546007853	0.4366827951	1.0039925
a_track[488]	2.028464e-01	0.34642574	-0.339488623	0.7623848873	1.0024134
a_track[489]	-5.611948e-02	0.30679931	-0.540950618	0.4331255516	1.0016360
a_track[490]	-6.308560e-02	0.30891855	-0.553248264	0.4261000777	1.0015107
a_track[491]	7.556832e-02	0.32541268	-0.448122510	0.5993457132	1.0033402
a_track[492]	8.113421e-02	0.32115306	-0.436298040	0.5911962285	1.0009889
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a_track[497]	-3.352685e-02	0.21626165	-0.378815617	0.3090535516	1.0022409
a_track[498]	-2.043378e-01	0.20936711	-0.533453796	0.1336158489	1.0019241
a_track[499]	-3.056105e-02	0.22441517	-0.374144583	0.3433856863	1.0012569
a_track[500]	3.544282e-01	0.25488278	-0.046461431	0.7896676403	1.0037007
a_track[501]	-3.304785e-02	0.25984518	-0.456777572	0.3862072985	1.0013080
a_track[502]	4.940679e-01	0.33220670	-0.045983805	1.0191415350	1.0030547
a_track[503]	-7.958169e-01	0.21660056	-1.139561874	-0.4433120205	1.0001625
a_track[504]	-1.024248e-01	0.24536884	-0.486582523	0.3035087998	1.0020585
a_track[505]	-5.759898e-01	0.22567018	-0.929784690	-0.2132279086	1.0019200
a_track[506]	-4.036152e-01	0.23350363	-0.780328361	-0.0314023748	1.0006719
a_track[507]	4.024040e-01	0.32958674	-0.135534985	0.9287749440	1.0002307
a_track[508]	1.258088e-01	0.23601903	-0.245732826	0.5017184260	0.9996962
a_track[509]	6.555910e-01	0.28804836	0.198393948	1.1167634630	1.0005793
a_track[510]	-5.339179e-01	0.22138899	-0.880974283	-0.1766503472	1.0014887
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a_track[513]	-2.984147e-01	0.21006334	-0.625118886	0.0416551468	1.0011117
a_track[514]	-3.098611e-01	0.20558812	-0.634093325	0.0226653459	1.0001928
a_track[515]	-3.011412e-01	0.20729155	-0.628269117	0.0266754191	1.0008393

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a_track[518]	-7.476606e-02	0.21423917	-0.408744788	0.2729941553	1.0014466
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a_track[520]	-1.428062e-01	0.21524040	-0.485400875	0.2064319652	1.0019173
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a_track[523]	-3.823964e-01	0.22261037	-0.729703317	-0.0205482935	1.0025388
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a_track[527]	-4.507160e-02	0.22440977	-0.403107257	0.3286745287	1.0023723
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a_track[542]	-1.008418e-01	0.20129885	-0.417187338	0.2297518363	0.9996086
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a_track[544]	1.144849e-01	0.21862714	-0.227091048	0.4648941908	1.0007177
a_track[545]	4.533255e-02	0.21471804	-0.281169192	0.3864568884	1.0003247
a_track[546]	-2.794520e-01	0.19672343	-0.600621283	0.0344735479	1.0028363
a_track[547]	5.300006e-01	0.32540116	0.015176042	1.0523120760	1.0045549
a_track[548]	3.260384e-01	0.24074580	-0.058884385	0.7106690485	0.9999257
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a_track[552]	1.058959e-01	0.20704468	-0.226070902	0.4326215914	1.0022189
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a_track[554]	-1.131025e-01	0.22633756	-0.468493688	0.2493713772	1.0013966
a_track[555]	-2.590484e-02	0.23579466	-0.406604839	0.3604643737	1.0017234
a_track[556]	2.521938e-01	0.25365053	-0.143416550	0.6683440567	1.0013912
a_track[557]	-1.915308e-02	0.22501420	-0.378530546	0.3387793617	0.9999725
a_track[558]	6.291818e-01	0.31008891	0.122841739	1.1166485005	1.0024137

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a_track[560]	3.599711e-01	0.25991267	-0.043773245	0.7742545734	1.0016172
a_track[561]	-4.452146e-01	0.19949872	-0.758227405	-0.1200831160	1.0014303
a_track[562]	4.441579e-01	0.28910782	-0.018618189	0.9064854095	0.9999157
a_track[563]	-1.673051e-01	0.21416307	-0.500387309	0.1827740937	1.0006775
a_track[564]	7.539632e-02	0.22689037	-0.277688222	0.4380025237	0.9996697
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a_track[567]	3.005638e-01	0.25277870	-0.097097377	0.7014100349	1.0001867
a_track[568]	3.118728e-01	0.36606710	-0.267325343	0.8827184140	1.0001905
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a_track[570]	-4.749363e-01	0.21879812	-0.823613423	-0.1236804059	1.0006294
a_track[571]	-1.246570e-01	0.23342823	-0.493853546	0.2555479359	1.0010053
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a_track[574]	-2.547628e-01	0.22159318	-0.613155786	0.0947678281	1.0042278
a_track[575]	-3.301937e-01	0.20403637	-0.653513453	0.0003389617	1.0010476
a_track[576]	-1.813747e-01	0.21443986	-0.519110389	0.1594490538	1.0014528
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a_track[578]	6.110184e-02	0.23272819	-0.304040613	0.4354039984	1.0000174
a_track[579]	-1.156370e-01	0.22806416	-0.470531320	0.2484036558	1.0058669
a_track[580]	-2.048909e-02	0.22889356	-0.371474828	0.3468854907	1.0021879
a_track[581]	-4.566362e-01	0.19852061	-0.770212278	-0.1316742956	0.9996569
a_track[582]	4.991354e-01	0.29865096	0.034825210	0.9838811993	1.0023610
a_track[583]	-9.613310e-02	0.22224047	-0.446633938	0.2646791343	1.0008908
a_track[584]	-3.238037e-01	0.21664131	-0.669843530	0.0344948058	1.0011051
a_track[585]	-1.804164e-01	0.21344934	-0.510585771	0.1684579395	1.0006690
a_track[586]	-3.251809e-01	0.19219353	-0.626650023	-0.0179372111	0.9999865
a_track[587]	-2.364574e-01	0.27551569	-0.659395801	0.2106430176	1.0004331
a_track[588]	-3.516495e-01	0.26724821	-0.775686728	0.0796827387	1.0010707
a_track[589]	3.099423e-01	0.29097474	-0.138881864	0.7926045334	1.0013095
a_track[590]	1.462067e-01	0.29337039	-0.325795000	0.6214690080	1.0023850
a_track[591]	2.786464e-01	0.30789435	-0.202839669	0.7780911519	1.0017858
a_track[592]	4.471806e-01	0.30616844	-0.030605476	0.9493005776	1.0034417
a_cell[1]	6.114834e-03	0.06638085	-0.094967653	0.1137114708	1.1213479
a_cell[2]	-3.258942e-03	0.06541130	-0.105553277	0.0958877628	1.1143802
a_cell[3]	-4.538739e-03	0.06955815	-0.113893110	0.1006577363	1.1106277
a_cell[4]	8.005086e-04	0.06713809	-0.103312007	0.1110032126	1.1307884
a_cell[5]	-2.879782e-03	0.06750252	-0.108711611	0.0983666078	1.1196901
a_cell[6]	-5.444761e-03	0.06705916	-0.114183648	0.0984787876	1.1342039
a_cell[7]	8.072592e-03	0.06689545	-0.096349239	0.1162266966	1.1142509
a_cell[8]	5.347109e-03	0.06676406	-0.097428742	0.1161656992	1.1186980
a_cell[9]	2.168102e-02	0.06654433	-0.071034312	0.1423569582	1.1212516

a_cell[10]	1.175397e-03	0.06635782	-0.102696820	0.1068340163	1.1185049
a_cell[11]	2.069311e-03	0.06368895	-0.100161900	0.1030553399	1.1203584
a_cell[12]	6.938460e-04	0.06829169	-0.104603548	0.1091514743	1.1361632
a_cell[13]	-1.683939e-02	0.06953240	-0.136443057	0.0823384559	1.1115880
a_cell[14]	-1.391393e-02	0.06870246	-0.130695703	0.0858667924	1.1191990
a_cell[15]	3.167352e-02	0.07102111	-0.060204712	0.1563528300	1.1053809
a_cell[16]	-2.148286e-03	0.06856925	-0.114052935	0.1026054340	1.1447571
a_cell[17]	-5.472103e-03	0.07098021	-0.119128049	0.0974040734	1.1254337
a_cell[18]	2.285080e-02	0.06988181	-0.074395225	0.1407581742	1.1150475
a_cell[19]	6.927260e-03	0.06414257	-0.090753373	0.1113457770	1.1154067
a_cell[20]	-2.121069e-03	0.06487225	-0.109569465	0.1043282172	1.1179813
a_cell[21]	4.568065e-03	0.06350900	-0.090086330	0.1056570947	1.1245368
a_cell[22]	-1.452940e-02	0.06261467	-0.118659427	0.0775643695	1.0945238
a_cell[23]	-1.336925e-02	0.06623055	-0.124523491	0.0824432628	1.1256867
a_cell[24]	2.131525e-02	0.06554978	-0.067287537	0.1371473226	1.1123457
a_cell[25]	-1.157612e-02	0.06310170	-0.119203560	0.0827842790	1.0953805
a_cell[26]	1.072269e-02	0.06285374	-0.082599364	0.1171781800	1.1250231
a_cell[27]	1.410441e-03	0.06662291	-0.103614191	0.1033996973	1.1256592
a_cell[28]	3.972351e-03	0.06506153	-0.093081108	0.1098789022	1.1225889
a_cell[29]	8.858875e-04	0.06634580	-0.101814831	0.1075065241	1.1359840
a_cell[30]	-8.693342e-03	0.06917832	-0.120175652	0.0928944831	1.1162486
a_cell[31]	3.627394e-03	0.06649929	-0.098561583	0.1120610300	1.1186114
a_cell[32]	-5.429987e-04	0.06674680	-0.107795283	0.1058173925	1.1236601
a_cell[33]	-5.218943e-03	0.06836479	-0.117651239	0.0969481097	1.1127980
a_cell[34]	-3.354874e-03	0.06686714	-0.108016136	0.1038683938	1.1118401
a_cell[35]	8.435115e-03	0.06694154	-0.094456679	0.1216775821	1.1293031
a_cell[36]	3.580040e-03	0.06465950	-0.096548956	0.1078937237	1.1320008
a_cell[37]	1.634222e-03	0.06295561	-0.095523656	0.1071771442	1.1461605
a_cell[38]	-1.641358e-02	0.06569100	-0.130523355	0.0782997879	1.1155058
a_cell[39]	1.160481e-02	0.06317738	-0.080287692	0.1179506624	1.1122069
a_cell[40]	7.296157e-03	0.06375923	-0.089763049	0.1129903226	1.1261111
a_cell[41]	1.046302e-02	0.06262909	-0.081352316	0.1159756229	1.1241811
a_cell[42]	-3.065975e-02	0.06671723	-0.150997757	0.0574536838	1.0850008
a_cell[43]	1.285890e-02	0.06267346	-0.081463115	0.1224931750	1.1048244
a_cell[44]	-1.371355e-02	0.06170882	-0.118512120	0.0798581134	1.0895580
a_cell[45]	-3.764224e-03	0.06104660	-0.101915810	0.0913426476	1.1105076
a_cell[46]	-2.562289e-05	0.06410436	-0.101541737	0.0989796681	1.1138056
a_cell[47]	2.944156e-04	0.06518456	-0.103963121	0.1027197082	1.1415966
a_cell[48]	1.297854e-02	0.06506300	-0.083612765	0.1242372737	1.1107562
a_cell[49]	-1.051609e-02	0.06637926	-0.130717217	0.0878599489	1.1202722
a_cell[50]	7.509719e-03	0.06416647	-0.088996859	0.1133853229	1.1256611
a_cell[51]	1.294642e-02	0.06743754	-0.081869618	0.1278940651	1.1294629
a_cell[52]	4.250544e-03	0.06480433	-0.097876271	0.1106462427	1.1256896

a_cell[53]	-3.611012e-03	0.06658812	-0.111155317	0.0966556899	1.1384929
a_cell[54]	5.991370e-03	0.06587191	-0.095590593	0.1115645904	1.1262038
a_cell[55]	-3.810821e-03	0.06385933	-0.105394941	0.0950230695	1.1116040
a_cell[56]	-1.101177e-02	0.06316696	-0.115103288	0.0802446291	1.1347925
a_cell[57]	-5.191452e-03	0.06421724	-0.109566021	0.0931463809	1.1042292
a_cell[58]	1.333343e-02	0.06234884	-0.079977135	0.1209863809	1.1119807
a_cell[59]	-2.721888e-02	0.06536553	-0.142801753	0.0574038103	1.1002666
a_cell[60]	2.170217e-02	0.06385680	-0.070045904	0.1337865721	1.0984533
a_cell[61]	-2.550565e-02	0.06532020	-0.136305636	0.0636770760	1.1057162
a_cell[62]	2.385469e-04	0.06491116	-0.102153977	0.1036305075	1.1366357
a_cell[63]	1.891365e-02	0.06536003	-0.072403253	0.1356825142	1.1052094
a_cell[64]	-3.105886e-02	0.06602042	-0.145540472	0.0545254042	1.0719063
a_cell[65]	-3.572623e-03	0.06396544	-0.109553249	0.0955868204	1.1038209
a_cell[66]	-9.908770e-03	0.06472527	-0.119924018	0.0880761946	1.1065072
a_cell[67]	3.451319e-03	0.06290944	-0.094751276	0.1074808263	1.1062075
a_cell[68]	4.405901e-03	0.06552537	-0.095793235	0.1116806015	1.1261742
a_cell[69]	-1.104925e-02	0.06406116	-0.117024510	0.0827333492	1.1176491
a_cell[70]	-2.569436e-03	0.06534890	-0.110051230	0.0994334654	1.1224134
a_cell[71]	5.939862e-03	0.06472196	-0.095899806	0.1120165107	1.1204485
a_cell[72]	-6.631688e-03	0.06866541	-0.111347129	0.0922513265	1.1129424
a_cell[73]	-2.430348e-03	0.06680104	-0.109329666	0.1037099940	1.1359340
a_cell[74]	-1.092480e-02	0.07003493	-0.124292979	0.0909589685	1.1040819
a_cell[75]	1.290429e-02	0.06603835	-0.083660334	0.1239927136	1.1279929
a_cell[76]	-5.149669e-03	0.06410106	-0.108985540	0.0892058598	1.1244146
a_cell[77]	-4.629640e-03	0.06390675	-0.106472693	0.0927697118	1.1115657
a_cell[78]	2.222082e-02	0.06450814	-0.065163988	0.1353011978	1.1101189
a_cell[79]	1.225166e-02	0.06711398	-0.087478122	0.1273712859	1.1177871
a_cell[80]	-1.498042e-02	0.06552419	-0.122778375	0.0793369656	1.1170031
a_cell[81]	-1.932567e-02	0.06496334	-0.129978407	0.0725861446	1.0929322
a_cell[82]	1.562898e-03	0.06232837	-0.097308131	0.1045039001	1.1312883
a_cell[83]	6.751076e-03	0.06017418	-0.088831820	0.1080951091	1.1072067
a_cell[84]	-3.817521e-02	0.07053216	-0.169826545	0.0477365389	1.0948380
a_cell[85]	-1.014581e-02	0.06069271	-0.114154550	0.0820679581	1.1039216
a_cell[86]	-5.489510e-03	0.06048996	-0.105501617	0.0892085820	1.1087354
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a_cell[88]	-8.577389e-03	0.06664931	-0.117165596	0.0899388964	1.1156917
a_cell[89]	2.272107e-02	0.06555779	-0.066356679	0.1395677239	1.1174720
a_cell[90]	-1.608011e-02	0.06779539	-0.131219919	0.0804509773	1.0935522
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a_cell[92]	-7.325623e-03	0.06316459	-0.111101111	0.0898231233	1.1074776
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a_cell[95]	-8.088233e-03	0.06659804	-0.116965077	0.0941492410	1.1019915

a_cell[96]	5.403049e-03	0.06218430	-0.088301680	0.1065764456	1.1129198
a_cell[97]	-2.388937e-03	0.06456228	-0.105366787	0.0993278330	1.1109967
a_cell[98]	-1.224560e-02	0.06648887	-0.122834251	0.0851523230	1.1368758
a_cell[99]	3.812168e-02	0.07306122	-0.047345712	0.1695008423	1.0936799
a_cell[100]	-2.797184e-03	0.06817680	-0.111255690	0.1039425759	1.1368882
a_cell[101]	-6.943490e-03	0.06433338	-0.109282813	0.0914049772	1.1174622
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a_cell[103]	-1.668657e-03	0.06621958	-0.107015974	0.0998456268	1.1159740
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a_cell[105]	1.963691e-03	0.06083226	-0.094605939	0.1015190271	1.0946200
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a_cell[107]	-2.497700e-02	0.06899328	-0.149777781	0.0693306790	1.0968828
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a_cell[109]	-2.033870e-03	0.06173573	-0.102029747	0.0939241066	1.1179723
a_cell[110]	-1.642489e-02	0.06277500	-0.125787256	0.0749096339	1.1082691
a_cell[111]	-2.327585e-02	0.06503335	-0.139043162	0.0664242260	1.0907338
a_cell[112]	-9.361721e-03	0.06689831	-0.124483457	0.0907151627	1.1117686
a_cell[113]	-1.213552e-02	0.06827260	-0.129368355	0.0856785270	1.1165801
a_cell[114]	3.271590e-03	0.06386179	-0.093049161	0.1091961386	1.1264566
a_cell[115]	-4.615153e-03	0.06575549	-0.111027153	0.0960474541	1.1291455
a_cell[116]	-1.313463e-02	0.06765248	-0.127456959	0.0829323317	1.0873507
a_cell[117]	-4.536528e-03	0.06665398	-0.117918206	0.0985088350	1.1040103
a_cell[118]	-4.814938e-03	0.06890295	-0.113128769	0.1046804145	1.1226459
a_cell[119]	3.480494e-02	0.07016157	-0.053447729	0.1574104489	1.0904862
a_cell[120]	1.795238e-03	0.06401557	-0.099173984	0.1052079567	1.1120856
a_cell[121]	5.685613e-03	0.06090599	-0.090414554	0.1065751326	1.1228576
a_cell[122]	1.638266e-02	0.06567211	-0.074799447	0.1279953158	1.1113961
a_cell[123]	-6.509948e-03	0.06731442	-0.111704979	0.0953721279	1.1216441
a_cell[124]	-8.310259e-03	0.06378641	-0.116101489	0.0870017023	1.1034376
a_cell[125]	-1.343391e-02	0.06569912	-0.123219347	0.0822022929	1.1007533
a_cell[126]	1.268008e-02	0.06311469	-0.079724363	0.1147021377	1.1007295
a_cell[127]	8.260417e-03	0.06214527	-0.085731091	0.1103768176	1.1078867
a_cell[128]	-1.640540e-02	0.06579091	-0.131366477	0.0730076198	1.0960309
a_cell[129]	1.790157e-02	0.06629690	-0.077182307	0.1323873565	1.1193887
a_cell[130]	8.695065e-03	0.06394703	-0.087196137	0.1144313324	1.1163517
a_cell[131]	-2.697758e-02	0.06970496	-0.149474377	0.0626589435	1.0950346
a_cell[132]	7.400030e-03	0.06286422	-0.087574385	0.1120703425	1.1218736
a_cell[133]	-5.534500e-03	0.06167261	-0.106906627	0.0926787567	1.1164232
a_cell[134]	-5.338868e-03	0.06330971	-0.106643776	0.0922904241	1.1067877
a_cell[135]	-2.168833e-02	0.06435645	-0.132350817	0.0686422004	1.1282085
a_cell[136]	1.504446e-02	0.06618046	-0.079738933	0.1272632579	1.1233712
a_cell[137]	7.881342e-03	0.06506708	-0.093311946	0.1127118305	1.1158637
a_cell[138]	4.864327e-03	0.06279356	-0.094246509	0.1079343618	1.1089044

a_cell[139]	-8.593260e-03	0.06572622	-0.117251614	0.0918868890	1.1067508
a_cell[140]	1.201012e-02	0.06616830	-0.085167194	0.1235387118	1.1278360
a_cell[141]	-1.181908e-02	0.06860189	-0.124376147	0.0899946122	1.1190654
a_cell[142]	-3.901634e-03	0.06588737	-0.109759856	0.1003755061	1.1262719
a_cell[143]	-4.865060e-04	0.06702711	-0.107733235	0.1062360683	1.1326904
a_cell[144]	2.072625e-03	0.06376062	-0.098325809	0.1079302572	1.1094290
a_cell[145]	-1.235502e-02	0.06599202	-0.129045170	0.0864292346	1.1217233
a_cell[146]	7.371911e-03	0.06460266	-0.091370092	0.1132714542	1.1464835
a_cell[147]	3.747731e-03	0.06436245	-0.095282757	0.1099110162	1.1099706
a_cell[148]	-2.521070e-03	0.06661291	-0.108841955	0.1004479768	1.1214273
a_cell[149]	-7.709053e-03	0.06615183	-0.120405942	0.0918789340	1.1143669
a_cell[150]	1.374387e-02	0.06682436	-0.081869652	0.1269893241	1.1243274
a_cell[151]	-3.260504e-03	0.06538078	-0.109261825	0.1011660223	1.1132869
a_cell[152]	1.285893e-02	0.06525204	-0.081061002	0.1201914624	1.1142057
a_cell[153]	-7.832123e-04	0.06790482	-0.109928679	0.1065909171	1.1140109
a_cell[154]	1.910046e-02	0.06657491	-0.077492415	0.1347044485	1.1099108
a_cell[155]	6.311358e-03	0.06442801	-0.093901849	0.1155195727	1.1098170
a_cell[156]	-1.765565e-02	0.06280534	-0.125802192	0.0683168273	1.0960776
a_cell[157]	-7.559746e-03	0.06232388	-0.108646389	0.0887542862	1.1110415
a_cell[158]	6.567129e-03	0.06193104	-0.085055941	0.1066753150	1.0997887
a_cell[159]	3.679496e-02	0.06834026	-0.046930149	0.1566352718	1.0934247
a_cell[160]	-3.780484e-03	0.06340123	-0.105102652	0.0958712275	1.1182882
a_cell[161]	-1.020239e-02	0.06206317	-0.112943517	0.0806869537	1.1017305
a_cell[162]	-2.529574e-03	0.06347236	-0.105461591	0.0935378845	1.1188826
a_cell[163]	-3.797403e-03	0.06173531	-0.101962555	0.0904046395	1.1060609
a_cell[164]	-4.215602e-03	0.06463257	-0.111430000	0.0958497320	1.1094865
a_cell[165]	-9.868780e-03	0.06433175	-0.119168776	0.0891452383	1.1194624
a_cell[166]	4.633252e-04	0.06726416	-0.103619813	0.1023463171	1.1181690
a_cell[167]	-3.386348e-03	0.06852356	-0.109476553	0.1008036679	1.1386828
a_cell[168]	-1.430919e-02	0.06802250	-0.129870653	0.0814271243	1.1057224
a_cell[169]	4.824342e-02	0.07601966	-0.035698860	0.1891139257	1.0811015
a_cell[170]	2.796131e-03	0.06881238	-0.106895865	0.1130384067	1.1396349
a_cell[171]	7.106017e-03	0.06718567	-0.093103422	0.1121340194	1.1184332
a_cell[172]	-9.752110e-04	0.06675910	-0.108279199	0.1048919541	1.1426854
a_cell[173]	9.123268e-03	0.06478443	-0.091272861	0.1163890481	1.1127230
a_cell[174]	-1.129687e-03	0.06588615	-0.105200419	0.1032723767	1.1430583
a_cell[175]	-2.000533e-03	0.06479960	-0.102791098	0.1001333074	1.1238220
a_cell[176]	1.182078e-03	0.06720135	-0.104496130	0.1078402233	1.1228716
a_cell[177]	-5.514946e-03	0.06587743	-0.108120344	0.0984797356	1.1149273
a_cell[178]	-1.414680e-02	0.06634046	-0.127206651	0.0831608027	1.1007314
a_cell[179]	-6.292775e-03	0.06316021	-0.108099478	0.0918757211	1.1229731
a_cell[180]	1.399030e-02	0.06432498	-0.080457382	0.1229714627	1.1252011
a_cell[181]	9.389689e-03	0.06220211	-0.080702481	0.1143971427	1.1021401

a_cell[182]	-1.658214e-02	0.06393178	-0.130191001	0.0747098343	1.1091057
a_cell[183]	-1.182230e-02	0.06433615	-0.120915129	0.0796169460	1.1144480
a_cell[184]	5.686375e-03	0.06559207	-0.096779234	0.1097742881	1.1247197
a_cell[185]	6.627099e-03	0.06522071	-0.094831083	0.1171844755	1.1138673
a_cell[186]	-6.269054e-03	0.06328662	-0.109907747	0.0901226137	1.1051983
a_cell[187]	-1.485536e-03	0.06309695	-0.099096024	0.0945458324	1.1164461
a_cell[188]	-2.555784e-04	0.06570621	-0.102453495	0.1052804965	1.1475878
a_cell[189]	9.873310e-04	0.06461002	-0.104080611	0.1035576559	1.1466577
a_cell[190]	-2.621670e-03	0.06636742	-0.110046567	0.1027771052	1.1241127
a_cell[191]	3.388670e-03	0.06762773	-0.102979519	0.1149554504	1.1272036
a_cell[192]	1.273396e-02	0.06522126	-0.082706481	0.1230832668	1.1238701
a_cell[193]	-2.169283e-03	0.06665013	-0.106932754	0.1018659069	1.1269062
a_cell[194]	7.328290e-03	0.06626710	-0.091221937	0.1152016123	1.1115243
a_cell[195]	3.067702e-02	0.06689432	-0.058239817	0.1512087654	1.1220233
a_cell[196]	-1.066496e-02	0.06787482	-0.126145534	0.0908478923	1.1111111
a_cell[197]	2.659170e-02	0.06899076	-0.070393197	0.1478576425	1.1164828
a_cell[198]	1.491489e-03	0.06741388	-0.104426950	0.1123020686	1.1345158
a_cell[199]	-1.305973e-03	0.06530592	-0.102685477	0.0993125894	1.1218658
a_cell[200]	-6.909870e-03	0.06701414	-0.119504969	0.0948048552	1.1292848
a_cell[201]	-4.519994e-03	0.06789466	-0.114692898	0.1019135568	1.1409657
a_cell[202]	1.426823e-02	0.06705744	-0.083305827	0.1309504655	1.1292497
a_cell[203]	1.261382e-02	0.06589291	-0.081693708	0.1230892274	1.1138541
a_cell[204]	5.124499e-02	0.07703484	-0.039069968	0.1964915093	1.0830832
a_cell[205]	2.588651e-02	0.06663450	-0.062864466	0.1450278243	1.0974134
a_cell[206]	-8.051614e-03	0.06769207	-0.122619617	0.0957648810	1.1176316
a_cell[207]	-1.182010e-02	0.06101849	-0.114405221	0.0775734961	1.1068885
a_cell[208]	-2.720801e-02	0.06791929	-0.149656026	0.0616467226	1.0878485
a_cell[209]	-6.124637e-03	0.06455563	-0.114842962	0.0929371500	1.1267066
a_cell[210]	7.369499e-03	0.06622006	-0.095162356	0.1182843591	1.1221645
a_cell[211]	5.367493e-02	0.07752264	-0.035355999	0.2007049519	1.1050056
a_cell[212]	-1.778005e-02	0.06495983	-0.132194356	0.0735902319	1.1104739
a_cell[213]	-6.555457e-03	0.06580932	-0.114508308	0.0913960941	1.1020902
a_cell[214]	1.752450e-02	0.06744344	-0.076081186	0.1340578277	1.1197964
a_cell[215]	-4.414881e-03	0.06588836	-0.107522951	0.0955537640	1.1176581
a_cell[216]	-6.156120e-03	0.06671549	-0.109434535	0.0950113059	1.1063691
a_cell[217]	-3.198524e-03	0.06778240	-0.111728813	0.0994338034	1.1350292
a_cell[218]	1.287567e-02	0.06672614	-0.080980862	0.1224086264	1.1236063
a_cell[219]	8.084379e-03	0.06506247	-0.089844813	0.1113368560	1.0949430
a_cell[220]	-5.192200e-03	0.06443489	-0.111924571	0.0944486413	1.1301761
a_cell[221]	6.847581e-03	0.06651369	-0.093003909	0.1122442654	1.1232808
a_cell[222]	2.788748e-03	0.06676960	-0.103129233	0.1150558696	1.1316825
a_cell[223]	1.361800e-03	0.06682776	-0.101727145	0.1096208977	1.1308721
a_cell[224]	-6.597513e-04	0.06195225	-0.100508606	0.0976605633	1.1011176

a_cell[225]	1.417209e-02	0.06733959	-0.081342300	0.1304396116	1.1011496
a_cell[226]	1.211969e-03	0.06495871	-0.100077385	0.1022303598	1.1152984
a_cell[227]	1.232088e-02	0.06556977	-0.084605085	0.1206920577	1.1097013
a_cell[228]	-4.262955e-03	0.06485801	-0.113173720	0.0970628686	1.1209764
a_cell[229]	-8.624446e-03	0.06458513	-0.122546832	0.0865653572	1.1150856
a_cell[230]	-2.659022e-02	0.06712986	-0.150542077	0.0625389620	1.0866567
a_cell[231]	1.094054e-03	0.06555660	-0.102385445	0.1058530109	1.1298337
a_cell[232]	-2.638778e-02	0.07083590	-0.150289900	0.0673800576	1.1069682
a_cell[233]	-2.722019e-03	0.06187058	-0.104013832	0.0930467412	1.1111603
a_cell[234]	6.846104e-03	0.06106001	-0.085739067	0.1071647578	1.1255845
a_cell[235]	8.852528e-04	0.06367509	-0.098417790	0.1047533507	1.1135870
a_cell[236]	1.215705e-02	0.06301874	-0.081471057	0.1208607260	1.1159157
a_cell[237]	-4.636932e-03	0.06465701	-0.108628266	0.0945196519	1.1119308
a_cell[238]	-9.440434e-03	0.06418517	-0.118682449	0.0861118600	1.1196906
a_cell[239]	-4.826907e-03	0.06556345	-0.112481418	0.0950553919	1.1152994
a_cell[240]	-3.434065e-03	0.06989507	-0.117220416	0.1019601752	1.1333303
a_cell[241]	2.757352e-02	0.07213412	-0.070172792	0.1565259556	1.1152015
a_cell[242]	-9.159225e-03	0.06691282	-0.119192861	0.0904099918	1.1134455
a_cell[243]	-6.150682e-03	0.06811700	-0.119827245	0.0976108861	1.1288577
a_cell[244]	-8.061599e-03	0.06817678	-0.120022374	0.0932837514	1.1134307
a_cell[245]	-5.527930e-03	0.06759717	-0.116258264	0.0963398904	1.1326664
a_cell[246]	2.140023e-02	0.06915942	-0.073230640	0.1409281044	1.0995866
a_cell[247]	-8.666692e-03	0.06806689	-0.120212642	0.0908929216	1.1415548
a_cell[248]	-2.856910e-03	0.06590070	-0.108189021	0.0997520293	1.1258631
a_cell[249]	2.511568e-03	0.06407533	-0.096118118	0.1051972456	1.1397487
a_cell[250]	1.709689e-03	0.06599070	-0.102997161	0.1081432423	1.1313407
a_cell[251]	-1.762730e-02	0.06723146	-0.129550852	0.0779971125	1.1266016
a_cell[252]	2.418997e-02	0.06950754	-0.069174449	0.1454273320	1.1164650
a_cell[253]	-1.741590e-02	0.06655111	-0.130055090	0.0749889967	1.1103114
a_cell[254]	-3.306413e-03	0.06432673	-0.108441832	0.0936627876	1.1234262
a_cell[255]	-1.143475e-03	0.06379875	-0.106836594	0.0981498014	1.1150024
a_cell[256]	9.367231e-04	0.06477893	-0.101940085	0.1066851083	1.1205979
a_cell[257]	-9.380528e-03	0.06245300	-0.110597104	0.0849935995	1.1259591
a_cell[258]	6.466051e-03	0.06153749	-0.087657880	0.1057847106	1.0995939
a_cell[259]	-1.477825e-04	0.06476379	-0.104895240	0.1067852996	1.1368804
a_cell[260]	-1.202965e-02	0.06442289	-0.125037499	0.0849261305	1.1134800
a_cell[261]	6.987239e-03	0.06418401	-0.090179933	0.1117649266	1.1263007
a_cell[262]	5.443467e-03	0.05783646	-0.085453385	0.1022818137	1.0924706
a_cell[263]	-1.357125e-02	0.06314854	-0.124171563	0.0767932609	1.0981776
a_cell[264]	1.128595e-02	0.06802753	-0.093746752	0.1274560897	1.1225388
a_cell[265]	-5.004818e-03	0.06914168	-0.115346628	0.0985676543	1.1481874
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a_cell[270]	-1.595601e-02	0.06972640	-0.136563730	0.0786672119	1.0911050
a_cell[271]	-1.322096e-02	0.06829709	-0.123854566	0.0838877413	1.1161073
a_cell[272]	-8.673691e-03	0.06781555	-0.121463249	0.0952265013	1.1209541
a_cell[273]	1.966565e-03	0.06856221	-0.105318374	0.1086061915	1.1226244
a_cell[274]	-3.757702e-03	0.06552869	-0.111324950	0.1001210619	1.1238094
a_cell[275]	-1.045054e-03	0.06777175	-0.108812017	0.1048838104	1.1430024
a_cell[276]	1.474816e-02	0.06591321	-0.083220816	0.1295395289	1.1079675
a_cell[277]	-1.454627e-02	0.06798210	-0.127619134	0.0814708058	1.1235113
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a_cell[282]	1.588713e-03	0.06515661	-0.099948558	0.1043970726	1.1052232
a_cell[283]	-4.607609e-03	0.06340719	-0.107089798	0.0951905762	1.1126888
a_cell[284]	2.307240e-02	0.06922999	-0.068663482	0.1426864224	1.1027565
a_cell[285]	-5.127230e-03	0.06500352	-0.112890332	0.0959253547	1.1251605
a_cell[286]	-2.204461e-02	0.07065290	-0.142001992	0.0714433753	1.1229584
a_cell[287]	-1.516528e-03	0.06396497	-0.104103732	0.1020029322	1.1260742
a_cell[288]	-2.153901e-02	0.06894520	-0.145549124	0.0702181373	1.1030752
a_cell[289]	-1.162480e-03	0.06393235	-0.107734150	0.0983134520	1.1213525
a_cell[290]	-3.187056e-03	0.06431470	-0.107660388	0.0966233508	1.0913883
a_cell[291]	-6.644308e-03	0.06869245	-0.118244365	0.0952879089	1.1181119
a_cell[292]	-1.151167e-03	0.06685673	-0.104221881	0.1057546927	1.1125852
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a_cell[294]	-8.017328e-03	0.06644239	-0.111488073	0.0909272078	1.1209979
a_cell[295]	-7.354090e-03	0.06790347	-0.116061482	0.0941624937	1.1425782
a_cell[296]	-6.428652e-03	0.06752186	-0.114474809	0.0969489994	1.1167370
a_cell[297]	-1.001743e-02	0.06766472	-0.125166548	0.0904204540	1.1175773
a_cell[298]	-8.408796e-03	0.06941137	-0.118244967	0.0921685728	1.1227606
a_cell[299]	5.219958e-03	0.06730331	-0.098065416	0.1164447833	1.1239684
a_cell[300]	1.677957e-03	0.06556664	-0.102023385	0.1047669569	1.1191819
a_cell[301]	1.286326e-02	0.06610776	-0.083344556	0.1249069862	1.1233920
a_cell[302]	5.074447e-04	0.06734559	-0.108043649	0.1089345970	1.1224084
a_cell[303]	1.086799e-02	0.06617219	-0.087018572	0.1244673791	1.1374924
a_cell[304]	-1.328373e-02	0.06665769	-0.129651630	0.0816006125	1.1367265
a_cell[305]	-1.339939e-02	0.06543959	-0.125217931	0.0870713678	1.1169130
a_cell[306]	-1.769550e-02	0.06765034	-0.135958648	0.0780240075	1.1117657
a_cell[307]	-4.423181e-03	0.06996639	-0.117903217	0.1013940439	1.1447565
a_cell[308]	4.391658e-04	0.06468047	-0.104689995	0.1019518032	1.1314832
a_cell[309]	-7.406646e-03	0.06607236	-0.119336516	0.0894645346	1.1274693
a_cell[310]	-9.337846e-03	0.06506569	-0.116242717	0.0852077339	1.1255639

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a_cell[312]	-2.659565e-02	0.06887099	-0.151974701	0.0619300069	1.0990811
a_cell[313]	-3.961616e-03	0.06669098	-0.109995838	0.0986131743	1.1535967
a_cell[314]	5.526688e-03	0.06398178	-0.094025662	0.1103119591	1.1262775
a_cell[315]	2.389068e-02	0.06541563	-0.063087380	0.1422365239	1.1118498
a_cell[316]	2.927510e-02	0.06884524	-0.060320787	0.1542590287	1.1036584
a_cell[317]	-2.038578e-02	0.07000746	-0.140710119	0.0775365711	1.0956966
a_cell[318]	8.149928e-04	0.06473237	-0.101305899	0.1072709876	1.1239993
a_cell[319]	-4.766703e-03	0.06630221	-0.112629937	0.1025157236	1.1288144
a_cell[320]	-1.267288e-03	0.06578168	-0.107097407	0.0993702603	1.1238469
a_cell[321]	1.079131e-02	0.06689067	-0.088562656	0.1242074330	1.1076916
a_cell[322]	1.223103e-03	0.06521585	-0.102884057	0.1071983415	1.1412402
a_cell[323]	3.390274e-03	0.06757498	-0.101372151	0.1119863388	1.1272072
a_cell[324]	6.015332e-03	0.06722005	-0.094994628	0.1200530843	1.1259397
a_cell[325]	2.242618e-02	0.06933140	-0.070307238	0.1468306058	1.1271159
a_cell[326]	-1.602203e-05	0.06900163	-0.109542549	0.1096038533	1.1407112
a_cell[327]	2.975656e-03	0.06557285	-0.099984962	0.1144052819	1.1263247
a_cell[328]	2.955370e-03	0.06597688	-0.101366301	0.1082502324	1.1011769
a_cell[329]	-3.938339e-03	0.06485948	-0.111088360	0.0906968745	1.1067364
a_cell[330]	6.272536e-04	0.06573377	-0.105575953	0.1056016844	1.1208259
a_cell[331]	5.655582e-05	0.06527396	-0.101383751	0.1018116813	1.1387906
a_cell[332]	6.867985e-03	0.07127597	-0.099494988	0.1193481726	1.1420285
a_cell[333]	-1.043191e-02	0.06676889	-0.120747171	0.0912019168	1.1440398
a_cell[334]	2.784451e-02	0.06941360	-0.062672302	0.1537397625	1.0962552
a_cell[335]	2.262617e-02	0.06538154	-0.066912415	0.1356622919	1.1082407
a_cell[336]	-6.069316e-03	0.06498861	-0.112048898	0.0914197130	1.1157510
a_cell[337]	-4.826312e-03	0.06531089	-0.108569360	0.0964571694	1.1203300
a_cell[338]	5.158833e-03	0.06786971	-0.096735272	0.1150284323	1.1207753
a_cell[339]	-8.883511e-03	0.06629814	-0.118801717	0.0928312159	1.1415822
a_cell[340]	-4.008945e-03	0.06500248	-0.108035988	0.0972371027	1.1286589
a_cell[341]	7.363074e-03	0.06411428	-0.088006707	0.1119793971	1.1117386
a_cell[342]	-1.193862e-03	0.06561076	-0.104671341	0.0975180600	1.1182623
a_cell[343]	-1.149405e-02	0.06833040	-0.127181968	0.0847761294	1.1346098
a_cell[344]	-1.388960e-02	0.06181331	-0.119473479	0.0766003190	1.1060430
a_cell[345]	-4.280996e-03	0.06607990	-0.111863497	0.0995278334	1.1438696
a_cell[346]	4.069974e-03	0.06671704	-0.099947093	0.1140893991	1.1155144
a_cell[347]	3.810986e-03	0.06623741	-0.099265538	0.1088993347	1.1233029
a_cell[348]	1.012465e-03	0.06573435	-0.103554985	0.1109945084	1.1226286
a_cell[349]	6.390210e-03	0.06794937	-0.097951458	0.1179716848	1.1374684
a_cell[350]	-2.162520e-04	0.06716241	-0.108512252	0.1063165599	1.1219943
a_cell[351]	9.719664e-03	0.06515383	-0.084995894	0.1175496294	1.1041756
a_cell[352]	1.499894e-02	0.06799905	-0.083312487	0.1355956149	1.1229240
a_cell[353]	-4.427741e-03	0.06481865	-0.113037538	0.0970281560	1.1295343

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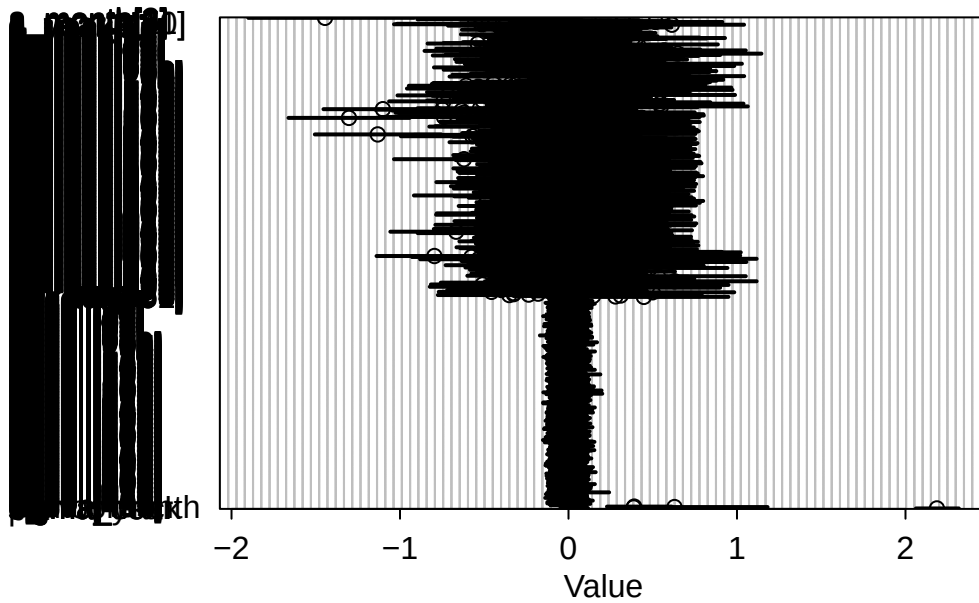
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a_cell[432]	4941.347192
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a_cell[438]	2390.643800
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a_cell[443]	3877.469428

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a_cell[446]    4241.224051
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a_cell[449]    3133.230921
a_cell[450]    1073.810902
a_cell[451]    2585.228582
a_cell[452]    4309.421761
a_cell[453]    4921.796514
a_cell[454]    5206.051395
b_tide         2604.526627
b_moon         2571.832820
sigma_month    1532.217165
sigma_year     2516.265460
sigma_track    809.026538
sigma_cell      6.840547
phi            2724.019384

```

```
plot(precis(m_space_use, depth = 2))
```



```
par(mfrow = c(2,2))
trankplot(m_space_use, ask = FALSE)
```



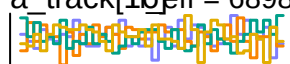
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a_track[15]eff = 7794



a_track[16]eff = 6898



a_track[17]eff = 7558



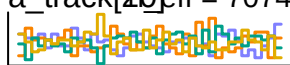
a_track[18]eff = 6996



a_track[19]eff = 6145



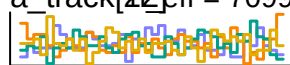
a_track[20]eff = 7074



a_track[21]eff = 6264



a_track[22]eff = 7099



a_track[23]eff = 6214



a_track[24]eff = 9398



a_track[25]eff = 6124



a_track[26]eff = 5640



a_track[27]eff = 7338



a_track[28]eff = 7804



a_track[29]eff = 8632



a_track[30]eff = 6538



a_track[31]eff = 8218



a_track[32]eff = 6578



a_track[33]eff = 7924



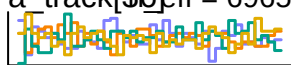
a_track[34]eff = 8033



a_track[35]eff = 7691



a_track[36]eff = 6965



a_track[37]eff = 6352



a_track[38]eff = 5792



a_track[39]eff = 6193



a_track[40]eff = 6314



a_track[41]eff = 6781



a_track[42]eff = 6543



a_track[43]eff = 7630



a_track[44]eff = 8259



a_track[45]eff = 9224



a_track[46]eff = 3928



a_track[47]eff = 5009



a_track[48]eff = 5031



a_track[49]eff = 4874



a_track[50]eff = 6600



a_track[51]eff = 5690



a_track[52]eff = 7500



a_track[53]eff = 9300



a_track[54]eff = 7183



a_track[55]eff = 6491



a_track[56]eff = 5347



a_track[57]eff = 6618



a_track[58]eff = 6476



a_track[59]eff = 6250



a_track[60]eff = 6909



a_track[61]eff = 6332



a_track[62]eff = 6806



a_track[63]eff = 6693



a_track[64]eff = 5474



a_track[65]eff = 8521



a_track[66]eff = 5443



a_track[67]eff = 8298



a_track[68]eff = 6008



a_track[69]eff = 5394



a_track[70]eff = 5618



a_track[71]eff = 5552



a_track[72]eff = 7749



a_track[73]eff = 7559



a_track[74]eff = 8128



a_track[75]eff = 8525



a_track[76]eff = 7100



a_track[77]eff = 7672



a_track[78]eff = 7548



a_track[79]eff = 6278



a_track[80]eff = 6246



a_track[81]eff = 7784



a_track[82]eff = 6569



a_track[83]eff = 7872



a_track[84]eff = 7090



a_track[85]eff = 7485



a_track[86]eff = 7286



a_track[87]eff = 7147



a_track[88]eff = 7653



a_track[89]eff = 7921



a_track[90]eff = 9696



a_track[91]eff = 6820



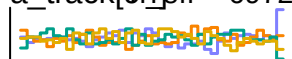
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a_track[93]eff = 6725



a_track[94]eff = 6972



a_track[95]eff = 7157



a_track[96]eff = 8809



a_track[97]eff = 6676



a_track[98]eff = 7877



a_track[99]eff = 7896



a_track[100]eff = 6286



a_track[101]eff = 7564



a_track[102]eff = 8701



a_track[103]eff = 8966





















a_track[374]f = 7428



a_track[375]f = 8931



a_track[376]f = 9234



a_track[377]f = 7661



a_track[378]f = 8406



a_track[379]f = 8166



a_track[380]f = 8704



a_track[381]f = 7818



a_track[382]f = 8363



a_track[383]f = 8617



a_track[384]f = 9736



a_track[385]f = 7648



a_track[386]f = 10862



a_track[387]f = 8817



a_track[388]f = 9310



a_track[389]f = 9012



a_track[390]f = 8309



a_track[391]f = 8988



a_track[392]f = 7141



a_track[393]f = 7638



a_track[394]f = 8384



a_track[395]f = 9044



a_track[396]f = 9237



a_track[397]f = 11129



a_track[398]f = 8761



a_track[399]f = 9355



a_track[400]f = 8775



a_track[401]f = 8081



a_track[402]f = 9835



a_track[403]f = 8831





a_track[434]f = 7999



a_track[435]f = 7850



a_track[436]f = 9285



a_track[437]f = 10323



a_track[438]f = 8277



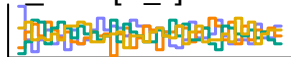
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a_track[442]f = 8116



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a_track[444]f = 9195



a_track[445]f = 6919



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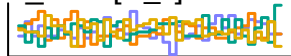
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a_track[449]f = 8431



a_track[450]f = 6817



a_track[451]f = 9101



a_track[452]f = 9552



a_track[453]f = 6628



a_track[454]f = 9614



a_track[455]f = 8849



a_track[456]f = 8361



a_track[457]f = 9321



a_track[458]f = 9343



a_track[459]f = 9372



a_track[460]f = 7849



a_track[461]f = 8221



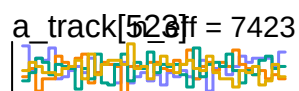
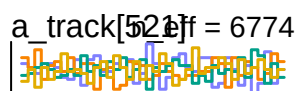
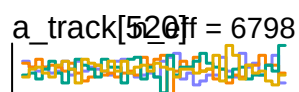
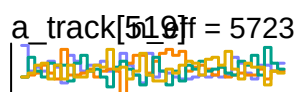
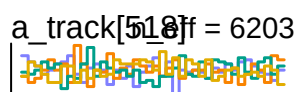
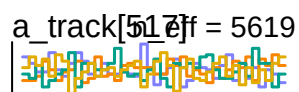
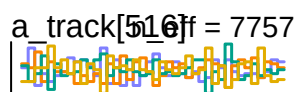
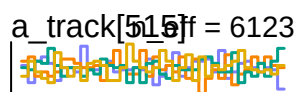
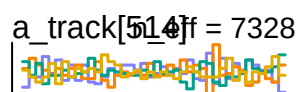
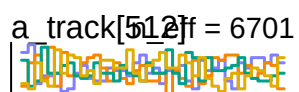
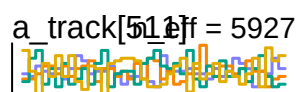
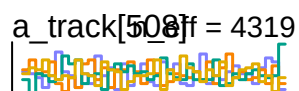
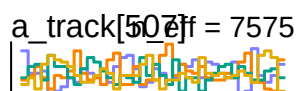
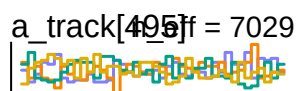
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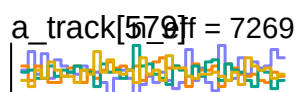
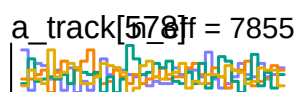
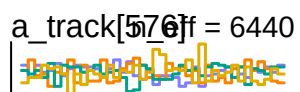
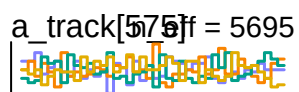
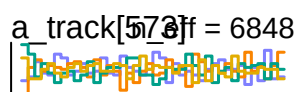
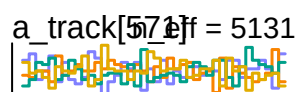
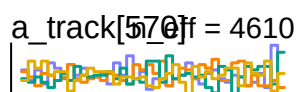
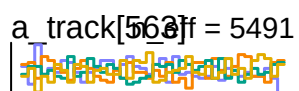
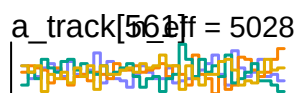
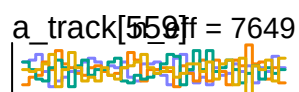
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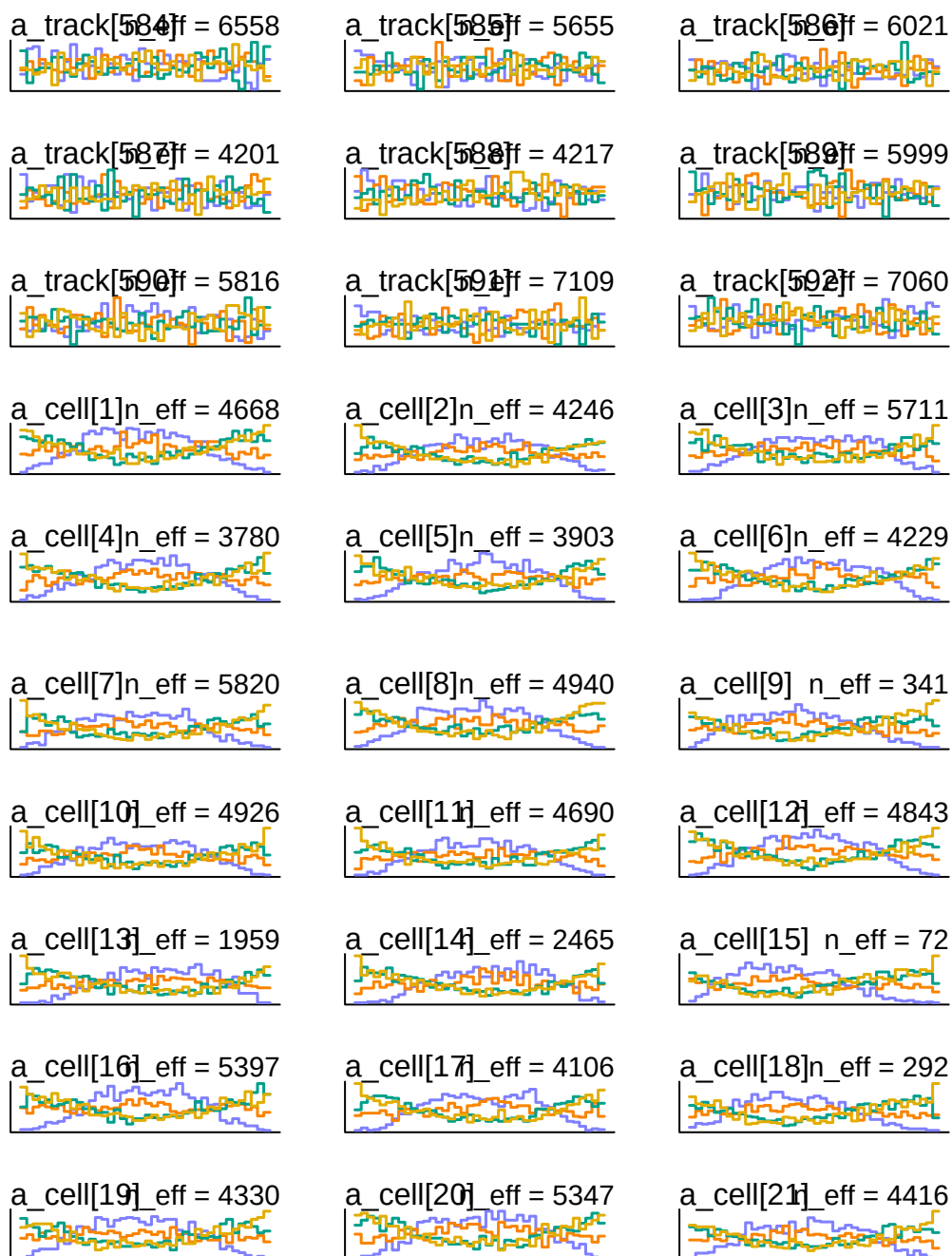




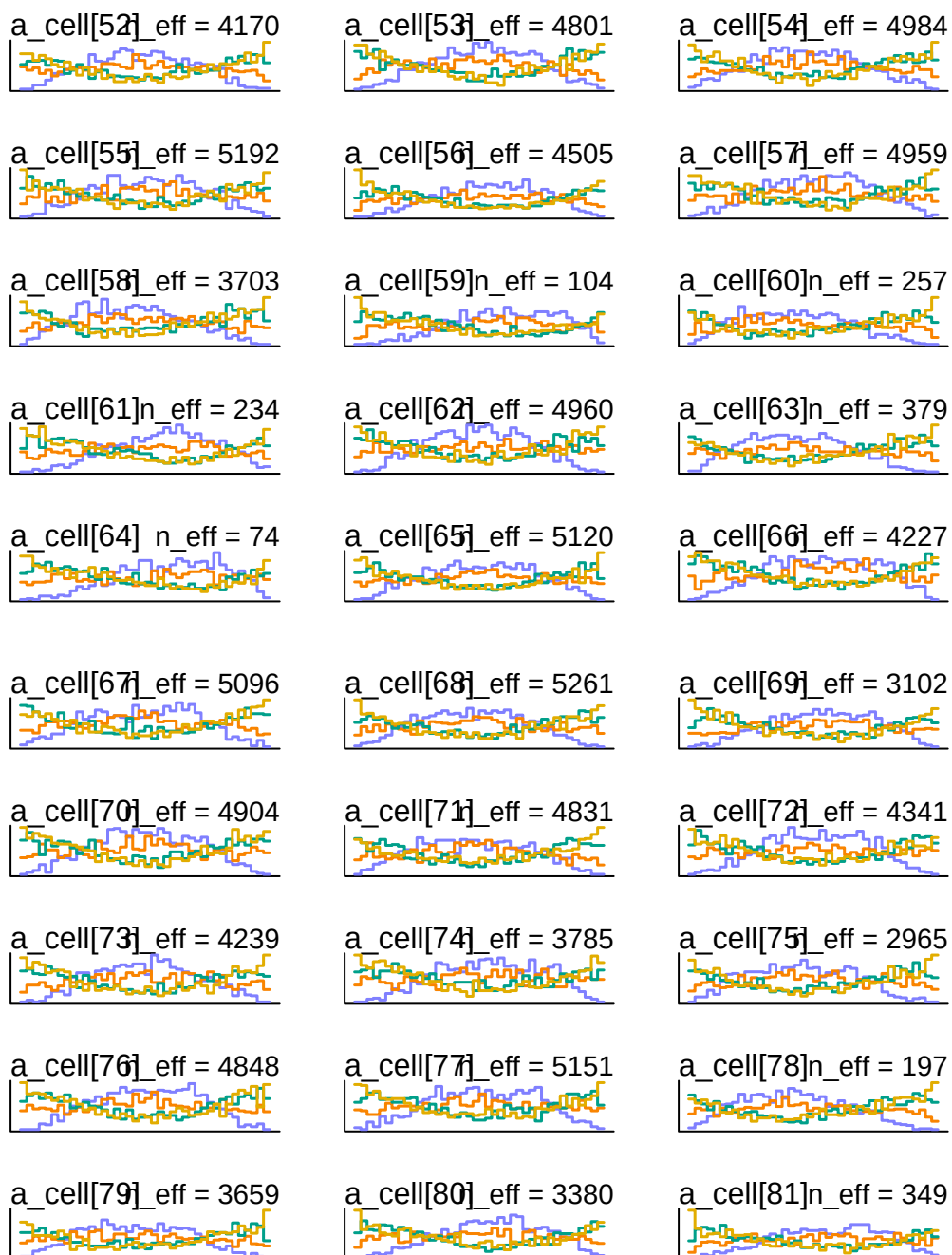


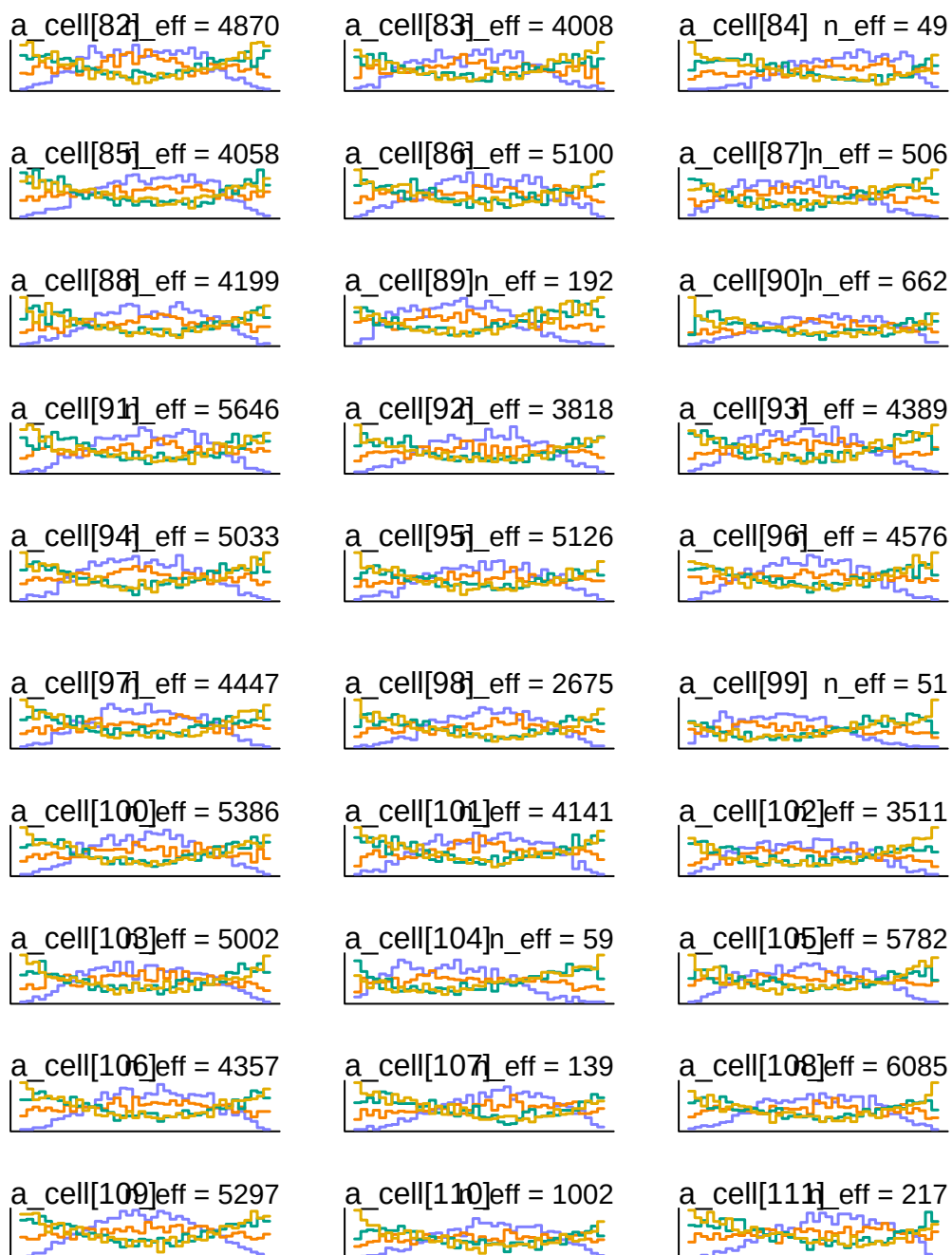


















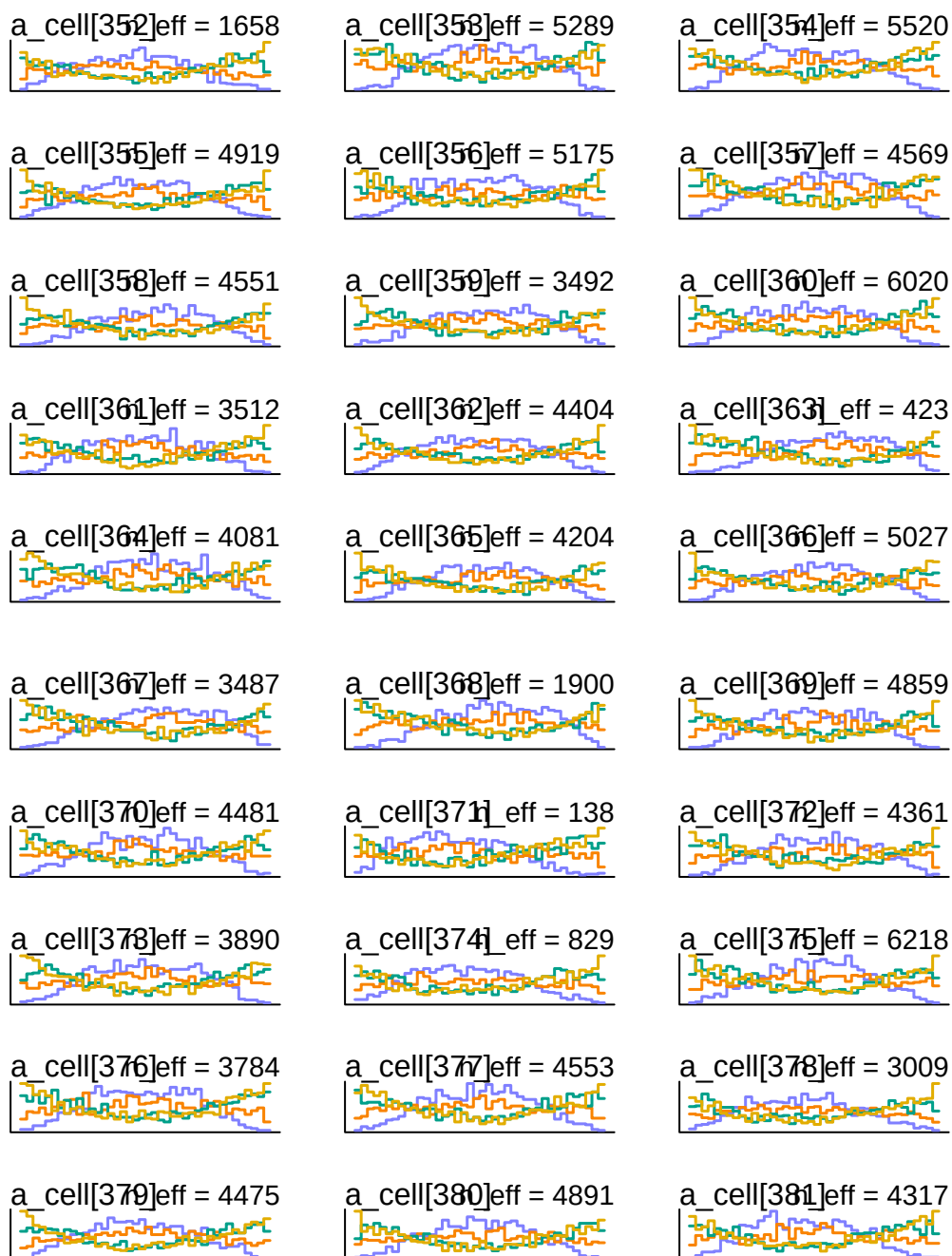




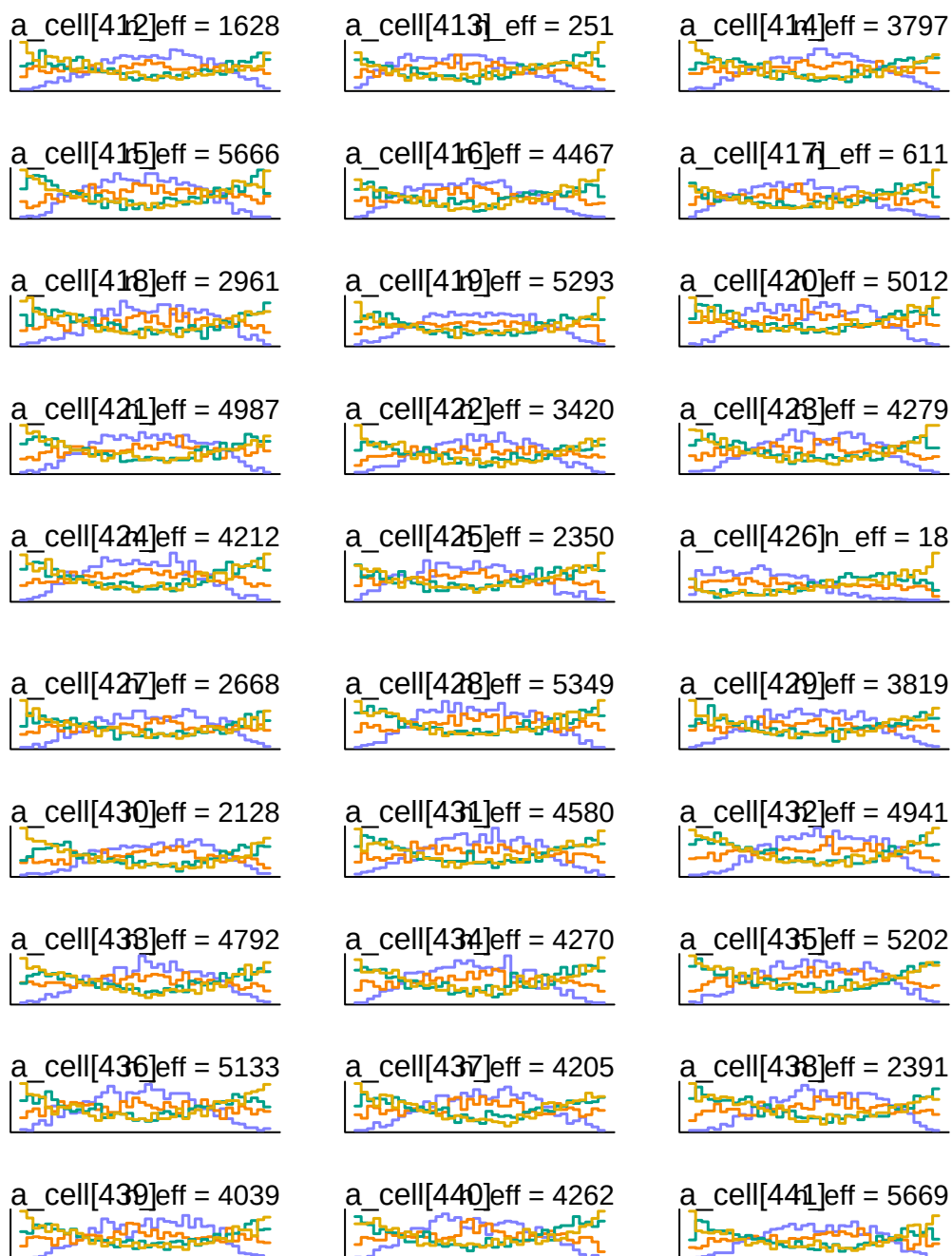


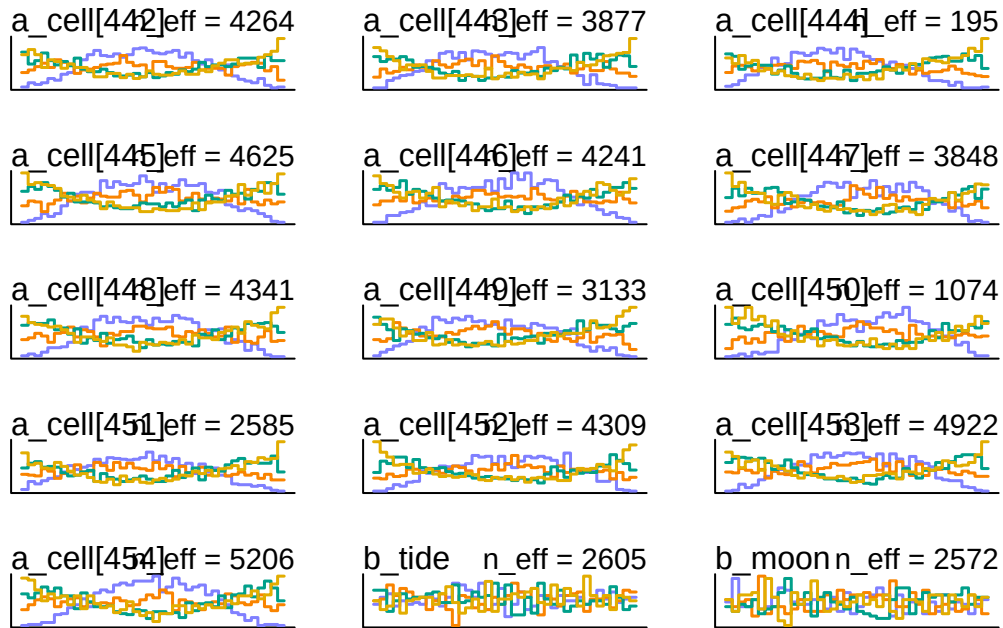




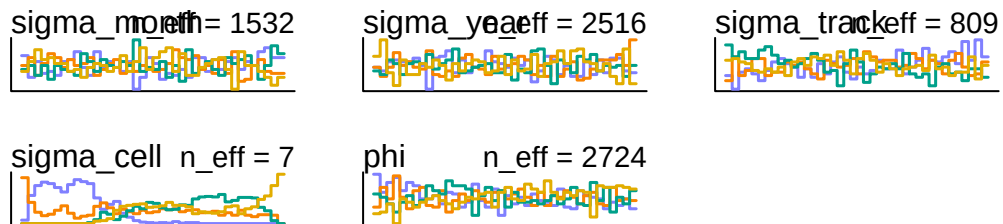








```
par(mfrow = c(1,1))
```



```
## convergence check
summary(m_space_use)
```

	mean	sd	5.5%	94.5%	rhat
a	-1.445107e+00	0.31631276	-1.900561296	-0.9155428765	1.0014583
a_month[1]	-1.450397e-01	0.14408829	-0.375085352	0.0713376433	1.0018410
a_month[2]	-3.903910e-01	0.15795403	-0.653530632	-0.1619841807	1.0021315
a_month[3]	-5.546982e-01	0.20456904	-0.897087022	-0.2498652388	1.0007412
a_month[4]	-2.798650e-01	0.22095437	-0.640181237	0.0527249246	1.0008298
a_month[5]	1.191828e-01	0.18908293	-0.184192797	0.4179400641	1.0008336
a_month[6]	-4.314699e-02	0.19116577	-0.349128444	0.2533594574	1.0028893
a_month[7]	3.205005e-01	0.17812703	0.046532757	0.5991325840	1.0014763
a_month[8]	3.267421e-01	0.18186829	0.048351504	0.6152736918	1.0017435
a_month[9]	5.164152e-01	0.24186211	0.145352136	0.9132331801	1.0016556
a_month[10]	-6.738535e-02	0.16620927	-0.328675068	0.1704865476	1.0017021
a_month[11]	-5.010591e-02	0.18407372	-0.352275596	0.2199075737	1.0008470
a_year[1]	-6.317773e-02	0.31522494	-0.580959059	0.3985824100	1.0007860
a_year[2]	-5.180224e-01	0.29611075	-1.018284925	-0.0868758523	1.0009025
a_year[3]	-5.134087e-01	0.30264673	-1.035817570	-0.0789229332	1.0009667
a_year[4]	6.136916e-01	0.29540679	0.119406844	1.0420436260	1.0008166
a_year[5]	-1.010020e-01	0.30080328	-0.616880550	0.3404886892	1.0010135
a_track[1]	-3.774271e-01	0.21463855	-0.715267461	-0.0361673556	1.0004256
a_track[2]	-1.513528e-01	0.22874739	-0.517373768	0.2203682608	1.0003257
a_track[3]	-1.519733e-01	0.22350905	-0.510056014	0.2169170725	1.0008599
a_track[4]	-1.560206e-01	0.23975716	-0.532252780	0.2356238555	1.0024700
a_track[5]	2.187780e-01	0.24817429	-0.174083680	0.6106338374	0.9997928
a_track[6]	5.997229e-04	0.29615740	-0.472377454	0.4693242881	1.0021253
a_track[7]	-2.993517e-02	0.21354067	-0.365755222	0.3139148400	1.0015800
a_track[8]	-2.930947e-02	0.22058278	-0.381812629	0.3285467349	1.0010143
a_track[9]	4.924901e-02	0.22887747	-0.308373965	0.4282147746	1.0008254
a_track[10]	-3.095314e-01	0.20226536	-0.632637269	0.0198071653	1.0006322
a_track[11]	-2.626772e-02	0.22058400	-0.379814310	0.3328285692	1.0016180
a_track[12]	-1.112962e-01	0.21145261	-0.438745543	0.2329777120	1.0007688
a_track[13]	-1.068260e-01	0.20588692	-0.430412519	0.2270574750	1.0001234
a_track[14]	-2.416467e-02	0.21833020	-0.369412183	0.3272963581	1.0024537
a_track[15]	4.462505e-02	0.22737978	-0.318552429	0.4084420554	1.0005367
a_track[16]	1.364397e-01	0.22437160	-0.211134726	0.4954251475	1.0005266
a_track[17]	1.344523e-01	0.23773527	-0.239146214	0.5161973113	1.0012231
a_track[18]	-9.718829e-02	0.21095085	-0.430826523	0.2478937895	1.0024682
a_track[19]	-1.097754e-01	0.20457734	-0.431276509	0.2227252344	1.0023201
a_track[20]	-3.134196e-01	0.20381789	-0.638807626	0.0113686458	0.9998787
a_track[21]	-2.544562e-01	0.20281725	-0.574682139	0.0784966580	1.0018813

a_track[22]	-1.784411e-01	0.20970216	-0.506752731	0.1535231680	1.0022010
a_track[23]	-1.780253e-01	0.20291461	-0.490903304	0.1428172924	1.0001018
a_track[24]	9.594414e-02	0.27474753	-0.344659260	0.5278794709	0.9997055
a_track[25]	-1.119906e-02	0.22466655	-0.370036999	0.3575149989	1.0006620
a_track[26]	-1.586679e-01	0.21571064	-0.496146355	0.1894792325	0.9998200
a_track[27]	3.302092e-01	0.28095297	-0.118299708	0.7782228128	1.0017717
a_track[28]	4.020375e-01	0.32906702	-0.125282060	0.9291091514	1.0026442
a_track[29]	3.806747e-01	0.26916143	-0.040720697	0.8199914913	1.0031243
a_track[30]	1.919295e-01	0.23724194	-0.181165323	0.5802338413	1.0022129
a_track[31]	1.025104e-01	0.22845368	-0.257703496	0.4741303092	0.9998107
a_track[32]	-1.531372e-01	0.20880159	-0.472631110	0.1870050772	1.0023912
a_track[33]	6.114854e-03	0.21964827	-0.343605634	0.3540807539	1.0009908
a_track[34]	9.316732e-03	0.21672908	-0.336785698	0.3575737409	1.0004305
a_track[35]	-7.175232e-02	0.21486852	-0.409800713	0.2740337440	1.0013324
a_track[36]	9.577877e-03	0.21804449	-0.342277825	0.3628314771	1.0004801
a_track[37]	-2.900415e-01	0.20835038	-0.628053769	0.0416242425	1.0016262
a_track[38]	-3.004489e-01	0.20745226	-0.619863927	0.0400266414	1.0009531
a_track[39]	-4.273982e-01	0.19430079	-0.742736722	-0.1143658036	1.0012302
a_track[40]	-5.389111e-01	0.18709765	-0.836368544	-0.2442831712	1.0019361
a_track[41]	-8.017437e-02	0.21808598	-0.427555565	0.2775385915	0.9997728
a_track[42]	4.112451e-01	0.29434672	-0.056656042	0.8865955464	1.0012939
a_track[43]	-2.169618e-01	0.21374691	-0.553372749	0.1276494903	1.0020116
a_track[44]	-1.428218e-01	0.22282677	-0.496672909	0.2214939170	1.0014126
a_track[45]	1.908462e-01	0.24867705	-0.198097758	0.5970768479	0.9997872
a_track[46]	-5.048056e-03	0.22398291	-0.364439851	0.3420170322	0.9997040
a_track[47]	7.426245e-02	0.22816408	-0.289724509	0.4444869775	1.0025439
a_track[48]	1.480431e-01	0.23378217	-0.221148327	0.5266830129	1.0012657
a_track[49]	2.306244e-01	0.23791349	-0.153156751	0.6129891219	1.0002112
a_track[50]	2.351405e-02	0.25291693	-0.370987984	0.4312343387	1.0009874
a_track[51]	-4.227226e-01	0.22521639	-0.784385405	-0.0591706299	1.0004801
a_track[52]	1.230974e-01	0.26940721	-0.310358558	0.5537006030	1.0017198
a_track[53]	3.123883e-01	0.29592723	-0.156076552	0.7822956366	1.0008125
a_track[54]	-7.129497e-02	0.25696808	-0.484844116	0.3431122381	1.0007508
a_track[55]	5.677176e-01	0.30688848	0.083836379	1.0523500900	1.0028559
a_track[56]	-4.937936e-01	0.23067083	-0.854675251	-0.1292452959	1.0021594
a_track[57]	1.253291e-01	0.25690402	-0.283145209	0.5460303700	1.0000049
a_track[58]	-2.538452e-01	0.24364794	-0.638667112	0.1293854806	1.0000596
a_track[59]	-2.477426e-01	0.23727655	-0.623345609	0.1287593844	1.0011344
a_track[60]	4.497221e-01	0.29666094	-0.009776617	0.9409100676	1.0002893
a_track[61]	-6.912851e-02	0.23683692	-0.444975031	0.3105336325	1.0002332
a_track[62]	6.488030e-01	0.31124886	0.159411264	1.1451094520	1.0007917
a_track[63]	1.155743e-01	0.24818475	-0.273281853	0.5143066604	1.0007083
a_track[64]	-3.718035e-01	0.22741953	-0.729571273	-0.0046385047	1.0015041

a_track[65]	2.868191e-01	0.36514110	-0.301627420	0.9004310400	1.0001767
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a_track[67]	2.700169e-01	0.30226620	-0.205084114	0.7714086953	1.0042992
a_track[68]	-5.936497e-02	0.24198468	-0.442629443	0.3357419757	0.9997807
a_track[69]	-1.403057e-01	0.23232155	-0.504811240	0.2388121522	1.0017241
a_track[70]	-1.415362e-01	0.23266234	-0.516247922	0.2337659566	1.0009577
a_track[71]	-1.351790e-01	0.23630318	-0.503519547	0.2433145495	1.0012388
a_track[72]	-8.815708e-03	0.26772118	-0.431280188	0.4203308669	1.0003713
a_track[73]	2.306715e-01	0.28802183	-0.220081998	0.6823459670	0.9994940
a_track[74]	-2.921764e-01	0.23913785	-0.674462709	0.0896851840	1.0000765
a_track[75]	2.341845e-01	0.28954965	-0.213873835	0.7042186906	1.0005717
a_track[76]	-1.922246e-01	0.25249853	-0.580333559	0.2252204232	0.9998529
a_track[77]	-1.917814e-01	0.24105404	-0.576691532	0.2012058964	1.0014913
a_track[78]	1.488677e-01	0.26053343	-0.272837138	0.5692621372	1.0000796
a_track[79]	-1.911708e-01	0.24750609	-0.576140762	0.2073857005	1.0010885
a_track[80]	-2.915053e-01	0.23850552	-0.665400936	0.0958023317	1.0003016
a_track[81]	-4.920715e-03	0.26190168	-0.414404482	0.4090182660	1.0004475
a_track[82]	1.231300e-01	0.26567644	-0.288885464	0.5531985593	1.0029535
a_track[83]	-1.071756e-01	0.26622132	-0.536100810	0.3122705975	1.0015951
a_track[84]	5.314563e-01	0.30278556	0.056593635	1.0293253760	1.0022679
a_track[85]	-2.114112e-01	0.24904409	-0.615218866	0.1925977986	1.0017234
a_track[86]	-2.034533e-01	0.25291003	-0.603626653	0.1966761363	1.0010002
a_track[87]	-2.068497e-01	0.24344081	-0.602250394	0.1898252891	1.0025712
a_track[88]	3.542227e-01	0.29828887	-0.127037858	0.8392225843	1.0007652
a_track[89]	4.806669e-01	0.31181265	-0.005083657	0.9837989127	1.0021701
a_track[90]	2.390858e-01	0.28709166	-0.212481797	0.6903771254	1.0000809
a_track[91]	-1.225279e-01	0.24830447	-0.505039088	0.2832421179	1.0002490
a_track[92]	-4.246604e-01	0.23681085	-0.798629304	-0.0427812166	0.9998186
a_track[93]	-1.170474e-01	0.25467360	-0.526479919	0.2988130077	0.9999316
a_track[94]	3.376584e-01	0.31136277	-0.163466739	0.8406961291	1.0004806
a_track[95]	1.324405e-01	0.28064583	-0.309270364	0.5866103495	1.0004330
a_track[96]	-9.330665e-02	0.25480299	-0.502965692	0.3171970372	1.0013936
a_track[97]	-3.873428e-01	0.22898262	-0.741207913	-0.0196287252	1.0014343
a_track[98]	3.436386e-01	0.30772387	-0.148642170	0.8419288619	0.9999154
a_track[99]	2.342123e-02	0.26602847	-0.404605259	0.4478530454	1.0006456
a_track[100]	3.844023e-01	0.30116361	-0.083993339	0.8773763825	1.0012665
a_track[101]	2.367787e-01	0.28639474	-0.218597095	0.6914700587	1.0016810
a_track[102]	2.375828e-01	0.37090255	-0.358282615	0.8384218863	1.0017411
a_track[103]	1.328627e-01	0.27413213	-0.310797171	0.5792091195	1.0015171
a_track[104]	-8.851360e-02	0.24999826	-0.487927883	0.3178087548	1.0003235
a_track[105]	-1.968267e-01	0.24304752	-0.579430752	0.2048540721	1.0053463
a_track[106]	-1.984749e-01	0.24058053	-0.570019482	0.1854869610	1.0004842
a_track[107]	2.618159e-01	0.28033860	-0.181777433	0.7150286876	1.0002080

a_track[108]	-3.759422e-01	0.19750530	-0.688115509	-0.0731262642	1.0032986
a_track[109]	-1.053813e-01	0.20754610	-0.437170212	0.2262127456	1.0033952
a_track[110]	-2.527546e-01	0.21004596	-0.580173130	0.0809917355	1.0019776
a_track[111]	5.048634e-01	0.33874293	-0.050591041	1.0428100300	1.0007851
a_track[112]	-3.731965e-01	0.20782085	-0.706298360	-0.0373367774	1.0008232
a_track[113]	3.433166e-01	0.35837174	-0.224350955	0.9204572551	1.0011766
a_track[114]	2.844899e-01	0.32728953	-0.231988095	0.8126214996	1.0008308
a_track[115]	3.358970e-01	0.31474833	-0.166168898	0.8235449992	0.9995173
a_track[116]	3.037918e-01	0.32894874	-0.235729290	0.8386284340	1.0028984
a_track[117]	3.038085e-01	0.27583047	-0.125522771	0.7444708161	1.0052585
a_track[118]	1.949831e-01	0.27163511	-0.229501226	0.6328818458	1.0017060
a_track[119]	3.853224e-01	0.33980805	-0.163853023	0.9185865649	1.0002009
a_track[120]	1.410209e-01	0.28097441	-0.301670774	0.6056856093	1.0005855
a_track[121]	3.802135e-01	0.30510226	-0.096486822	0.8783850549	1.0013615
a_track[122]	2.422461e-01	0.28855173	-0.216758666	0.6980815340	1.0003491
a_track[123]	-1.925150e-01	0.24072473	-0.573321813	0.1920937358	0.9996342
a_track[124]	3.105122e-01	0.33241118	-0.228111732	0.8409512482	1.0034097
a_track[125]	2.634421e-01	0.35147948	-0.298535213	0.8273118843	1.0018892
a_track[126]	-4.465999e-01	0.22121516	-0.803083932	-0.0856936622	1.0011855
a_track[127]	-1.079626e-01	0.24474589	-0.503169899	0.2899684516	1.0019543
a_track[128]	-4.448501e-01	0.23021709	-0.810939472	-0.0751191781	1.0013516
a_track[129]	-3.451728e-01	0.22241791	-0.703208612	0.0170563610	1.0010343
a_track[130]	2.938505e-01	0.32056946	-0.214246134	0.8126887793	1.0031007
a_track[131]	1.591713e-01	0.26543094	-0.253444332	0.5867374099	1.0009322
a_track[132]	-6.075406e-01	0.21515944	-0.947725540	-0.2540510815	1.0005020
a_track[133]	1.200733e-01	0.28129020	-0.318742271	0.5690957895	1.0035744
a_track[134]	-5.342078e-01	0.21524990	-0.876343635	-0.1944445974	1.0031817
a_track[135]	3.197379e-01	0.30833302	-0.175324190	0.8144294615	1.0029224
a_track[136]	4.073590e-01	0.34526257	-0.148872112	0.9663797715	1.0005436
a_track[137]	3.566438e-02	0.26051085	-0.375641144	0.4451788581	0.9996317
a_track[138]	-6.117008e-01	0.21004306	-0.959135061	-0.2690592615	1.0013615
a_track[139]	1.514289e-01	0.26481305	-0.268931283	0.5761860634	1.0027244
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a_track[142]	6.884228e-02	0.25089977	-0.326624002	0.4677096680	1.0017980
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a_track[146]	-5.009761e-01	0.21138825	-0.839912723	-0.1593419772	1.0013786
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a_track[148]	1.834497e-01	0.25040407	-0.209268683	0.5957603545	1.0007449
a_track[149]	7.546228e-02	0.23884196	-0.299640779	0.4629692549	1.0008622
a_track[150]	-1.512076e-02	0.23291262	-0.378665952	0.3651090734	1.0033037

a_track[151]	2.945528e-01	0.26426416	-0.120090307	0.7278583774	1.0014736
a_track[152]	5.338221e-01	0.28453660	0.089565187	0.9885570799	1.0003997
a_track[153]	-7.154486e-01	0.19107805	-1.015031286	-0.4063169634	1.0001367
a_track[154]	-4.847507e-01	0.20416311	-0.807858705	-0.1557823792	1.0009319
a_track[155]	-4.157603e-01	0.20934097	-0.746063266	-0.0754613872	0.9998356
a_track[156]	-2.683910e-01	0.21040435	-0.599988617	0.0677263797	1.0009561
a_track[157]	9.066124e-02	0.24161329	-0.288400287	0.4741869103	1.0002069
a_track[158]	8.675946e-02	0.24331732	-0.304233079	0.4827793430	1.0022249
a_track[159]	-2.073020e-01	0.22432405	-0.559817739	0.1574147580	1.0016595
a_track[160]	2.962613e-02	0.24093362	-0.363588061	0.4125779720	1.0007935
a_track[161]	-4.906849e-02	0.23225985	-0.408292912	0.3240498400	1.0017523
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a_track[170]	1.705504e-02	0.25352070	-0.377834310	0.4231478370	1.0011630
a_track[171]	5.425362e-01	0.30387823	0.061353914	1.0387688570	1.0012335
a_track[172]	-8.616327e-02	0.24642476	-0.476684881	0.3122072824	1.0000257
a_track[173]	2.181789e-01	0.30882350	-0.277632855	0.7066100288	1.0011458
a_track[174]	-8.419796e-02	0.25280914	-0.485739405	0.3248132975	1.0005145
a_track[175]	1.384838e-01	0.27634988	-0.301488418	0.5875259976	1.0000610
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a_track[184]	-7.483703e-01	0.25394466	-1.147279756	-0.3420126621	1.0012644
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a_track[186]	-6.343022e-01	0.26110698	-1.046788088	-0.2131653413	1.0004282
a_track[187]	-2.284165e-01	0.28380681	-0.679028557	0.2327172970	1.0009176
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a_track[189]	-6.896811e-02	0.31835554	-0.573113750	0.4483075056	1.0034052
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a_track[195]	2.746172e-01	0.31205889	-0.210330204	0.7782600919	1.0005269
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a_track[267]	-7.786086e-02	0.29199171	-0.533122134	0.3965942926	1.0023844
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a_track[321]	2.172829e-01	0.34067050	-0.323585182	0.7566961860	1.0023489
a_track[322]	-1.996098e-01	0.28438686	-0.647405735	0.2638121024	1.0025211

a_track[323]	2.348115e-01	0.33057545	-0.279932870	0.7672180670	1.0038542
a_track[324]	8.056331e-02	0.32715432	-0.437894528	0.6162971286	1.0014484
a_track[325]	1.501991e-01	0.36605143	-0.453753558	0.7334427651	1.0008811
a_track[326]	-6.579679e-02	0.29723364	-0.542848266	0.4200014413	1.0002550
a_track[327]	7.552784e-02	0.31767484	-0.414052123	0.5954433632	1.0025429
a_track[328]	2.717304e-01	0.31835224	-0.224981478	0.7775794207	1.0011567
a_track[329]	2.481538e-01	0.33860284	-0.286820372	0.7989492033	1.0005818
a_track[330]	7.843672e-02	0.31433336	-0.419080084	0.5820098908	0.9996266
a_track[331]	2.428648e-01	0.32083711	-0.257711652	0.7538656584	1.0019269
a_track[332]	7.745293e-02	0.32313938	-0.442621129	0.5844362902	0.9993720
a_track[333]	-2.020406e-01	0.29212618	-0.655298874	0.2773110259	1.0008349
a_track[334]	9.127310e-02	0.31277497	-0.397494155	0.5801881979	1.0002989
a_track[335]	9.566226e-02	0.30458258	-0.378960942	0.5888686183	1.0024404
a_track[336]	7.805474e-02	0.32387243	-0.435566736	0.5898254020	1.0010671
a_track[337]	2.051444e-01	0.32704729	-0.311929216	0.7430914701	1.0029534
a_track[338]	2.117124e-01	0.33738882	-0.319698155	0.7444326640	1.0023494
a_track[339]	-5.829138e-02	0.29637521	-0.522511640	0.4193877328	1.0009997
a_track[340]	-2.082381e-01	0.28389536	-0.649893375	0.2540611796	1.0056866
a_track[341]	-6.830406e-02	0.29291841	-0.531997976	0.4080981691	1.0029104
a_track[342]	-3.525561e-01	0.27770743	-0.784296020	0.0932668123	1.0008060
a_track[343]	8.328056e-02	0.31584326	-0.415313034	0.5934849399	1.0015327
a_track[344]	-2.251294e-01	0.27838766	-0.665335216	0.2193665179	1.0003722
a_track[345]	2.213247e-01	0.32039411	-0.289841870	0.7382763959	1.0039849
a_track[346]	2.277580e-01	0.32654594	-0.292895433	0.7579448695	1.0011514
a_track[347]	-7.634480e-02	0.29610158	-0.554409870	0.3916434994	0.9999687
a_track[348]	6.790608e-02	0.33410147	-0.459427217	0.5964770515	1.0011092
a_track[349]	5.562585e-02	0.33744558	-0.481693469	0.5814365895	1.0010310
a_track[350]	2.148344e-01	0.32397755	-0.304143141	0.7308394281	1.0010408
a_track[351]	-2.406361e-01	0.28993975	-0.698399220	0.2247071889	1.0015615
a_track[352]	1.750020e-01	0.35212603	-0.376088088	0.7500373985	1.0011128
a_track[353]	7.594166e-02	0.30589570	-0.403530611	0.5702150988	1.0014235
a_track[354]	6.782043e-02	0.32218039	-0.436087999	0.5821809215	1.0025759
a_track[355]	-8.833760e-02	0.29433084	-0.562327127	0.3842779308	1.0020833
a_track[356]	-2.216985e-01	0.28998650	-0.681017616	0.2529493877	1.0013387
a_track[357]	4.905938e-02	0.33414316	-0.494314138	0.5925606255	1.0008984
a_track[358]	1.334263e-01	0.34445535	-0.411452675	0.6816662178	1.0013433
a_track[359]	4.841182e-02	0.33837609	-0.491580568	0.5827897266	1.0054337
a_track[360]	6.774407e-02	0.32575027	-0.457920686	0.5979570907	1.0006109
a_track[361]	1.882474e-01	0.34235860	-0.350764609	0.7455302234	1.0001532
a_track[362]	5.682773e-02	0.31664873	-0.451501501	0.5627092331	1.0010598
a_track[363]	-8.660365e-02	0.28710320	-0.527653257	0.3680823315	1.0059985
a_track[364]	-8.610437e-02	0.30034411	-0.562811781	0.3966854223	0.9999213
a_track[365]	-8.549036e-02	0.30164430	-0.559126727	0.4036500694	1.0002147

a_track[366]	3.573958e-02	0.35373036	-0.513978965	0.5983047995	1.0005209
a_track[367]	4.871982e-02	0.33314608	-0.497281782	0.5779720770	1.0011676
a_track[368]	1.572376e-01	0.34180199	-0.385623600	0.7124101653	1.0008106
a_track[369]	5.260883e-02	0.33436071	-0.475242711	0.5928505739	1.0008011
a_track[370]	6.471904e-02	0.32482838	-0.462383298	0.5967531602	1.0009585
a_track[371]	-4.898975e-01	0.27657476	-0.917612085	-0.0530642174	1.0008043
a_track[372]	-8.611840e-02	0.31285298	-0.592098619	0.4334767440	1.0027939
a_track[373]	-2.221513e-01	0.29057870	-0.677905148	0.2483448192	1.0019487
a_track[374]	5.212133e-02	0.32555093	-0.459387270	0.5803250072	1.0021774
a_track[375]	1.334823e-01	0.37041693	-0.436846728	0.7311519356	1.0030547
a_track[376]	-6.803274e-02	0.32892892	-0.591709929	0.4679887556	0.9998352
a_track[377]	2.131855e-01	0.32664567	-0.294703660	0.7304761253	1.0016493
a_track[378]	2.275840e-01	0.32266414	-0.289531715	0.7452255447	1.0016671
a_track[379]	1.867267e-01	0.34180808	-0.344694255	0.7306944664	0.9993370
a_track[380]	6.457890e-02	0.32519188	-0.457218847	0.5853935565	1.0015783
a_track[381]	8.069429e-02	0.31370859	-0.417627064	0.5863315306	1.0033007
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a_track[383]	2.730148e-01	0.32584112	-0.238831037	0.7995859647	1.0009062
a_track[384]	8.408141e-02	0.32051852	-0.433881189	0.6011953489	1.0013216
a_track[385]	-4.967922e-02	0.30085763	-0.529680349	0.4360408638	1.0021821
a_track[386]	-5.217783e-02	0.30990448	-0.546432643	0.4516529626	0.9999703
a_track[387]	2.209286e-01	0.34778629	-0.325062947	0.7797961919	1.0004103
a_track[388]	-5.297573e-02	0.29164361	-0.505963110	0.4217065524	1.0016172
a_track[389]	9.728841e-02	0.30965186	-0.390935456	0.5802125294	1.0019550
a_track[390]	-5.619617e-02	0.29389437	-0.524135343	0.4182931208	0.9997701
a_track[391]	-6.067755e-02	0.29921478	-0.540702656	0.4311168301	1.0007491
a_track[392]	-6.330116e-02	0.29086406	-0.523533365	0.4080564664	1.0028878
a_track[393]	2.119476e-01	0.33347153	-0.310791874	0.7566501774	1.0003481
a_track[394]	-2.170092e-01	0.27345265	-0.651860293	0.2261503847	1.0009531
a_track[395]	-5.921157e-02	0.29433093	-0.533166704	0.4182064793	1.0012072
a_track[396]	-6.361509e-02	0.29037653	-0.516899653	0.4126297375	1.0007208
a_track[397]	7.778720e-02	0.33494977	-0.460474395	0.6112189987	1.0031245
a_track[398]	8.653928e-02	0.30626969	-0.404539611	0.6007611207	1.0035412
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a_track[400]	9.221719e-02	0.30587308	-0.383304957	0.5888594068	1.0021867
a_track[401]	-5.515933e-02	0.30799706	-0.547018822	0.4363199225	1.0001511
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a_track[403]	-5.248129e-02	0.30665324	-0.541984103	0.4439994542	1.0010227
a_track[404]	-5.612388e-02	0.30522085	-0.547953524	0.4405837083	1.0007339
a_track[405]	7.843567e-02	0.32634440	-0.429511281	0.5925635139	1.0024159
a_track[406]	1.718752e-01	0.34535610	-0.379983478	0.7137357536	0.9998679
a_track[407]	-5.228440e-02	0.30185157	-0.520344764	0.4276032772	1.0024445
a_track[408]	-5.703783e-02	0.29647663	-0.536040397	0.4075789181	1.0005641

a_track[409]	-5.246237e-02	0.32176208	-0.573509722	0.4596086752	1.0008535
a_track[410]	-3.426099e-01	0.27882499	-0.782305995	0.1001700578	1.0010093
a_track[411]	4.453664e-02	0.34886905	-0.511782340	0.5968796686	1.0015370
a_track[412]	4.705963e-02	0.34570263	-0.516232720	0.5955449912	1.0012529
a_track[413]	-3.328219e-01	0.28469192	-0.786047515	0.1275214141	1.0073048
a_track[414]	1.911068e-01	0.33913933	-0.344675394	0.7309456298	1.0013370
a_track[415]	-5.460379e-02	0.28738790	-0.504812790	0.4008913219	1.0023895
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a_track[417]	2.205108e-01	0.33526914	-0.314538370	0.7467869780	1.0016656
a_track[418]	-6.173625e-02	0.29669133	-0.519987714	0.4064476304	1.0021176
a_track[419]	-6.345432e-02	0.30038586	-0.537257572	0.4221174765	1.0015520
a_track[420]	-5.984316e-02	0.30015292	-0.543559922	0.4221900008	1.0009579
a_track[421]	-6.133728e-02	0.29412401	-0.533886572	0.4060475762	1.0018302
a_track[422]	1.711072e-01	0.34976806	-0.377779115	0.7143120494	1.0008059
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a_track[425]	-5.815834e-02	0.30316186	-0.527117585	0.4208138365	1.0022313
a_track[426]	9.102838e-02	0.29989178	-0.377744961	0.5626787466	1.0015532
a_track[427]	-2.166795e-01	0.29460917	-0.677111932	0.2632050568	1.0006855
a_track[428]	7.629742e-02	0.31915907	-0.427955529	0.5781403564	1.0002931
a_track[429]	-6.492831e-02	0.31200757	-0.561755904	0.4230827560	1.0006877
a_track[430]	-7.311353e-02	0.30802578	-0.554371428	0.4294303170	1.0023315
a_track[431]	7.348518e-02	0.30588502	-0.413077739	0.5866907805	1.0014332
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a_track[434]	2.248842e-01	0.32928942	-0.292398178	0.7480018544	1.0000392
a_track[435]	-3.528867e-01	0.28329425	-0.790008951	0.0909338224	1.0016752
a_track[436]	1.920913e-01	0.35396633	-0.374138733	0.7494312136	1.0007557
a_track[437]	7.252952e-02	0.32697101	-0.449514666	0.6058798668	1.0018796
a_track[438]	-6.658935e-02	0.29823315	-0.534390457	0.4121417451	1.0019096
a_track[439]	8.448298e-02	0.30988334	-0.409985437	0.5856619013	1.0005784
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a_track[441]	-7.701863e-02	0.29797745	-0.554204218	0.3938856703	1.0002946
a_track[442]	6.166551e-02	0.32649787	-0.462367867	0.5837388814	1.0002185
a_track[443]	-2.206394e-01	0.29151056	-0.686230965	0.2416167517	0.9997590
a_track[444]	-3.592695e-01	0.27805431	-0.799344578	0.0916865509	1.0016391
a_track[445]	-6.225550e-02	0.31409971	-0.561495172	0.4321725475	0.9997249
a_track[446]	7.257796e-02	0.32913344	-0.442297840	0.5947168639	1.0018698
a_track[447]	1.881845e-01	0.34595247	-0.359403415	0.7330919666	1.0012131
a_track[448]	6.283079e-02	0.32204287	-0.437371153	0.5789023309	1.0018456
a_track[449]	8.615460e-02	0.30146405	-0.390626576	0.5690897149	1.0021751
a_track[450]	-6.667713e-01	0.24937552	-1.057502076	-0.2614085334	0.9999268
a_track[451]	5.443167e-02	0.34417061	-0.505881209	0.6108150392	1.0011325

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a_track[453]	-4.386690e-01	0.28601898	-0.895269967	0.0126120090	1.0014747
a_track[454]	4.598893e-02	0.34273889	-0.502176122	0.6031851039	1.0025197
a_track[455]	6.947185e-02	0.30281029	-0.412866091	0.5555537261	1.0020216
a_track[456]	9.676796e-02	0.30259768	-0.389882627	0.5875649339	1.0023329
a_track[457]	9.358108e-02	0.32769286	-0.420162017	0.6153393120	1.0044831
a_track[458]	-6.641774e-02	0.30086351	-0.546863671	0.4135156892	1.0044728
a_track[459]	8.528952e-02	0.31095380	-0.421679600	0.5761446953	1.0020428
a_track[460]	9.334251e-02	0.31611460	-0.403520040	0.6033261754	1.0024303
a_track[461]	1.842006e-01	0.34148075	-0.354332361	0.7275029187	1.0033890
a_track[462]	7.623601e-02	0.30720960	-0.424610222	0.5750064347	1.0015158
a_track[463]	-6.988524e-02	0.31292121	-0.561704952	0.4362809042	1.0014331
a_track[464]	-5.911151e-02	0.30819832	-0.545643215	0.4325315672	0.9997614
a_track[465]	7.013861e-02	0.32412164	-0.447479427	0.5804439590	1.0023968
a_track[466]	1.775218e-01	0.33816754	-0.373760974	0.7170319437	1.0012407
a_track[467]	-5.510199e-02	0.29675456	-0.522238647	0.4234210814	1.0013891
a_track[468]	1.661700e-01	0.35260633	-0.396419498	0.7306395922	1.0022114
a_track[469]	-6.300052e-02	0.29935700	-0.532988889	0.4189337690	1.0028670
a_track[470]	-6.331428e-02	0.32132302	-0.577103888	0.4544471873	1.0033423
a_track[471]	-2.015212e-01	0.28321694	-0.647765989	0.2518204492	1.0003313
a_track[472]	2.272927e-01	0.33489806	-0.306994424	0.7679323207	1.0006939
a_track[473]	8.527795e-02	0.30111203	-0.391508312	0.5610932980	1.0021335
a_track[474]	5.912602e-02	0.32493396	-0.464737783	0.5761953489	1.0010155
a_track[475]	1.025835e-01	0.30643910	-0.377050437	0.5863971852	1.0006343
a_track[476]	-5.667018e-02	0.30702373	-0.550093175	0.4319324428	1.0048375
a_track[477]	1.899829e-01	0.34432168	-0.341301303	0.7400636235	1.0030161
a_track[478]	7.242911e-02	0.31045403	-0.420756626	0.5773964696	1.0005681
a_track[479]	6.649104e-02	0.33506390	-0.462811764	0.6011847933	1.0000686
a_track[480]	2.061618e-01	0.34818056	-0.357729493	0.7711116221	1.0006749
a_track[481]	-5.905888e-02	0.30350523	-0.539500682	0.4258567462	1.0020256
a_track[482]	8.654067e-02	0.31814620	-0.418424010	0.6028446400	1.0004421
a_track[483]	8.700859e-02	0.31504061	-0.419944564	0.6036069705	1.0013781
a_track[484]	-1.910468e-01	0.29592739	-0.664987975	0.2901188520	1.0029422
a_track[485]	9.214872e-02	0.30826615	-0.403738865	0.5948629900	1.0051006
a_track[486]	-1.892167e-01	0.28555807	-0.642608069	0.2763014653	1.0027651
a_track[487]	-5.942398e-02	0.30365589	-0.546007853	0.4366827951	1.0039925
a_track[488]	2.028464e-01	0.34642574	-0.339488623	0.7623848873	1.0024134
a_track[489]	-5.611948e-02	0.30679931	-0.540950618	0.4331255516	1.0016360
a_track[490]	-6.308560e-02	0.30891855	-0.553248264	0.4261000777	1.0015107
a_track[491]	7.556832e-02	0.32541268	-0.448122510	0.5993457132	1.0033402
a_track[492]	8.113421e-02	0.32115306	-0.436298040	0.5911962285	1.0009889
a_track[493]	-3.597068e-02	0.21721748	-0.375490463	0.3270767941	0.9996494
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a_track[495]	5.191781e-01	0.31719982	0.018506118	1.0226266000	1.0008561
a_track[496]	-3.239729e-02	0.21994119	-0.371077669	0.3245180695	1.0021164
a_track[497]	-3.352685e-02	0.21626165	-0.378815617	0.3090535516	1.0022409
a_track[498]	-2.043378e-01	0.20936711	-0.533453796	0.1336158489	1.0019241
a_track[499]	-3.056105e-02	0.22441517	-0.374144583	0.3433856863	1.0012569
a_track[500]	3.544282e-01	0.25488278	-0.046461431	0.7896676403	1.0037007
a_track[501]	-3.304785e-02	0.25984518	-0.456777572	0.3862072985	1.0013080
a_track[502]	4.940679e-01	0.33220670	-0.045983805	1.0191415350	1.0030547
a_track[503]	-7.958169e-01	0.21660056	-1.139561874	-0.4433120205	1.0001625
a_track[504]	-1.024248e-01	0.24536884	-0.486582523	0.3035087998	1.0020585
a_track[505]	-5.759898e-01	0.22567018	-0.929784690	-0.2132279086	1.0019200
a_track[506]	-4.036152e-01	0.23350363	-0.780328361	-0.0314023748	1.0006719
a_track[507]	4.024040e-01	0.32958674	-0.135534985	0.9287749440	1.0002307
a_track[508]	1.258088e-01	0.23601903	-0.245732826	0.5017184260	0.9996962
a_track[509]	6.555910e-01	0.28804836	0.198393948	1.1167634630	1.0005793
a_track[510]	-5.339179e-01	0.22138899	-0.880974283	-0.1766503472	1.0014887
a_track[511]	-1.643391e-01	0.19283880	-0.472573506	0.1467580252	1.0030897
a_track[512]	2.757757e-01	0.23187734	-0.089325830	0.6445751673	1.0001802
a_track[513]	-2.984147e-01	0.21006334	-0.625118886	0.0416551468	1.0011117
a_track[514]	-3.098611e-01	0.20558812	-0.634093325	0.0226653459	1.0001928
a_track[515]	-3.011412e-01	0.20729155	-0.628269117	0.0266754191	1.0008393
a_track[516]	6.059993e-01	0.28610774	0.157770614	1.0558920605	1.0012359
a_track[517]	-2.248672e-01	0.21083749	-0.557015039	0.1134410874	1.0025182
a_track[518]	-7.476606e-02	0.21423917	-0.408744788	0.2729941553	1.0014466
a_track[519]	-7.644736e-02	0.22231264	-0.428428896	0.2822649742	1.0008558
a_track[520]	-1.428062e-01	0.21524040	-0.485400875	0.2064319652	1.0019173
a_track[521]	1.174402e-01	0.22998752	-0.245287949	0.4884545168	0.9998490
a_track[522]	1.910675e-01	0.23535665	-0.172987363	0.5739693662	1.0011502
a_track[523]	-3.823964e-01	0.22261037	-0.729703317	-0.0205482935	1.0025388
a_track[524]	3.816076e-01	0.25910879	-0.019229586	0.8032133140	1.0023071
a_track[525]	2.681674e-01	0.25386807	-0.134569328	0.6779703159	1.0006409
a_track[526]	-1.784988e-01	0.26572669	-0.607241887	0.2437050363	1.0009479
a_track[527]	-4.507160e-02	0.22440977	-0.403107257	0.3286745287	1.0023723
a_track[528]	-2.709609e-01	0.21108527	-0.612161266	0.0691158904	1.0027606
a_track[529]	4.854921e-01	0.28151396	0.045944840	0.9434553345	1.0012498
a_track[530]	9.284237e-02	0.23019253	-0.273746356	0.4615049162	0.9999267
a_track[531]	-4.684373e-02	0.21219663	-0.380125045	0.2960842010	1.0001549
a_track[532]	2.923280e-02	0.21356345	-0.310513962	0.3826263769	1.0005639
a_track[533]	-2.925523e-01	0.18855557	-0.584865748	0.0081572807	1.0015207
a_track[534]	2.558566e-02	0.21134641	-0.303269179	0.3683536925	1.0016289
a_track[535]	-1.734612e-01	0.19158544	-0.483908934	0.1304565150	0.9998243
a_track[536]	-2.358473e-01	0.19541144	-0.543029382	0.0806416241	1.0003101
a_track[537]	2.560254e-01	0.23354274	-0.110517969	0.6323182998	1.0006809

a_track[538]	-1.663798e-01	0.19904806	-0.473854943	0.1591267583	1.0005807
a_track[539]	-1.657575e-01	0.19641107	-0.471325043	0.1595739564	1.0007236
a_track[540]	-3.765273e-02	0.20138528	-0.354094406	0.2912376305	1.0007535
a_track[541]	-2.814808e-01	0.19010011	-0.576838687	0.0252387541	1.0039129
a_track[542]	-1.008418e-01	0.20129885	-0.417187338	0.2297518363	0.9996086
a_track[543]	-9.789572e-02	0.21202948	-0.431527763	0.2455908474	1.0025709
a_track[544]	1.144849e-01	0.21862714	-0.227091048	0.4648941908	1.0007177
a_track[545]	4.533255e-02	0.21471804	-0.281169192	0.3864568884	1.0003247
a_track[546]	-2.794520e-01	0.19672343	-0.600621283	0.0344735479	1.0028363
a_track[547]	5.300006e-01	0.32540116	0.015176042	1.0523120760	1.0045549
a_track[548]	3.260384e-01	0.24074580	-0.058884385	0.7106690485	0.9999257
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a_track[550]	-1.570790e-01	0.19978525	-0.464696506	0.1696843415	1.0016913
a_track[551]	-3.981078e-02	0.20280577	-0.357415561	0.2895426243	1.0013935
a_track[552]	1.058959e-01	0.20704468	-0.226070902	0.4326215914	1.0022189
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a_track[554]	-1.131025e-01	0.22633756	-0.468493688	0.2493713772	1.0013966
a_track[555]	-2.590484e-02	0.23579466	-0.406604839	0.3604643737	1.0017234
a_track[556]	2.521938e-01	0.25365053	-0.143416550	0.6683440567	1.0013912
a_track[557]	-1.915308e-02	0.22501420	-0.378530546	0.3387793617	0.9999725
a_track[558]	6.291818e-01	0.31008891	0.122841739	1.1166485005	1.0024137
a_track[559]	-9.102768e-02	0.21782843	-0.429368912	0.2534805701	1.0002063
a_track[560]	3.599711e-01	0.25991267	-0.043773245	0.7742545734	1.0016172
a_track[561]	-4.452146e-01	0.19949872	-0.758227405	-0.1200831160	1.0014303
a_track[562]	4.441579e-01	0.28910782	-0.018618189	0.9064854095	0.9999157
a_track[563]	-1.673051e-01	0.21416307	-0.500387309	0.1827740937	1.0006775
a_track[564]	7.539632e-02	0.22689037	-0.277688222	0.4380025237	0.9996697
a_track[565]	-5.094858e-01	0.19563254	-0.816952350	-0.2004337935	1.0014520
a_track[566]	4.097310e-01	0.34938662	-0.152774674	0.9500796398	1.0000661
a_track[567]	3.005638e-01	0.25277870	-0.097097377	0.7014100349	1.0001867
a_track[568]	3.118728e-01	0.36606710	-0.267325343	0.8827184140	1.0001905
a_track[569]	-2.461441e-01	0.22812156	-0.608635272	0.1143556676	1.0015975
a_track[570]	-4.749363e-01	0.21879812	-0.823613423	-0.1236804059	1.0006294
a_track[571]	-1.246570e-01	0.23342823	-0.493853546	0.2555479359	1.0010053
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a_track[573]	-2.622707e-02	0.22728425	-0.387547917	0.3414621091	1.0013919
a_track[574]	-2.547628e-01	0.22159318	-0.613155786	0.0947678281	1.0042278
a_track[575]	-3.301937e-01	0.20403637	-0.653513453	0.0003389617	1.0010476
a_track[576]	-1.813747e-01	0.21443986	-0.519110389	0.1594490538	1.0014528
a_track[577]	-3.982149e-01	0.20641093	-0.720211411	-0.0574491876	0.9996393
a_track[578]	6.110184e-02	0.23272819	-0.304040613	0.4354039984	1.0000174
a_track[579]	-1.156370e-01	0.22806416	-0.470531320	0.2484036558	1.0058669
a_track[580]	-2.048909e-02	0.22889356	-0.371474828	0.3468854907	1.0021879

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a_track[583]	-9.613310e-02	0.22224047	-0.446633938	0.2646791343	1.0008908
a_track[584]	-3.238037e-01	0.21664131	-0.669843530	0.0344948058	1.0011051
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a_track[586]	-3.251809e-01	0.19219353	-0.626650023	-0.0179372111	0.9999865
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a_track[588]	-3.516495e-01	0.26724821	-0.775686728	0.0796827387	1.0010707
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a_cell[1]	6.114834e-03	0.06638085	-0.094967653	0.1137114708	1.1213479
a_cell[2]	-3.258942e-03	0.06541130	-0.105553277	0.0958877628	1.1143802
a_cell[3]	-4.538739e-03	0.06955815	-0.113893110	0.1006577363	1.1106277
a_cell[4]	8.005086e-04	0.06713809	-0.103312007	0.1110032126	1.1307884
a_cell[5]	-2.879782e-03	0.06750252	-0.108711611	0.0983666078	1.1196901
a_cell[6]	-5.444761e-03	0.06705916	-0.114183648	0.0984787876	1.1342039
a_cell[7]	8.072592e-03	0.06689545	-0.096349239	0.1162266966	1.1142509
a_cell[8]	5.347109e-03	0.06676406	-0.097428742	0.1161656992	1.1186980
a_cell[9]	2.168102e-02	0.06654433	-0.071034312	0.1423569582	1.1212516
a_cell[10]	1.175397e-03	0.06635782	-0.102696820	0.1068340163	1.1185049
a_cell[11]	2.069311e-03	0.06368895	-0.100161900	0.1030553399	1.1203584
a_cell[12]	6.938460e-04	0.06829169	-0.104603548	0.1091514743	1.1361632
a_cell[13]	-1.683939e-02	0.06953240	-0.136443057	0.0823384559	1.1115880
a_cell[14]	-1.391393e-02	0.06870246	-0.130695703	0.0858667924	1.1191990
a_cell[15]	3.167352e-02	0.07102111	-0.060204712	0.1563528300	1.1053809
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a_cell[17]	-5.472103e-03	0.07098021	-0.119128049	0.0974040734	1.1254337
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a_cell[22]	-1.452940e-02	0.06261467	-0.118659427	0.0775643695	1.0945238
a_cell[23]	-1.336925e-02	0.06623055	-0.124523491	0.0824432628	1.1256867
a_cell[24]	2.131525e-02	0.06554978	-0.067287537	0.1371473226	1.1123457
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a_cell[26]	1.072269e-02	0.06285374	-0.082599364	0.1171781800	1.1250231
a_cell[27]	1.410441e-03	0.06662291	-0.103614191	0.1033996973	1.1256592
a_cell[28]	3.972351e-03	0.06506153	-0.093081108	0.1098789022	1.1225889
a_cell[29]	8.858875e-04	0.06634580	-0.101814831	0.1075065241	1.1359840
a_cell[30]	-8.693342e-03	0.06917832	-0.120175652	0.0928944831	1.1162486
a_cell[31]	3.627394e-03	0.06649929	-0.098561583	0.1120610300	1.1186114

a_cell[32]	-5.429987e-04	0.06674680	-0.107795283	0.1058173925	1.1236601
a_cell[33]	-5.218943e-03	0.06836479	-0.117651239	0.0969481097	1.1127980
a_cell[34]	-3.354874e-03	0.06686714	-0.108016136	0.1038683938	1.1118401
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a_cell[37]	1.634222e-03	0.06295561	-0.095523656	0.1071771442	1.1461605
a_cell[38]	-1.641358e-02	0.06569100	-0.130523355	0.0782997879	1.1155058
a_cell[39]	1.160481e-02	0.06317738	-0.080287692	0.1179506624	1.1122069
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a_cell[45]	-3.764224e-03	0.06104660	-0.101915810	0.0913426476	1.1105076
a_cell[46]	-2.562289e-05	0.06410436	-0.101541737	0.0989796681	1.1138056
a_cell[47]	2.944156e-04	0.06518456	-0.103963121	0.1027197082	1.1415966
a_cell[48]	1.297854e-02	0.06506300	-0.083612765	0.1242372737	1.1107562
a_cell[49]	-1.051609e-02	0.06637926	-0.130717217	0.0878599489	1.1202722
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a_cell[51]	1.294642e-02	0.06743754	-0.081869618	0.1278940651	1.1294629
a_cell[52]	4.250544e-03	0.06480433	-0.097876271	0.1106462427	1.1256896
a_cell[53]	-3.611012e-03	0.06658812	-0.111155317	0.0966556899	1.1384929
a_cell[54]	5.991370e-03	0.06587191	-0.095590593	0.1115645904	1.1262038
a_cell[55]	-3.810821e-03	0.06385933	-0.105394941	0.0950230695	1.1116040
a_cell[56]	-1.101177e-02	0.06316696	-0.115103288	0.0802446291	1.1347925
a_cell[57]	-5.191452e-03	0.06421724	-0.109566021	0.0931463809	1.1042292
a_cell[58]	1.333343e-02	0.06234884	-0.079977135	0.1209863809	1.1119807
a_cell[59]	-2.721888e-02	0.06536553	-0.142801753	0.0574038103	1.1002666
a_cell[60]	2.170217e-02	0.06385680	-0.070045904	0.1337865721	1.0984533
a_cell[61]	-2.550565e-02	0.06532020	-0.136305636	0.0636770760	1.1057162
a_cell[62]	2.385469e-04	0.06491116	-0.102153977	0.1036305075	1.1366357
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a_cell[65]	-3.572623e-03	0.06396544	-0.109553249	0.0955868204	1.1038209
a_cell[66]	-9.908770e-03	0.06472527	-0.119924018	0.0880761946	1.1065072
a_cell[67]	3.451319e-03	0.06290944	-0.094751276	0.1074808263	1.1062075
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a_cell[70]	-2.569436e-03	0.06534890	-0.110051230	0.0994334654	1.1224134
a_cell[71]	5.939862e-03	0.06472196	-0.095899806	0.1120165107	1.1204485
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a_cell[73]	-2.430348e-03	0.06680104	-0.109329666	0.1037099940	1.1359340
a_cell[74]	-1.092480e-02	0.07003493	-0.124292979	0.0909589685	1.1040819

a_cell[75]	1.290429e-02	0.06603835	-0.083660334	0.1239927136	1.1279929
a_cell[76]	-5.149669e-03	0.06410106	-0.108985540	0.0892058598	1.1244146
a_cell[77]	-4.629640e-03	0.06390675	-0.106472693	0.0927697118	1.1115657
a_cell[78]	2.222082e-02	0.06450814	-0.065163988	0.1353011978	1.1101189
a_cell[79]	1.225166e-02	0.06711398	-0.087478122	0.1273712859	1.1177871
a_cell[80]	-1.498042e-02	0.06552419	-0.122778375	0.0793369656	1.1170031
a_cell[81]	-1.932567e-02	0.06496334	-0.129978407	0.0725861446	1.0929322
a_cell[82]	1.562898e-03	0.06232837	-0.097308131	0.1045039001	1.1312883
a_cell[83]	6.751076e-03	0.06017418	-0.088831820	0.1080951091	1.1072067
a_cell[84]	-3.817521e-02	0.07053216	-0.169826545	0.0477365389	1.0948380
a_cell[85]	-1.014581e-02	0.06069271	-0.114154550	0.0820679581	1.1039216
a_cell[86]	-5.489510e-03	0.06048996	-0.105501617	0.0892085820	1.1087354
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a_cell[88]	-8.577389e-03	0.06664931	-0.117165596	0.0899388964	1.1156917
a_cell[89]	2.272107e-02	0.06555779	-0.066356679	0.1395677239	1.1174720
a_cell[90]	-1.608011e-02	0.06779539	-0.131219919	0.0804509773	1.0935522
a_cell[91]	-8.146875e-03	0.06426474	-0.113507271	0.0904937213	1.1207374
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a_cell[94]	2.285006e-03	0.06836503	-0.102474572	0.1125780033	1.1235589
a_cell[95]	-8.088233e-03	0.06659804	-0.116965077	0.0941492410	1.1019915
a_cell[96]	5.403049e-03	0.06218430	-0.088301680	0.1065764456	1.1129198
a_cell[97]	-2.388937e-03	0.06456228	-0.105366787	0.0993278330	1.1109967
a_cell[98]	-1.224560e-02	0.06648887	-0.122834251	0.0851523230	1.1368758
a_cell[99]	3.812168e-02	0.07306122	-0.047345712	0.1695008423	1.0936799
a_cell[100]	-2.797184e-03	0.06817680	-0.111255690	0.1039425759	1.1368882
a_cell[101]	-6.943490e-03	0.06433338	-0.109282813	0.0914049772	1.1174622
a_cell[102]	1.140654e-02	0.06580775	-0.085876047	0.1205096353	1.1107146
a_cell[103]	-1.668657e-03	0.06621958	-0.107015974	0.0998456268	1.1159740
a_cell[104]	3.381872e-02	0.06309918	-0.047237107	0.1478085070	1.0745710
a_cell[105]	1.963691e-03	0.06083226	-0.094605939	0.1015190271	1.0946200
a_cell[106]	-1.110979e-02	0.06407708	-0.120145563	0.0843863252	1.1185749
a_cell[107]	-2.497700e-02	0.06899328	-0.149777781	0.0693306790	1.0968828
a_cell[108]	-3.486290e-03	0.06201205	-0.102365940	0.0907589010	1.1126199
a_cell[109]	-2.033870e-03	0.06173573	-0.102029747	0.0939241066	1.1179723
a_cell[110]	-1.642489e-02	0.06277500	-0.125787256	0.0749096339	1.1082691
a_cell[111]	-2.327585e-02	0.06503335	-0.139043162	0.0664242260	1.0907338
a_cell[112]	-9.361721e-03	0.06689831	-0.124483457	0.0907151627	1.1117686
a_cell[113]	-1.213552e-02	0.06827260	-0.129368355	0.0856785270	1.1165801
a_cell[114]	3.271590e-03	0.06386179	-0.093049161	0.1091961386	1.1264566
a_cell[115]	-4.615153e-03	0.06575549	-0.111027153	0.0960474541	1.1291455
a_cell[116]	-1.313463e-02	0.06765248	-0.127456959	0.0829323317	1.0873507
a_cell[117]	-4.536528e-03	0.06665398	-0.117918206	0.0985088350	1.1040103

a_cell[118]	-4.814938e-03	0.06890295	-0.113128769	0.1046804145	1.1226459
a_cell[119]	3.480494e-02	0.07016157	-0.053447729	0.1574104489	1.0904862
a_cell[120]	1.795238e-03	0.06401557	-0.099173984	0.1052079567	1.1120856
a_cell[121]	5.685613e-03	0.06090599	-0.090414554	0.1065751326	1.1228576
a_cell[122]	1.638266e-02	0.06567211	-0.074799447	0.1279953158	1.1113961
a_cell[123]	-6.509948e-03	0.06731442	-0.111704979	0.0953721279	1.1216441
a_cell[124]	-8.310259e-03	0.06378641	-0.116101489	0.0870017023	1.1034376
a_cell[125]	-1.343391e-02	0.06569912	-0.123219347	0.0822022929	1.1007533
a_cell[126]	1.268008e-02	0.06311469	-0.079724363	0.1147021377	1.1007295
a_cell[127]	8.260417e-03	0.06214527	-0.085731091	0.1103768176	1.1078867
a_cell[128]	-1.640540e-02	0.06579091	-0.131366477	0.0730076198	1.0960309
a_cell[129]	1.790157e-02	0.06629690	-0.077182307	0.1323873565	1.1193887
a_cell[130]	8.695065e-03	0.06394703	-0.087196137	0.1144313324	1.1163517
a_cell[131]	-2.697758e-02	0.06970496	-0.149474377	0.0626589435	1.0950346
a_cell[132]	7.400030e-03	0.06286422	-0.087574385	0.1120703425	1.1218736
a_cell[133]	-5.534500e-03	0.06167261	-0.106906627	0.0926787567	1.1164232
a_cell[134]	-5.338868e-03	0.06330971	-0.106643776	0.0922904241	1.1067877
a_cell[135]	-2.168833e-02	0.06435645	-0.132350817	0.0686422004	1.1282085
a_cell[136]	1.504446e-02	0.06618046	-0.079738933	0.1272632579	1.1233712
a_cell[137]	7.881342e-03	0.06506708	-0.093311946	0.1127118305	1.1158637
a_cell[138]	4.864327e-03	0.06279356	-0.094246509	0.1079343618	1.1089044
a_cell[139]	-8.593260e-03	0.06572622	-0.117251614	0.0918868890	1.1067508
a_cell[140]	1.201012e-02	0.06616830	-0.085167194	0.1235387118	1.1278360
a_cell[141]	-1.181908e-02	0.06860189	-0.124376147	0.0899946122	1.1190654
a_cell[142]	-3.901634e-03	0.06588737	-0.109759856	0.1003755061	1.1262719
a_cell[143]	-4.865060e-04	0.06702711	-0.107733235	0.1062360683	1.1326904
a_cell[144]	2.072625e-03	0.06376062	-0.098325809	0.1079302572	1.1094290
a_cell[145]	-1.235502e-02	0.06599202	-0.129045170	0.0864292346	1.1217233
a_cell[146]	7.371911e-03	0.06460266	-0.091370092	0.1132714542	1.1464835
a_cell[147]	3.747731e-03	0.06436245	-0.095282757	0.1099110162	1.1099706
a_cell[148]	-2.521070e-03	0.06661291	-0.108841955	0.1004479768	1.1214273
a_cell[149]	-7.709053e-03	0.06615183	-0.120405942	0.0918789340	1.1143669
a_cell[150]	1.374387e-02	0.06682436	-0.081869652	0.1269893241	1.1243274
a_cell[151]	-3.260504e-03	0.06538078	-0.109261825	0.1011660223	1.1132869
a_cell[152]	1.285893e-02	0.06525204	-0.081061002	0.1201914624	1.1142057
a_cell[153]	-7.832123e-04	0.06790482	-0.109928679	0.1065909171	1.1140109
a_cell[154]	1.910046e-02	0.06657491	-0.077492415	0.1347044485	1.1099108
a_cell[155]	6.311358e-03	0.06442801	-0.093901849	0.1155195727	1.1098170
a_cell[156]	-1.765565e-02	0.06280534	-0.125802192	0.0683168273	1.0960776
a_cell[157]	-7.559746e-03	0.06232388	-0.108646389	0.0887542862	1.1110415
a_cell[158]	6.567129e-03	0.06193104	-0.085055941	0.1066753150	1.0997887
a_cell[159]	3.679496e-02	0.06834026	-0.046930149	0.1566352718	1.0934247
a_cell[160]	-3.780484e-03	0.06340123	-0.105102652	0.0958712275	1.1182882

a_cell[161]	-1.020239e-02	0.06206317	-0.112943517	0.0806869537	1.1017305
a_cell[162]	-2.529574e-03	0.06347236	-0.105461591	0.0935378845	1.1188826
a_cell[163]	-3.797403e-03	0.06173531	-0.101962555	0.0904046395	1.1060609
a_cell[164]	-4.215602e-03	0.06463257	-0.111430000	0.0958497320	1.1094865
a_cell[165]	-9.868780e-03	0.06433175	-0.119168776	0.0891452383	1.1194624
a_cell[166]	4.633252e-04	0.06726416	-0.103619813	0.1023463171	1.1181690
a_cell[167]	-3.386348e-03	0.06852356	-0.109476553	0.1008036679	1.1386828
a_cell[168]	-1.430919e-02	0.06802250	-0.129870653	0.0814271243	1.1057224
a_cell[169]	4.824342e-02	0.07601966	-0.035698860	0.1891139257	1.0811015
a_cell[170]	2.796131e-03	0.06881238	-0.106895865	0.1130384067	1.1396349
a_cell[171]	7.106017e-03	0.06718567	-0.093103422	0.1121340194	1.1184332
a_cell[172]	-9.752110e-04	0.06675910	-0.108279199	0.1048919541	1.1426854
a_cell[173]	9.123268e-03	0.06478443	-0.091272861	0.1163890481	1.1127230
a_cell[174]	-1.129687e-03	0.06588615	-0.105200419	0.1032723767	1.1430583
a_cell[175]	-2.000533e-03	0.06479960	-0.102791098	0.1001333074	1.1238220
a_cell[176]	1.182078e-03	0.06720135	-0.104496130	0.1078402233	1.1228716
a_cell[177]	-5.514946e-03	0.06587743	-0.108120344	0.0984797356	1.1149273
a_cell[178]	-1.414680e-02	0.06634046	-0.127206651	0.0831608027	1.1007314
a_cell[179]	-6.292775e-03	0.06316021	-0.108099478	0.0918757211	1.1229731
a_cell[180]	1.399030e-02	0.06432498	-0.080457382	0.1229714627	1.1252011
a_cell[181]	9.389689e-03	0.06220211	-0.080702481	0.1143971427	1.1021401
a_cell[182]	-1.658214e-02	0.06393178	-0.130191001	0.0747098343	1.1091057
a_cell[183]	-1.182230e-02	0.06433615	-0.120915129	0.0796169460	1.1144480
a_cell[184]	5.686375e-03	0.06559207	-0.096779234	0.1097742881	1.1247197
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a_cell[186]	-6.269054e-03	0.06328662	-0.109907747	0.0901226137	1.1051983
a_cell[187]	-1.485536e-03	0.06309695	-0.099096024	0.0945458324	1.1164461
a_cell[188]	-2.555784e-04	0.06570621	-0.102453495	0.1052804965	1.1475878
a_cell[189]	9.873310e-04	0.06461002	-0.104080611	0.1035576559	1.1466577
a_cell[190]	-2.621670e-03	0.06636742	-0.110046567	0.1027771052	1.1241127
a_cell[191]	3.388670e-03	0.06762773	-0.102979519	0.1149554504	1.1272036
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a_cell[193]	-2.169283e-03	0.06665013	-0.106932754	0.1018659069	1.1269062
a_cell[194]	7.328290e-03	0.06626710	-0.091221937	0.1152016123	1.1115243
a_cell[195]	3.067702e-02	0.06689432	-0.058239817	0.1512087654	1.1220233
a_cell[196]	-1.066496e-02	0.06787482	-0.126145534	0.0908478923	1.1111111
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a_cell[199]	-1.305973e-03	0.06530592	-0.102685477	0.0993125894	1.1218658
a_cell[200]	-6.909870e-03	0.06701414	-0.119504969	0.0948048552	1.1292848
a_cell[201]	-4.519994e-03	0.06789466	-0.114692898	0.1019135568	1.1409657
a_cell[202]	1.426823e-02	0.06705744	-0.083305827	0.1309504655	1.1292497
a_cell[203]	1.261382e-02	0.06589291	-0.081693708	0.1230892274	1.1138541

a_cell[204]	5.124499e-02	0.07703484	-0.039069968	0.1964915093	1.0830832
a_cell[205]	2.588651e-02	0.06663450	-0.062864466	0.1450278243	1.0974134
a_cell[206]	-8.051614e-03	0.06769207	-0.122619617	0.0957648810	1.1176316
a_cell[207]	-1.182010e-02	0.06101849	-0.114405221	0.0775734961	1.1068885
a_cell[208]	-2.720801e-02	0.06791929	-0.149656026	0.0616467226	1.0878485
a_cell[209]	-6.124637e-03	0.06455563	-0.114842962	0.0929371500	1.1267066
a_cell[210]	7.369499e-03	0.06622006	-0.095162356	0.1182843591	1.1221645
a_cell[211]	5.367493e-02	0.07752264	-0.035355999	0.2007049519	1.1050056
a_cell[212]	-1.778005e-02	0.06495983	-0.132194356	0.0735902319	1.1104739
a_cell[213]	-6.555457e-03	0.06580932	-0.114508308	0.0913960941	1.1020902
a_cell[214]	1.752450e-02	0.06744344	-0.076081186	0.1340578277	1.1197964
a_cell[215]	-4.414881e-03	0.06588836	-0.107522951	0.0955537640	1.1176581
a_cell[216]	-6.156120e-03	0.06671549	-0.109434535	0.0950113059	1.1063691
a_cell[217]	-3.198524e-03	0.06778240	-0.111728813	0.0994338034	1.1350292
a_cell[218]	1.287567e-02	0.06672614	-0.080980862	0.1224086264	1.1236063
a_cell[219]	8.084379e-03	0.06506247	-0.089844813	0.1113368560	1.0949430
a_cell[220]	-5.192200e-03	0.06443489	-0.111924571	0.0944486413	1.1301761
a_cell[221]	6.847581e-03	0.06651369	-0.093003909	0.1122442654	1.1232808
a_cell[222]	2.788748e-03	0.06676960	-0.103129233	0.1150558696	1.1316825
a_cell[223]	1.361800e-03	0.06682776	-0.101727145	0.1096208977	1.1308721
a_cell[224]	-6.597513e-04	0.06195225	-0.100508606	0.0976605633	1.1011176
a_cell[225]	1.417209e-02	0.06733959	-0.081342300	0.1304396116	1.1011496
a_cell[226]	1.211969e-03	0.06495871	-0.100077385	0.1022303598	1.1152984
a_cell[227]	1.232088e-02	0.06556977	-0.084605085	0.1206920577	1.1097013
a_cell[228]	-4.262955e-03	0.06485801	-0.113173720	0.0970628686	1.1209764
a_cell[229]	-8.624446e-03	0.06458513	-0.122546832	0.0865653572	1.1150856
a_cell[230]	-2.659022e-02	0.06712986	-0.150542077	0.0625389620	1.0866567
a_cell[231]	1.094054e-03	0.06555660	-0.102385445	0.1058530109	1.1298337
a_cell[232]	-2.638778e-02	0.07083590	-0.150289900	0.0673800576	1.1069682
a_cell[233]	-2.722019e-03	0.06187058	-0.104013832	0.0930467412	1.1111603
a_cell[234]	6.846104e-03	0.06106001	-0.085739067	0.1071647578	1.1255845
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a_cell[238]	-9.440434e-03	0.06418517	-0.118682449	0.0861118600	1.1196906
a_cell[239]	-4.826907e-03	0.06556345	-0.112481418	0.0950553919	1.1152994
a_cell[240]	-3.434065e-03	0.06989507	-0.117220416	0.1019601752	1.1333303
a_cell[241]	2.757352e-02	0.07213412	-0.070172792	0.1565259556	1.1152015
a_cell[242]	-9.159225e-03	0.06691282	-0.119192861	0.0904099918	1.1134455
a_cell[243]	-6.150682e-03	0.06811700	-0.119827245	0.0976108861	1.1288577
a_cell[244]	-8.061599e-03	0.06817678	-0.120022374	0.0932837514	1.1134307
a_cell[245]	-5.527930e-03	0.06759717	-0.116258264	0.0963398904	1.1326664
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a_cell[250]	1.709689e-03	0.06599070	-0.102997161	0.1081432423	1.1313407
a_cell[251]	-1.762730e-02	0.06723146	-0.129550852	0.0779971125	1.1266016
a_cell[252]	2.418997e-02	0.06950754	-0.069174449	0.1454273320	1.1164650
a_cell[253]	-1.741590e-02	0.06655111	-0.130055090	0.0749889967	1.1103114
a_cell[254]	-3.306413e-03	0.06432673	-0.108441832	0.0936627876	1.1234262
a_cell[255]	-1.143475e-03	0.06379875	-0.106836594	0.0981498014	1.1150024
a_cell[256]	9.367231e-04	0.06477893	-0.101940085	0.1066851083	1.1205979
a_cell[257]	-9.380528e-03	0.06245300	-0.110597104	0.0849935995	1.1259591
a_cell[258]	6.466051e-03	0.06153749	-0.087657880	0.1057847106	1.0995939
a_cell[259]	-1.477825e-04	0.06476379	-0.104895240	0.1067852996	1.1368804
a_cell[260]	-1.202965e-02	0.06442289	-0.125037499	0.0849261305	1.1134800
a_cell[261]	6.987239e-03	0.06418401	-0.090179933	0.1117649266	1.1263007
a_cell[262]	5.443467e-03	0.05783646	-0.085453385	0.1022818137	1.0924706
a_cell[263]	-1.357125e-02	0.06314854	-0.124171563	0.0767932609	1.0981776
a_cell[264]	1.128595e-02	0.06802753	-0.093746752	0.1274560897	1.1225388
a_cell[265]	-5.004818e-03	0.06914168	-0.115346628	0.0985676543	1.1481874
a_cell[266]	7.385683e-03	0.06684102	-0.092839989	0.1192887840	1.1430807
a_cell[267]	-1.271801e-02	0.06692266	-0.128938632	0.0883155931	1.1204924
a_cell[268]	1.243824e-03	0.06692447	-0.103244141	0.1057317759	1.1291133
a_cell[269]	-3.909630e-03	0.06859520	-0.114604784	0.1033690912	1.1325933
a_cell[270]	-1.595601e-02	0.06972640	-0.136563730	0.0786672119	1.0911050
a_cell[271]	-1.322096e-02	0.06829709	-0.123854566	0.0838877413	1.1161073
a_cell[272]	-8.673691e-03	0.06781555	-0.121463249	0.0952265013	1.1209541
a_cell[273]	1.966565e-03	0.06856221	-0.105318374	0.1086061915	1.1226244
a_cell[274]	-3.757702e-03	0.06552869	-0.111324950	0.1001210619	1.1238094
a_cell[275]	-1.045054e-03	0.06777175	-0.108812017	0.1048838104	1.1430024
a_cell[276]	1.474816e-02	0.06591321	-0.083220816	0.1295395289	1.1079675
a_cell[277]	-1.454627e-02	0.06798210	-0.127619134	0.0814708058	1.1235113
a_cell[278]	9.692981e-03	0.06544983	-0.081044785	0.1170760672	1.1072392
a_cell[279]	2.014511e-03	0.06574113	-0.100857268	0.1040811933	1.1339815
a_cell[280]	-1.084732e-03	0.06916025	-0.108659417	0.1057973937	1.1227159
a_cell[281]	-1.411858e-02	0.06641135	-0.126805108	0.0848755229	1.1245505
a_cell[282]	1.588713e-03	0.06515661	-0.099948558	0.1043970726	1.1052232
a_cell[283]	-4.607609e-03	0.06340719	-0.107089798	0.0951905762	1.1126888
a_cell[284]	2.307240e-02	0.06922999	-0.068663482	0.1426864224	1.1027565
a_cell[285]	-5.127230e-03	0.06500352	-0.112890332	0.0959253547	1.1251605
a_cell[286]	-2.204461e-02	0.07065290	-0.142001992	0.0714433753	1.1229584
a_cell[287]	-1.516528e-03	0.06396497	-0.104103732	0.1020029322	1.1260742
a_cell[288]	-2.153901e-02	0.06894520	-0.145549124	0.0702181373	1.1030752
a_cell[289]	-1.162480e-03	0.06393235	-0.107734150	0.0983134520	1.1213525

a_cell[290]	-3.187056e-03	0.06431470	-0.107660388	0.0966233508	1.0913883
a_cell[291]	-6.644308e-03	0.06869245	-0.118244365	0.0952879089	1.1181119
a_cell[292]	-1.151167e-03	0.06685673	-0.104221881	0.1057546927	1.1125852
a_cell[293]	1.171735e-02	0.06774114	-0.081645033	0.1253437814	1.1027533
a_cell[294]	-8.017328e-03	0.06644239	-0.111488073	0.0909272078	1.1209979
a_cell[295]	-7.354090e-03	0.06790347	-0.116061482	0.0941624937	1.1425782
a_cell[296]	-6.428652e-03	0.06752186	-0.114474809	0.0969489994	1.1167370
a_cell[297]	-1.001743e-02	0.06766472	-0.125166548	0.0904204540	1.1175773
a_cell[298]	-8.408796e-03	0.06941137	-0.118244967	0.0921685728	1.1227606
a_cell[299]	5.219958e-03	0.06730331	-0.098065416	0.1164447833	1.1239684
a_cell[300]	1.677957e-03	0.06556664	-0.102023385	0.1047669569	1.1191819
a_cell[301]	1.286326e-02	0.06610776	-0.083344556	0.1249069862	1.1233920
a_cell[302]	5.074447e-04	0.06734559	-0.108043649	0.1089345970	1.1224084
a_cell[303]	1.086799e-02	0.06617219	-0.087018572	0.1244673791	1.1374924
a_cell[304]	-1.328373e-02	0.06665769	-0.129651630	0.0816006125	1.1367265
a_cell[305]	-1.339939e-02	0.06543959	-0.125217931	0.0870713678	1.1169130
a_cell[306]	-1.769550e-02	0.06765034	-0.135958648	0.0780240075	1.1117657
a_cell[307]	-4.423181e-03	0.06996639	-0.117903217	0.1013940439	1.1447565
a_cell[308]	4.391658e-04	0.06468047	-0.104689995	0.1019518032	1.1314832
a_cell[309]	-7.406646e-03	0.06607236	-0.119336516	0.0894645346	1.1274693
a_cell[310]	-9.337846e-03	0.06506569	-0.116242717	0.0852077339	1.1255639
a_cell[311]	3.194617e-03	0.06790994	-0.102983349	0.1116425591	1.1185671
a_cell[312]	-2.659565e-02	0.06887099	-0.151974701	0.0619300069	1.0990811
a_cell[313]	-3.961616e-03	0.06669098	-0.109995838	0.0986131743	1.1535967
a_cell[314]	5.526688e-03	0.06398178	-0.094025662	0.1103119591	1.1262775
a_cell[315]	2.389068e-02	0.06541563	-0.063087380	0.1422365239	1.1118498
a_cell[316]	2.927510e-02	0.06884524	-0.060320787	0.1542590287	1.1036584
a_cell[317]	-2.038578e-02	0.07000746	-0.140710119	0.0775365711	1.0956966
a_cell[318]	8.149928e-04	0.06473237	-0.101305899	0.1072709876	1.1239993
a_cell[319]	-4.766703e-03	0.06630221	-0.112629937	0.1025157236	1.1288144
a_cell[320]	-1.267288e-03	0.06578168	-0.107097407	0.0993702603	1.1238469
a_cell[321]	1.079131e-02	0.06689067	-0.088562656	0.1242074330	1.1076916
a_cell[322]	1.223103e-03	0.06521585	-0.102884057	0.1071983415	1.1412402
a_cell[323]	3.390274e-03	0.06757498	-0.101372151	0.1119863388	1.1272072
a_cell[324]	6.015332e-03	0.06722005	-0.094994628	0.1200530843	1.1259397
a_cell[325]	2.242618e-02	0.06933140	-0.070307238	0.1468306058	1.1271159
a_cell[326]	-1.602203e-05	0.06900163	-0.109542549	0.1096038533	1.1407112
a_cell[327]	2.975656e-03	0.06557285	-0.099984962	0.1144052819	1.1263247
a_cell[328]	2.955370e-03	0.06597688	-0.101366301	0.1082502324	1.1011769
a_cell[329]	-3.938339e-03	0.06485948	-0.111088360	0.0906968745	1.1067364
a_cell[330]	6.272536e-04	0.06573377	-0.105575953	0.1056016844	1.1208259
a_cell[331]	5.655582e-05	0.06527396	-0.101383751	0.1018116813	1.1387906
a_cell[332]	6.867985e-03	0.07127597	-0.099494988	0.1193481726	1.1420285

a_cell[333]	-1.043191e-02	0.06676889	-0.120747171	0.0912019168	1.1440398
a_cell[334]	2.784451e-02	0.06941360	-0.062672302	0.1537397625	1.0962552
a_cell[335]	2.262617e-02	0.06538154	-0.066912415	0.1356622919	1.1082407
a_cell[336]	-6.069316e-03	0.06498861	-0.112048898	0.0914197130	1.1157510
a_cell[337]	-4.826312e-03	0.06531089	-0.108569360	0.0964571694	1.1203300
a_cell[338]	5.158833e-03	0.06786971	-0.096735272	0.1150284323	1.1207753
a_cell[339]	-8.883511e-03	0.06629814	-0.118801717	0.0928312159	1.1415822
a_cell[340]	-4.008945e-03	0.06500248	-0.108035988	0.0972371027	1.1286589
a_cell[341]	7.363074e-03	0.06411428	-0.088006707	0.1119793971	1.1117386
a_cell[342]	-1.193862e-03	0.06561076	-0.104671341	0.0975180600	1.1182623
a_cell[343]	-1.149405e-02	0.06833040	-0.127181968	0.0847761294	1.1346098
a_cell[344]	-1.388960e-02	0.06181331	-0.119473479	0.0766003190	1.1060430
a_cell[345]	-4.280996e-03	0.06607990	-0.111863497	0.0995278334	1.1438696
a_cell[346]	4.069974e-03	0.06671704	-0.099947093	0.1140893991	1.1155144
a_cell[347]	3.810986e-03	0.06623741	-0.099265538	0.1088993347	1.1233029
a_cell[348]	1.012465e-03	0.06573435	-0.103554985	0.1109945084	1.1226286
a_cell[349]	6.390210e-03	0.06794937	-0.097951458	0.1179716848	1.1374684
a_cell[350]	-2.162520e-04	0.06716241	-0.108512252	0.1063165599	1.1219943
a_cell[351]	9.719664e-03	0.06515383	-0.084995894	0.1175496294	1.1041756
a_cell[352]	1.499894e-02	0.06799905	-0.083312487	0.1355956149	1.1229240
a_cell[353]	-4.427741e-03	0.06481865	-0.113037538	0.0970281560	1.1295343
a_cell[354]	2.829717e-03	0.06429116	-0.096157953	0.1044580347	1.1166602
a_cell[355]	5.014347e-03	0.06783072	-0.096378580	0.1168007714	1.1241325
a_cell[356]	-2.126495e-03	0.06759569	-0.111156808	0.1045203665	1.1115416
a_cell[357]	-4.480389e-03	0.06591159	-0.111666064	0.0953060780	1.1040777
a_cell[358]	-9.535537e-03	0.06901932	-0.120638501	0.0918037627	1.1287983
a_cell[359]	-1.092472e-02	0.06709156	-0.126465654	0.0899613800	1.1100193
a_cell[360]	1.094016e-03	0.06754378	-0.106951145	0.1118161428	1.1281703
a_cell[361]	1.430851e-03	0.06582281	-0.100609231	0.1028415248	1.1269927
a_cell[362]	-3.331471e-03	0.06441757	-0.107064715	0.0947397349	1.1139709
a_cell[363]	-1.939689e-02	0.06612733	-0.134648035	0.0713103052	1.1172123
a_cell[364]	-1.032633e-02	0.06402650	-0.120144452	0.0866489436	1.1234315
a_cell[365]	-9.294278e-03	0.06520609	-0.116625230	0.0898915840	1.1246423
a_cell[366]	-5.006936e-03	0.06696229	-0.114844368	0.1011727325	1.1161386
a_cell[367]	-1.681931e-02	0.06941413	-0.137004236	0.0806176568	1.1042305
a_cell[368]	-1.663210e-02	0.06615371	-0.131577976	0.0796069482	1.1055859
a_cell[369]	-2.773241e-03	0.06578172	-0.108390833	0.1010420866	1.1144804
a_cell[370]	-7.448357e-04	0.06308356	-0.098660051	0.0981496037	1.1171505
a_cell[371]	2.661174e-02	0.06784376	-0.064117017	0.1456189689	1.0881053
a_cell[372]	-6.584905e-03	0.06680850	-0.116351348	0.0930165703	1.1170423
a_cell[373]	-2.620406e-03	0.06727220	-0.110291550	0.1047757219	1.1144581
a_cell[374]	1.829322e-02	0.06796008	-0.081572040	0.1368834350	1.1166745
a_cell[375]	-2.273066e-03	0.06437794	-0.102489101	0.0992748120	1.1302581

a_cell[376]	6.319193e-04	0.06769531	-0.105951082	0.1120451799	1.1235826
a_cell[377]	-3.301090e-04	0.06561061	-0.107903295	0.1037659250	1.1267427
a_cell[378]	1.357299e-02	0.06879245	-0.084583806	0.1272125259	1.1123832
a_cell[379]	5.447874e-03	0.07117665	-0.102726608	0.1147415856	1.1197013
a_cell[380]	-5.675462e-04	0.06783439	-0.111254709	0.1090082761	1.1366986
a_cell[381]	9.226299e-03	0.06719297	-0.090671428	0.1236583755	1.1274235
a_cell[382]	3.257822e-02	0.07276823	-0.059796219	0.1614961150	1.0995322
a_cell[383]	1.356895e-02	0.06463789	-0.083303316	0.1230137050	1.1089543
a_cell[384]	5.653849e-03	0.06194676	-0.089270639	0.1075571559	1.1019094
a_cell[385]	-1.672448e-02	0.06652484	-0.132045955	0.0753401781	1.1059311
a_cell[386]	-1.015564e-02	0.06905148	-0.124999715	0.0940712726	1.1059870
a_cell[387]	-6.923708e-03	0.06483382	-0.110408677	0.0957402697	1.1239818
a_cell[388]	1.157315e-02	0.06574562	-0.083162039	0.1194335592	1.1236022
a_cell[389]	4.333962e-03	0.06465849	-0.092497034	0.1057487568	1.1066162
a_cell[390]	-1.768517e-02	0.06562833	-0.130614427	0.0732930337	1.1142791
a_cell[391]	1.005419e-02	0.06632547	-0.086944187	0.1162443661	1.1253440
a_cell[392]	-5.074337e-03	0.06401776	-0.110276262	0.0952506235	1.1100686
a_cell[393]	-9.068171e-03	0.06257595	-0.114271271	0.0828368358	1.1213577
a_cell[394]	9.194752e-03	0.06219616	-0.086579027	0.1126618379	1.1204746
a_cell[395]	-1.396347e-02	0.06632985	-0.125944672	0.0834299412	1.1190540
a_cell[396]	-1.063497e-02	0.07117955	-0.126736597	0.0910516459	1.1364099
a_cell[397]	-7.224470e-04	0.06376380	-0.103036963	0.0980602902	1.1178237
a_cell[398]	5.936396e-03	0.06851140	-0.104113882	0.1208225918	1.1339901
a_cell[399]	-4.135940e-03	0.06594059	-0.108662178	0.0962381499	1.1141732
a_cell[400]	-6.143317e-03	0.06734244	-0.111059270	0.0914152326	1.1314154
a_cell[401]	-7.855801e-03	0.06976858	-0.120121787	0.0987204234	1.1145279
a_cell[402]	1.624130e-03	0.06702385	-0.104391829	0.1063727118	1.1161727
a_cell[403]	-8.819273e-03	0.06675948	-0.119700264	0.0949173775	1.1176456
a_cell[404]	4.739377e-03	0.06529664	-0.098911538	0.1100375474	1.1080252
a_cell[405]	-1.135846e-03	0.06568527	-0.106754039	0.1041901757	1.1240368
a_cell[406]	-1.015056e-02	0.06776941	-0.122620770	0.0898367418	1.1223280
a_cell[407]	2.285205e-03	0.06416450	-0.094572448	0.1043416878	1.1149639
a_cell[408]	3.164563e-02	0.07005006	-0.061265585	0.1566271573	1.0946632
a_cell[409]	2.552445e-03	0.06622233	-0.102936047	0.1065026931	1.1235745
a_cell[410]	6.887078e-03	0.06426728	-0.088580385	0.1126426872	1.1048454
a_cell[411]	-3.243926e-03	0.06601375	-0.111449175	0.1003598158	1.1231206
a_cell[412]	-1.379133e-02	0.06543970	-0.124599478	0.0802744924	1.1175194
a_cell[413]	2.075637e-02	0.06518600	-0.069354155	0.1339094302	1.0967477
a_cell[414]	-5.646478e-03	0.06328997	-0.109105109	0.0890142396	1.1195574
a_cell[415]	3.130533e-03	0.06559776	-0.101377734	0.1110754315	1.1348224
a_cell[416]	1.267859e-02	0.06656189	-0.086796286	0.1277392400	1.1255850
a_cell[417]	1.581553e-02	0.06686114	-0.077784943	0.1310012611	1.1122265
a_cell[418]	-1.054842e-02	0.06509044	-0.116264432	0.0854826421	1.1198139

a_cell[419]	5.990114e-04	0.06846876	-0.107699084	0.1094401217	1.1119142
a_cell[420]	2.139276e-04	0.06612439	-0.100953173	0.1039358794	1.0987686
a_cell[421]	-6.577008e-03	0.06631513	-0.117893340	0.0947881829	1.1134956
a_cell[422]	-6.152592e-03	0.06761023	-0.118134500	0.0967715436	1.1416855
a_cell[423]	-5.038144e-03	0.06798511	-0.116335416	0.0979504776	1.1317824
a_cell[424]	4.098491e-03	0.06904338	-0.099923880	0.1137405831	1.1361596
a_cell[425]	1.338450e-02	0.06647414	-0.081903125	0.1274577302	1.1109865
a_cell[426]	7.143782e-02	0.09222957	-0.027013854	0.2411704928	1.1478197
a_cell[427]	-1.351207e-02	0.06627894	-0.127478912	0.0865680804	1.1075258
a_cell[428]	-8.782841e-04	0.06589456	-0.104695437	0.1014696595	1.1180451
a_cell[429]	2.408445e-04	0.06664466	-0.104767881	0.1078537118	1.1085992
a_cell[430]	-1.500355e-02	0.06875013	-0.134412420	0.0862882462	1.1180376
a_cell[431]	1.096232e-03	0.06559649	-0.102519914	0.1063116122	1.1013675
a_cell[432]	-8.990315e-03	0.06661693	-0.119904103	0.0947684214	1.1208085
a_cell[433]	-8.155450e-03	0.06811453	-0.117875139	0.0905645943	1.1433020
a_cell[434]	5.095777e-03	0.06536822	-0.096398308	0.1104722487	1.1118409
a_cell[435]	-5.612435e-03	0.06701208	-0.118906058	0.0992455613	1.1128123
a_cell[436]	3.596138e-03	0.06509656	-0.096153248	0.1093193160	1.1258079
a_cell[437]	-4.630097e-03	0.06568508	-0.108271493	0.0902395093	1.1357793
a_cell[438]	-1.551520e-02	0.07061597	-0.138292613	0.0865822935	1.1328954
a_cell[439]	-5.703541e-03	0.06846477	-0.113740977	0.0992482286	1.1211710
a_cell[440]	5.121841e-03	0.06667544	-0.098907708	0.1154796525	1.1254450
a_cell[441]	-6.054926e-03	0.06809301	-0.114248532	0.0969730363	1.1283239
a_cell[442]	3.049824e-03	0.06552620	-0.096825289	0.1100918867	1.1095744
a_cell[443]	1.176746e-02	0.06698614	-0.085512750	0.1270751009	1.1187947
a_cell[444]	2.138867e-02	0.06960547	-0.076654841	0.1444557604	1.1150260
a_cell[445]	-5.619810e-04	0.06881505	-0.109791959	0.1093949516	1.1103357
a_cell[446]	-9.723377e-04	0.06641249	-0.104502662	0.1039992920	1.1103856
a_cell[447]	-9.770247e-03	0.06936603	-0.125460910	0.0973498208	1.1328772
a_cell[448]	4.041814e-03	0.06779260	-0.102364493	0.1109086142	1.1259303
a_cell[449]	1.329360e-02	0.06711285	-0.081591475	0.1279004495	1.1031108
a_cell[450]	-1.364185e-02	0.06691656	-0.125908019	0.0834847938	1.1285305
a_cell[451]	1.303475e-02	0.06838390	-0.083933399	0.1295157776	1.1161701
a_cell[452]	-4.642429e-03	0.06880067	-0.111285196	0.0992222450	1.1085109
a_cell[453]	-5.342326e-04	0.06650997	-0.109164842	0.1090793561	1.1327839
a_cell[454]	-1.100078e-03	0.06605135	-0.105514909	0.1044052729	1.1366515
b_tide	3.539911e-02	0.02952091	-0.011997336	0.0827005592	1.0001506
b_moon	-4.252085e-03	0.02885194	-0.050111561	0.0412940497	1.0000301
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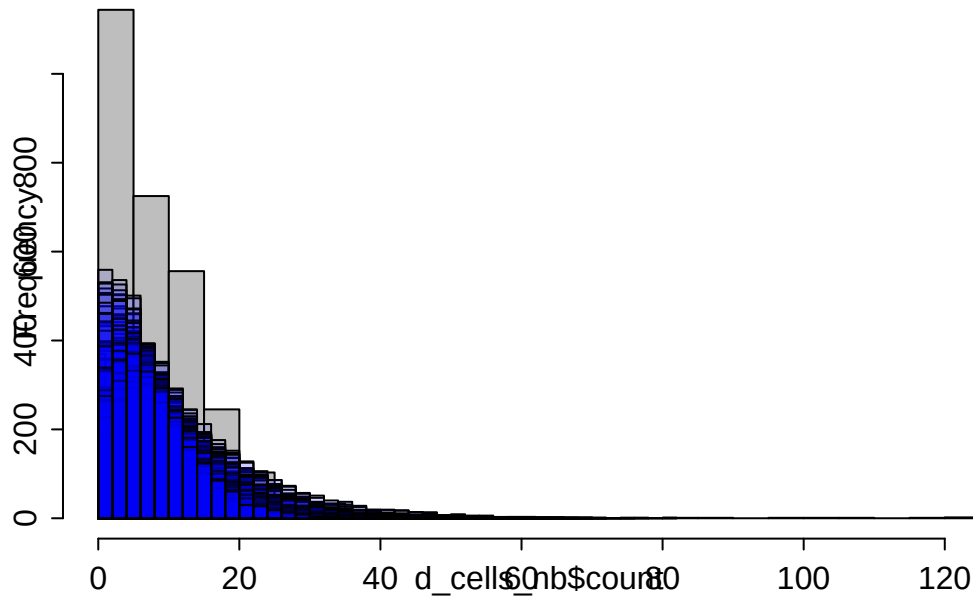
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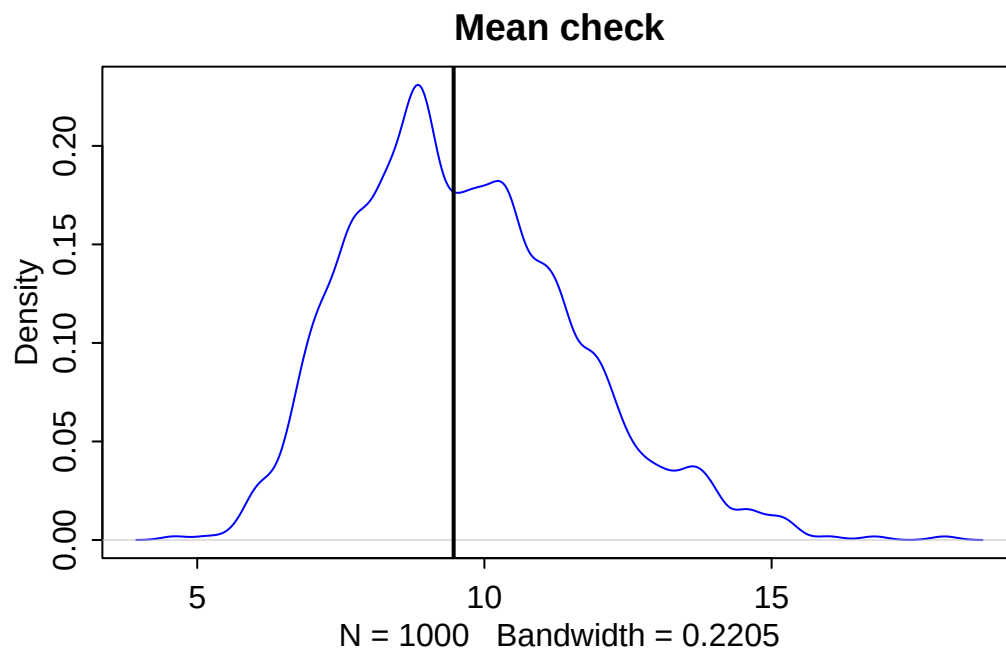
```
## posterior predictive check
x_sim <- sim(m_space_use, n = 1000)
```

```
hist(d_cells_nb$count, breaks = 30, col = "grey",
     main = "Observed vs simulated counts")
for(i in 1:50) {
  hist(x_sim[i,], breaks = 30,
       col = col.alpha("blue", 0.1), add = TRUE)
}
```

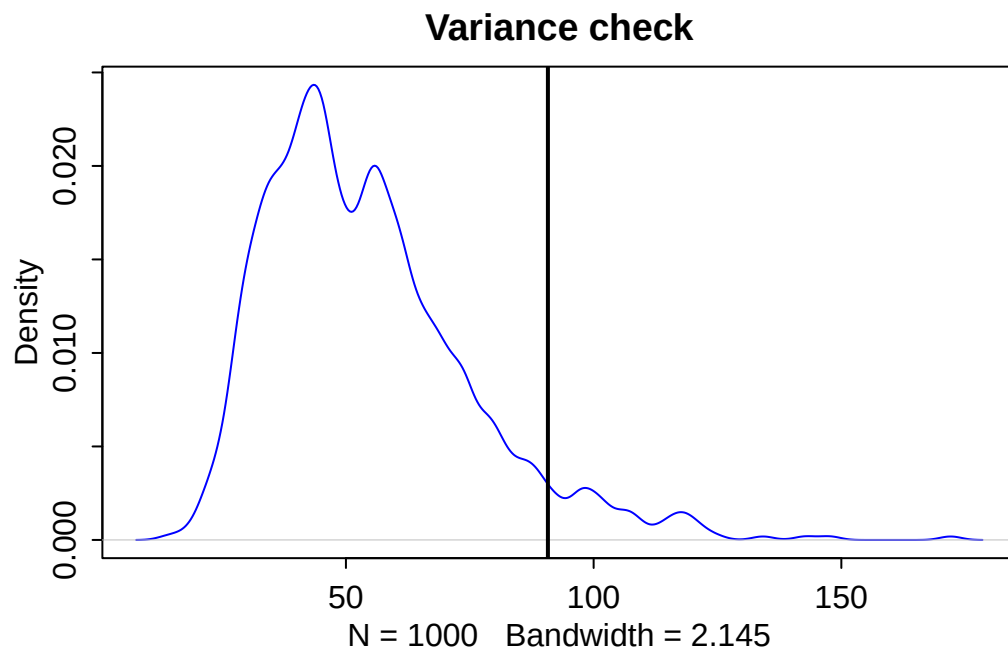
Observed vs simulated counts



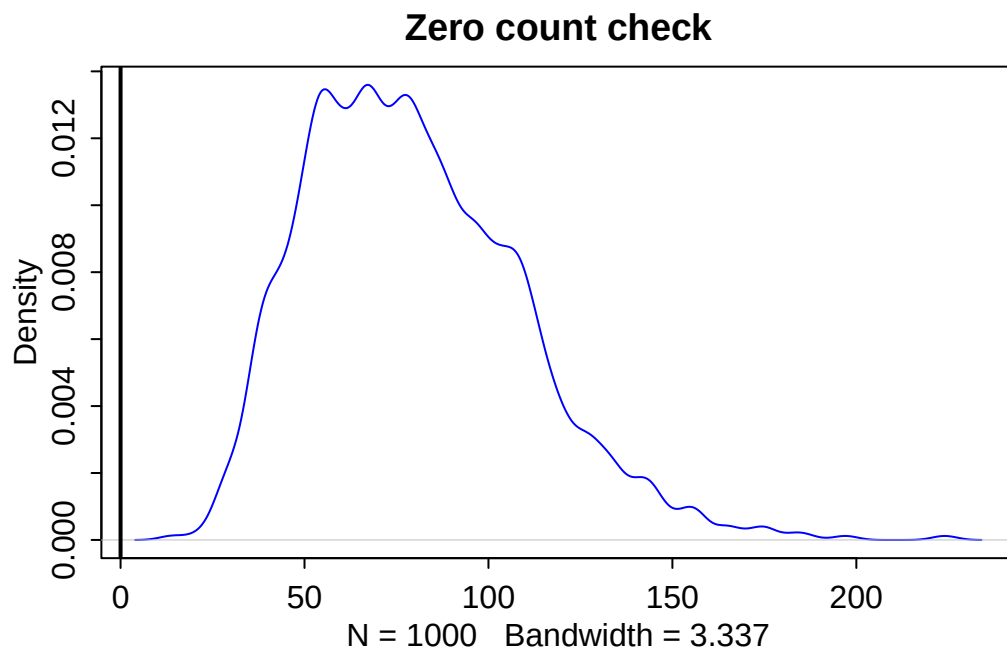
```
mean_sim <- apply(x_sim, 1, mean)
dens(mean_sim, col = "blue")
abline(v = mean(d_cells_nb$count), lwd = 2)
title("Mean check")
```



```
var_sim <- apply(x_sim, 1, var)
dens(var_sim, col = "blue")
abline(v = var(d_cells_nb$count), lwd = 2)
title("Variance check")
```

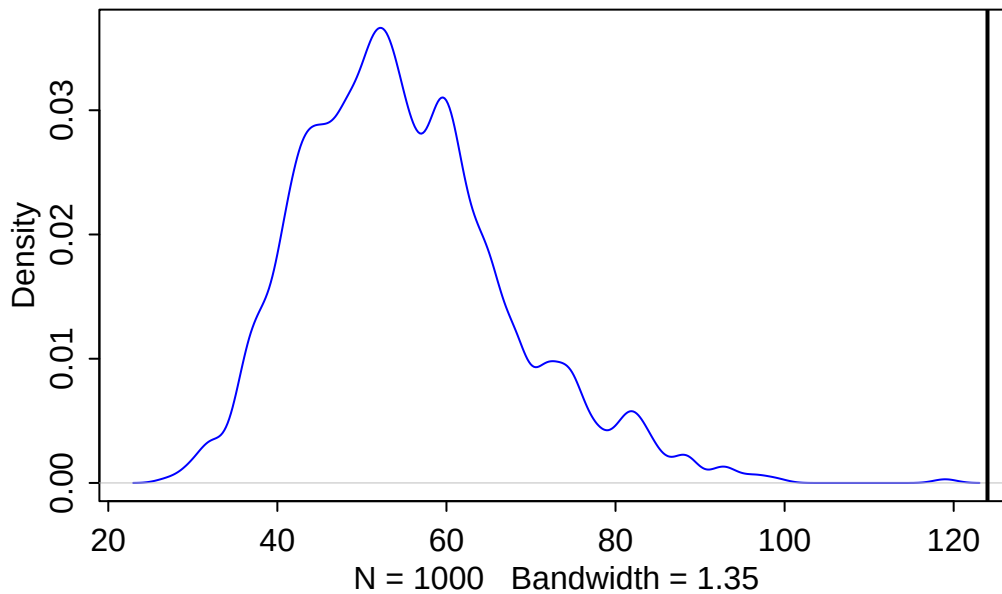


```
zero_sim <- apply(x_sim, 1, function(x) sum(x == 0))  
dens(zero_sim, col = "blue")  
abline(v = sum(d_cells_nb$count == 0), lwd = 2)  
title("Zero count check")
```



```
max_sim <- apply(x_sim, 1, max)
dens(max_sim, col = "blue")
abline(v = max(d_cells_nb$count), lwd = 2)
title("Maximum count check")
```

Maximum count check



```
post <- extract.samples(m_space_use)

cell_hotspots <- gps_cells %>%
  distinct(cell_id, x_cell, y_cell) %>%
  arrange(cell_id) %>%
  mutate(
    mean_cell_effect = apply(post$a_cell, 2, mean),
    lpi_cell_effect = apply(post$a_cell, 2, PI, prob = 0.89)[1, ],
    upi_cell_effect = apply(post$a_cell, 2, PI, prob = 0.89)[2, ]
  ) %>%
  arrange(desc(mean_cell_effect))

head(cell_hotspots, 20)
```

A tibble: 20 x 6

	cell_id	x_cell	y_cell	mean_cell_effect	lpi_cell_effect	upi_cell_effect
	<int>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>
1	426	22566	382894	0.0714	-0.0270	0.241
2	211	22557	382903	0.0537	-0.0354	0.201
3	204	22557	382896	0.0512	-0.0391	0.196
4	169	22556	382886	0.0482	-0.0357	0.189
5	99	22553	382888	0.0381	-0.0473	0.170

6	159	22555	382899	0.0368	-0.0469	0.157
7	119	22554	382884	0.0348	-0.0534	0.157
8	104	22553	382893	0.0338	-0.0472	0.148
9	382	22564	382892	0.0326	-0.0598	0.161
10	15	22548	382903	0.0317	-0.0602	0.156
11	408	22565	382895	0.0316	-0.0613	0.157
12	195	22557	382887	0.0307	-0.0582	0.151
13	316	22561	382905	0.0293	-0.0603	0.154
14	334	22562	382894	0.0278	-0.0627	0.154
15	241	22559	382883	0.0276	-0.0702	0.157
16	371	22563	382905	0.0266	-0.0641	0.146
17	197	22557	382889	0.0266	-0.0704	0.148
18	205	22557	382897	0.0259	-0.0629	0.145
19	252	22559	382894	0.0242	-0.0692	0.145
20	315	22561	382904	0.0239	-0.0631	0.142

```
### Extreme hotspot
```

```
# Extract posterior samples
```

```
post <- extract.samples(m_space_use)
```

```
# compute mean cell effect
```

```
cell_effect <- apply(post$a_cell, 2, mean)
```

```
cell_low <- apply(post$a_cell, 2, PI, prob = 0.89)[1, ]
```

```
cell_high <- apply(post$a_cell, 2, PI, prob = 0.89)[2, ]
```

```
# attach back to your spatial data
```

```
cell_map <- gps_cells %>%
```

```
  distinct(cell_id, x_cell, y_cell) %>%
```

```
  arrange(cell_id) %>%
```

```
  mutate(
```

```
    effect = cell_effect,
```

```
    low = cell_low,
```

```
    high = cell_high
```

```
)
```

```
cell_map <- cell_map %>%
```

```
  mutate(
```

```
    intensity = exp(effect)
```

```
)
```

```
library(ggplot2)
```

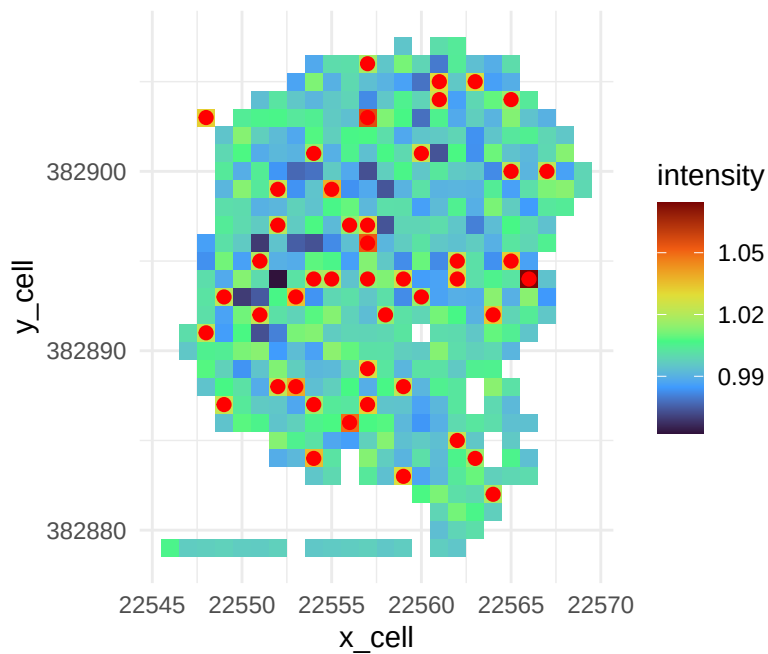
```

threshold <- quantile(cell_map$intensity, 0.9)

cell_map <- cell_map %>%
  mutate(
    hotspot = intensity > threshold
  )

ggplot(cell_map, aes(x = x_cell, y = y_cell)) +
  geom_tile(aes(fill = intensity)) +
  geom_point(data = subset(cell_map, hotspot),
            color = "red", size = 2) +
  scale_fill_viridis_c(option = "turbo") +
  coord_equal() +
  theme_minimal()

```



```

### Consecutive points in the same cell

# create bouts
library(dplyr)

cell_size <- 50

```

```

gps_bouts <- gps_env %>%
  arrange(file_name, datetime_local) %>% # IMPORTANT: time order
  mutate(
    x_cell = floor(x_true_m / cell_size),
    y_cell = floor(y_true_m / cell_size),
    cell_id = paste(x_cell, y_cell, sep = "_")
  ) %>%
  group_by(file_name) %>%
  mutate(
    new_bout = cell_id != lag(cell_id, default = first(cell_id)),
    bout_id = cumsum(new_bout)
  ) %>%
  ungroup()

#calculate bout duration
bouts_summary <- gps_bouts %>%
  group_by(file_name, cell_id, bout_id) %>%
  summarise(
    n_points = n(),
    duration_sec = n_points * 2,
    .groups = "drop"
  )

# summarise per track-cell
cell_behavior <- bouts_summary %>%
  group_by(file_name, cell_id) %>%
  summarise(
    n_visits = n(), # number of bouts
    total_time = sum(duration_sec), # total time in cell
    mean_duration = mean(duration_sec), # average stay
    max_duration = max(duration_sec), # longest stay
    .groups = "drop"
  )

View(cell_behavior)

```

Map

```

library(tidyr)

# Attach spatial coordinates
cell_behavior <- cell_behavior %>%

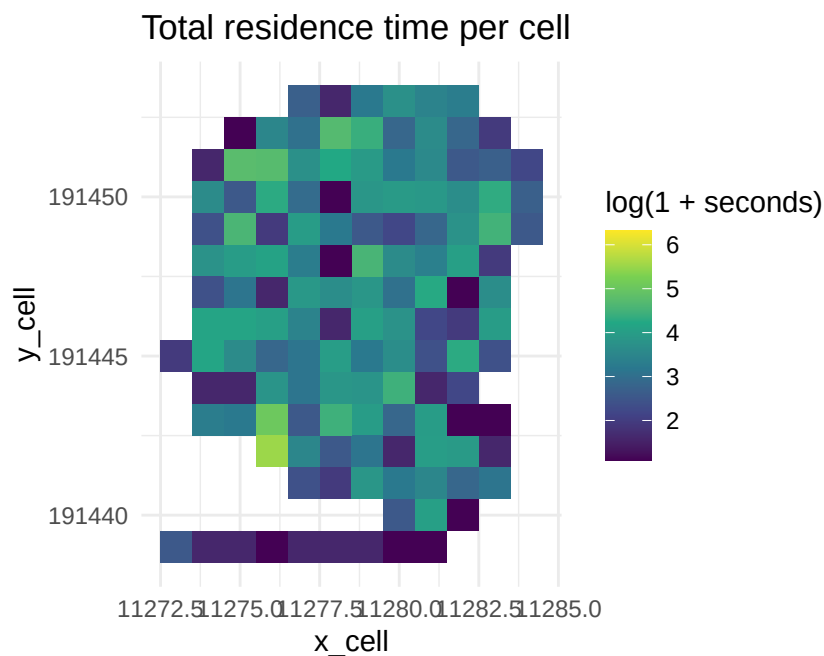
```

```

separate(cell_id, into = c("x_cell", "y_cell"), sep = "_") %>%
mutate(
  x_cell = as.numeric(x_cell),
  y_cell = as.numeric(y_cell)
)

## Total time (log scale)
ggplot(cell_behavior, aes(x = x_cell, y = y_cell, fill = log1p(total_time))) +
  geom_tile() +
  scale_fill_viridis_c() +
  coord_equal() +
  labs(
    title = "Total residence time per cell",
    fill = "log(1 + seconds)"
  ) +
  theme_minimal()

```



```

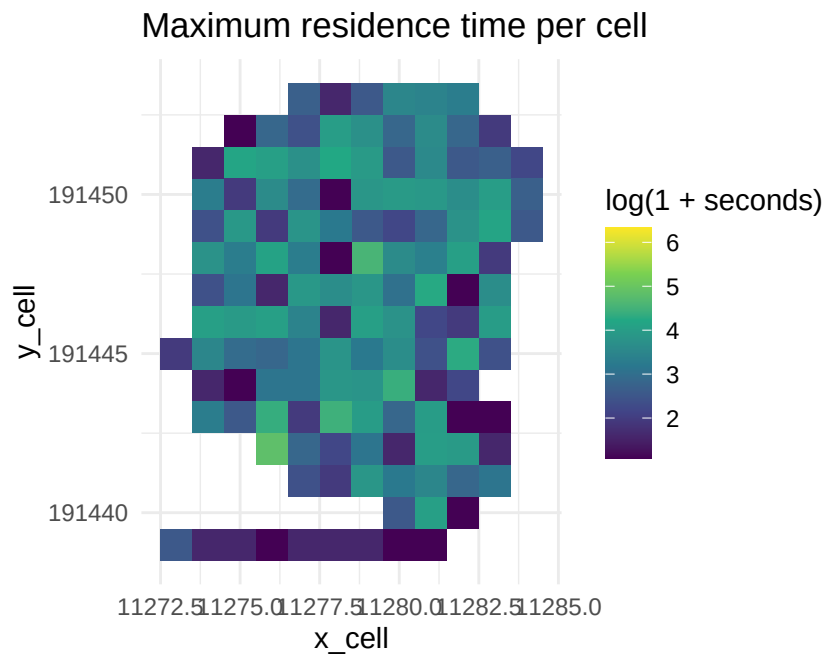
## Maximum duration (log scale)
ggplot(cell_behavior, aes(x = x_cell, y = y_cell, fill = log1p(max_duration))) +
  geom_tile() +
  scale_fill_viridis_c() +
  coord_equal() +

```

```

labs(
  title = "Maximum residence time per cell",
  fill = "log(1 + seconds)"
) +
theme_minimal()

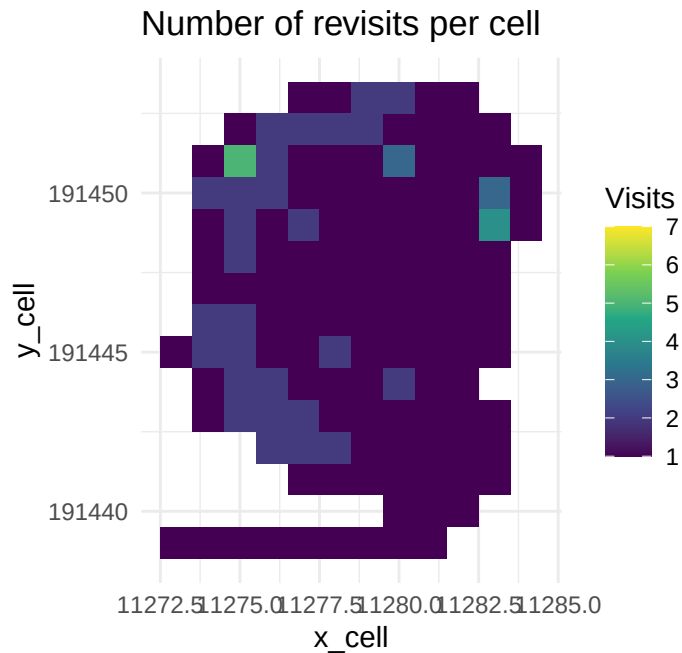
```



```

## number of visits
ggplot(cell_behavior, aes(x = x_cell, y = y_cell, fill = n_visits)) +
  geom_tile() +
  scale_fill_viridis_c() +
  coord_equal() +
  labs(
    title = "Number of revisits per cell",
    fill = "Visits"
  ) +
  theme_minimal()

```



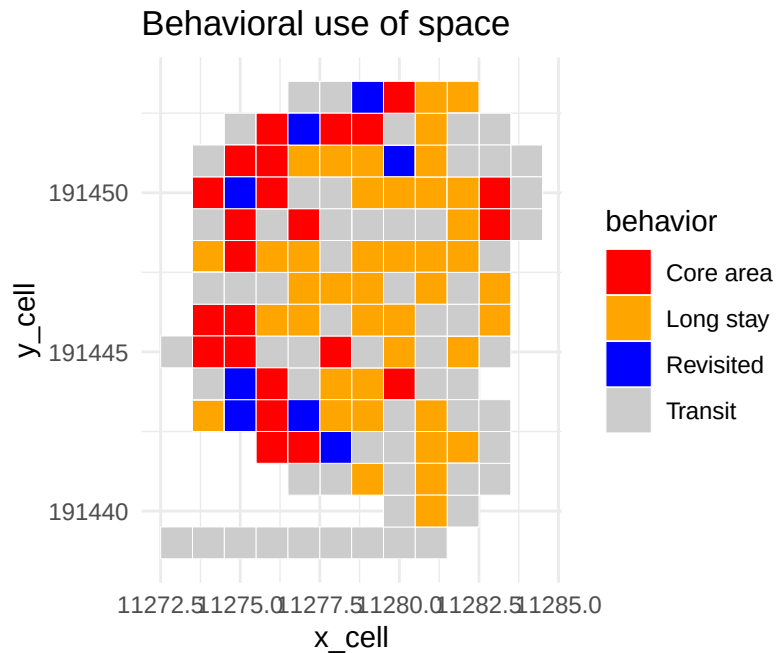
```
### Classify behavior per cell
```

```
cell_behavior <- cell_behavior %>%
  mutate(
    high_time = total_time > median(total_time),
    high_visit = n_visits > median(n_visits),

    behavior = case_when(
      high_time & high_visit ~ "Core area",
      high_time & !high_visit ~ "Long stay",
      !high_time & high_visit ~ "Revisited",
      TRUE ~ "Transit"
    )
  )

ggplot(cell_behavior, aes(x = x_cell, y = y_cell, fill = behavior)) +
  geom_tile(color = "white", size = 0.1) +
  scale_fill_manual(values = c(
    "Core area" = "red",
    "Long stay" = "orange",
    "Revisited" = "blue",
    "Transit" = "grey80"
  )) +
```

```
coord_equal() +
theme_minimal() +
labs(title = "Behavioral use of space")
```



```
## Aggregate Bayesian intensity map into 50m cells
cell_map_agg <- cell_map %>%
  mutate(
    x_cell_agg = floor(x_cell / 2),
    y_cell_agg = floor(y_cell / 2)
  ) %>%
  group_by(x_cell_agg, y_cell_agg) %>%
  summarise(
    intensity = mean(intensity, na.rm = TRUE),
    effect = mean(effect, na.rm = TRUE),
    .groups = "drop"
  ) %>%
  rename(
    x_cell = x_cell_agg,
    y_cell = y_cell_agg
  )
```

```

### compare datasets and test agreement
compare_map <- cell_behavior %>%
  left_join(cell_map_agg, by = c("x_cell", "y_cell"))

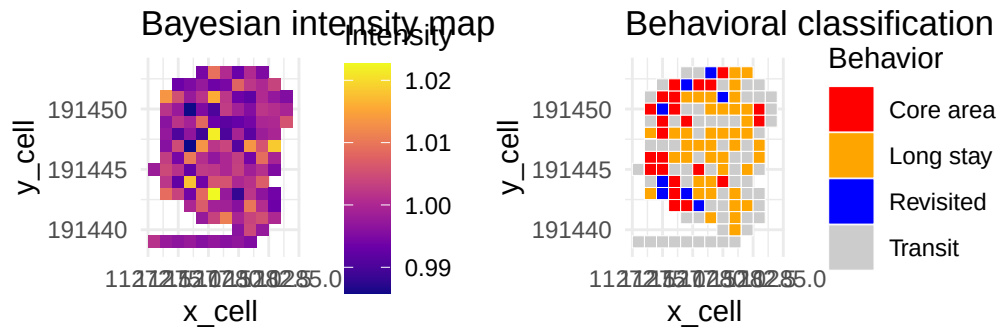
## side by side comparison
library(ggplot2)
library(patchwork)

p1 <- ggplot(cell_map_agg, aes(x = x_cell, y = y_cell, fill = intensity)) +
  geom_tile() +
  scale_fill_viridis_c(option = "plasma") +
  coord_equal() +
  theme_minimal() +
  labs(
    title = "Bayesian intensity map",
    fill = "Intensity"
  )

p2 <- ggplot(cell_behavior, aes(x = x_cell, y = y_cell, fill = behavior)) +
  geom_tile(color = "white", linewidth = 0.1) +
  scale_fill_manual(values = c(
    "Core area" = "red",
    "Long stay" = "orange",
    "Revisited" = "blue",
    "Transit" = "grey80"
  )) +
  coord_equal() +
  theme_minimal() +
  labs(
    title = "Behavioral classification",
    fill = "Behavior"
  )

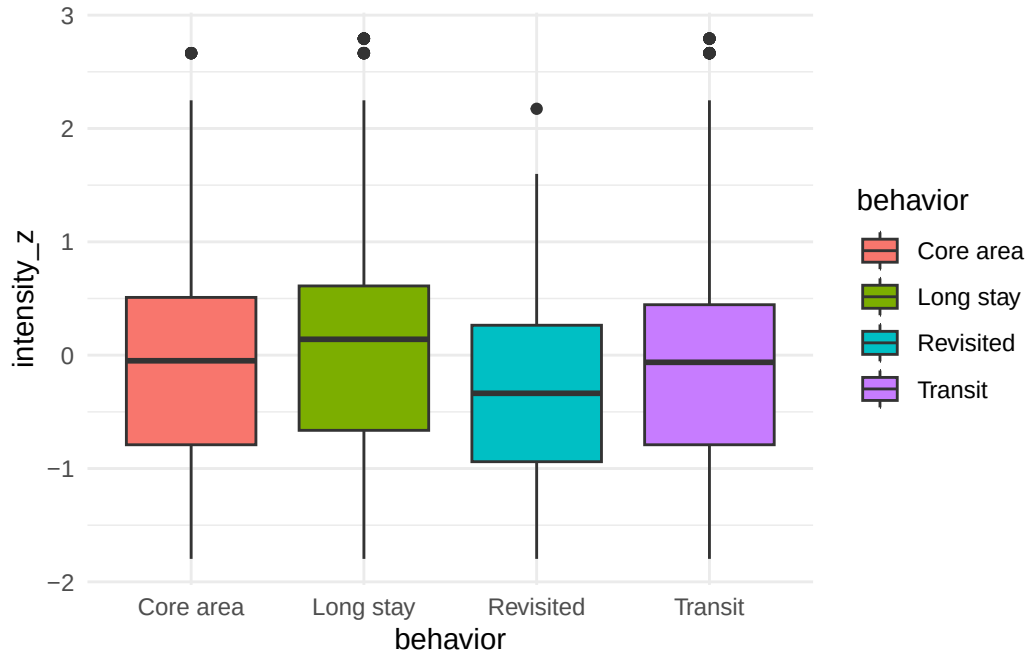
p1 + p2

```



```
## comparison map
compare_map <- compare_map %>%
  mutate(
    intensity_z = scale(intensity)
  )

ggplot(compare_map, aes(x = behavior, y = intensity_z, fill = behavior)) +
  geom_boxplot() +
  theme_minimal()
```



```
### Bout duration Gaussian model

# First extract one row per bout with environmental data
bout_covariates <- gps_bouts %>%
  group_by(file_name, bout_id) %>%
  summarise(
    tide_height_m = first(tide_height_m),
    moon_illumination = first(moon_illumination),
    month = first(month),
    year = first(year),
    .groups = "drop"
  )

# Join with bout durations
bout_data <- bouts_summary %>%
  left_join(bout_covariates, by = c("file_name", "bout_id"))

# prepare variables
bout_data <- bout_data %>%
  filter(duration_sec > 0) %>% # just in case
  mutate(
    log_duration = log(duration_sec),
```

```

# IDs for hierarchical effects
month_id = as.integer(factor(month)),
year_id = as.integer(factor(year)),
track_id = as.integer(factor(file_name)),

# standardize predictors
tide_s = as.numeric(scale(tide_height_m)),
moon_s = as.numeric(scale(moon_illumination))
)

# build data luist for ulam
d_bout <- list(
  N = nrow(bout_data),
  Y = bout_data$log_duration,
  tide = bout_data$tide_s,
  moon = bout_data$moon_s,
  month = bout_data$month_id,
  year = bout_data$year_id,
  track = bout_data$track_id,

  n_month = length(unique(bout_data$month_id)),
  n_year = length(unique(bout_data$year_id)),
  n_track = length(unique(bout_data$track_id))
)

# Full bout model
library(rethinking)

m_bout <- ulam(
  alist(
    Y ~ dnorm(mu, sigma),

    mu <- a +
      a_month[month] +
      a_year[year] +
      a_track[track] +
      b_tide * tide +
      b_moon * moon,

    # priors
    a ~ dnorm(0, 1),

```

```

a_month[month] ~ dnorm(0, sigma_month),
a_year[year] ~ dnorm(0, sigma_year),
a_track[track] ~ dnorm(0, sigma_track),

b_tide ~ dnorm(0, 1),
b_moon ~ dnorm(0, 1),

sigma ~ dexp(1),
sigma_month ~ dexp(1),
sigma_year ~ dexp(1),
sigma_track ~ dexp(1)
),
data = d_bout,
chains = 4,
cores = 4,
iter = 2000
)

```

Running MCMC with 4 parallel chains, with 1 thread(s) per chain...

Chain 1 Iteration: 1 / 2000 [0%] (Warmup)

Chain 2 Iteration: 1 / 2000 [0%] (Warmup)

Chain 3 Iteration: 1 / 2000 [0%] (Warmup)

Chain 4 Iteration: 1 / 2000 [0%] (Warmup)

Chain 1 Iteration: 100 / 2000 [5%] (Warmup)

Chain 2 Iteration: 100 / 2000 [5%] (Warmup)

Chain 4 Iteration: 100 / 2000 [5%] (Warmup)

Chain 1 Iteration: 200 / 2000 [10%] (Warmup)

Chain 3 Iteration: 100 / 2000 [5%] (Warmup)

Chain 4 Iteration: 200 / 2000 [10%] (Warmup)

Chain 1 Iteration: 300 / 2000 [15%] (Warmup)

Chain 3 Iteration: 200 / 2000 [10%] (Warmup)

Chain 4 Iteration: 300 / 2000 [15%] (Warmup)

Chain 1 Iteration: 400 / 2000 [20%] (Warmup)

Chain 4 Iteration: 400 / 2000 [20%] (Warmup)

Chain 3 Iteration: 300 / 2000 [15%] (Warmup)

Chain 2 Iteration: 200 / 2000 [10%] (Warmup)

Chain 1 Iteration: 500 / 2000 [25%] (Warmup)

Chain 3 Iteration: 400 / 2000 [20%] (Warmup)
 Chain 4 Iteration: 500 / 2000 [25%] (Warmup)
 Chain 1 Iteration: 600 / 2000 [30%] (Warmup)
 Chain 2 Iteration: 300 / 2000 [15%] (Warmup)
 Chain 3 Iteration: 500 / 2000 [25%] (Warmup)
 Chain 4 Iteration: 600 / 2000 [30%] (Warmup)
 Chain 2 Iteration: 400 / 2000 [20%] (Warmup)
 Chain 4 Iteration: 700 / 2000 [35%] (Warmup)
 Chain 1 Iteration: 700 / 2000 [35%] (Warmup)
 Chain 3 Iteration: 600 / 2000 [30%] (Warmup)
 Chain 2 Iteration: 500 / 2000 [25%] (Warmup)
 Chain 4 Iteration: 800 / 2000 [40%] (Warmup)
 Chain 2 Iteration: 600 / 2000 [30%] (Warmup)
 Chain 1 Iteration: 800 / 2000 [40%] (Warmup)
 Chain 4 Iteration: 900 / 2000 [45%] (Warmup)
 Chain 2 Iteration: 700 / 2000 [35%] (Warmup)
 Chain 3 Iteration: 700 / 2000 [35%] (Warmup)
 Chain 2 Iteration: 800 / 2000 [40%] (Warmup)
 Chain 1 Iteration: 900 / 2000 [45%] (Warmup)
 Chain 4 Iteration: 1000 / 2000 [50%] (Warmup)
 Chain 4 Iteration: 1001 / 2000 [50%] (Sampling)
 Chain 2 Iteration: 900 / 2000 [45%] (Warmup)
 Chain 3 Iteration: 800 / 2000 [40%] (Warmup)
 Chain 4 Iteration: 1100 / 2000 [55%] (Sampling)
 Chain 1 Iteration: 1000 / 2000 [50%] (Warmup)
 Chain 1 Iteration: 1001 / 2000 [50%] (Sampling)
 Chain 4 Iteration: 1200 / 2000 [60%] (Sampling)
 Chain 2 Iteration: 1000 / 2000 [50%] (Warmup)
 Chain 2 Iteration: 1001 / 2000 [50%] (Sampling)
 Chain 3 Iteration: 900 / 2000 [45%] (Warmup)
 Chain 1 Iteration: 1100 / 2000 [55%] (Sampling)
 Chain 4 Iteration: 1300 / 2000 [65%] (Sampling)
 Chain 2 Iteration: 1100 / 2000 [55%] (Sampling)
 Chain 1 Iteration: 1200 / 2000 [60%] (Sampling)
 Chain 4 Iteration: 1400 / 2000 [70%] (Sampling)
 Chain 3 Iteration: 1000 / 2000 [50%] (Warmup)
 Chain 2 Iteration: 1200 / 2000 [60%] (Sampling)
 Chain 3 Iteration: 1001 / 2000 [50%] (Sampling)
 Chain 1 Iteration: 1300 / 2000 [65%] (Sampling)
 Chain 4 Iteration: 1500 / 2000 [75%] (Sampling)
 Chain 2 Iteration: 1300 / 2000 [65%] (Sampling)
 Chain 3 Iteration: 1100 / 2000 [55%] (Sampling)
 Chain 1 Iteration: 1400 / 2000 [70%] (Sampling)

```

Chain 4 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 2 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 3 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 1 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 4 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 2 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 3 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 1 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 4 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 2 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 1 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 3 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 4 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 2 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 1 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 3 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 4 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 4 finished in 37.6 seconds.
Chain 2 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 1 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 2 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 3 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 1 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 1 finished in 39.8 seconds.
Chain 2 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 2 finished in 40.0 seconds.
Chain 3 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 3 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 3 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 3 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 3 finished in 42.5 seconds.

```

All 4 chains finished successfully.
 Mean chain execution time: 40.0 seconds.
 Total execution time: 42.6 seconds.

```

# predicted effect of tide
tide_seq_m <- seq(min(bout_data$tide_height_m),
                  max(bout_data$tide_height_m),
                  length.out = 100)

tide_seq_s <- (tide_seq_m - mean(bout_data$tide_height_m)) /

```

```

        sd(bout_data$tide_height_m)

pred_data <- list(
  tide = tide_seq_s,
  moon = rep(0, length(tide_seq_s)),
  month = rep(1, length(tide_seq_s)),
  year = rep(1, length(tide_seq_s)),
  track = rep(1, length(tide_seq_s))
)

mu <- link(m_bout, data = pred_data)

mu_mean <- apply(mu, 2, function(x) mean(x, na.rm = TRUE))

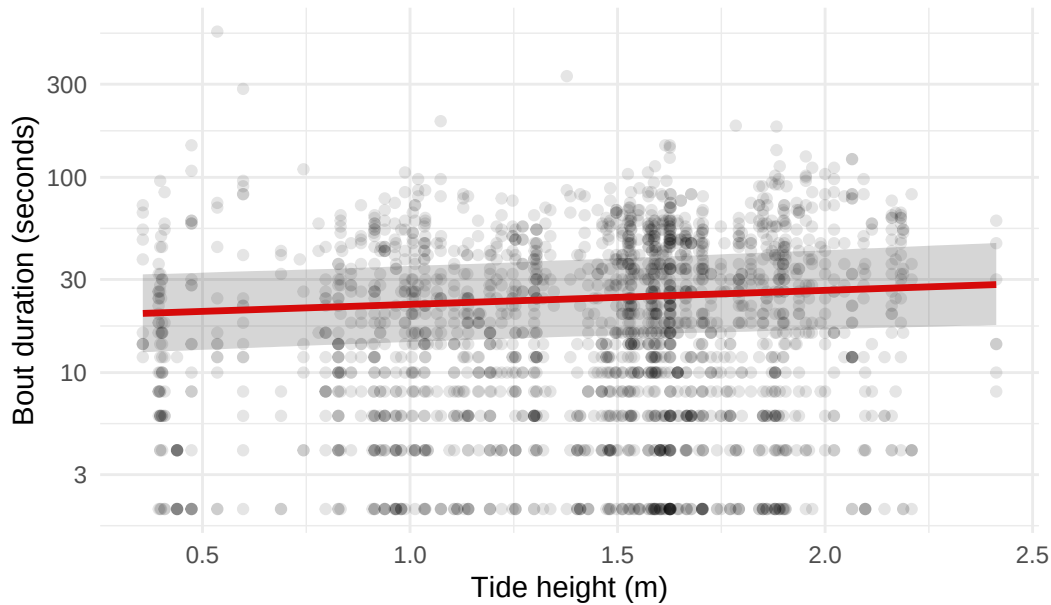
mu_PI <- apply(mu, 2, function(x)
  quantile(x, probs = c(0.055, 0.945), na.rm = TRUE)
)

pred_df <- data.frame(
  tide_m = tide_seq_m,
  duration = exp(mu_mean),
  low = exp(mu_PI[1, ]),
  high = exp(mu_PI[2, ])
)

ggplot(bout_data, aes(x = tide_height_m, y = duration_sec)) +
  geom_point(alpha = 0.1) +
  geom_line(data = pred_df,
    aes(x = tide_m, y = duration),
    color = "red", size = 1.2) +
  geom_ribbon(data = pred_df,
    aes(x = tide_m, ymin = low, ymax = high),
    alpha = 0.2,
    inherit.aes = FALSE) +
  scale_y_log10() +
  labs(
    x = "Tide height (m)",
    y = "Bout duration (seconds)",
    title = "Effect of tide on shark residence time"
  ) +
  theme_minimal()

```

Effect of tide on shark residence time



```
precis(m_bout, depth = 2)
```

	mean	sd	5.5%	94.5%	rhat
a	2.7973404681	0.19873585	2.460981935	3.061101085	1.0023729
a_month[1]	-0.0420122147	0.05603908	-0.145961398	0.026036246	1.0016554
a_month[2]	0.0136605888	0.05381218	-0.068083593	0.102580995	1.0055086
a_month[3]	0.0026824653	0.06425496	-0.098975444	0.106093191	1.0076780
a_month[4]	0.0146720001	0.06720902	-0.079489563	0.131362273	1.0059148
a_month[5]	-0.0175102212	0.06922322	-0.139103713	0.073716101	1.0054115
a_month[6]	-0.0344089175	0.07586676	-0.169132467	0.053854078	1.0018573
a_month[7]	0.0249356564	0.06933211	-0.063272894	0.144741836	1.0082289
a_month[8]	0.0253256255	0.06971603	-0.064327537	0.149828078	1.0019419
a_month[9]	0.0095914885	0.07481999	-0.094973174	0.128151011	1.0097326
a_month[10]	0.0054940012	0.05189508	-0.074821600	0.089085979	1.0049487
a_month[11]	-0.0018019697	0.05808892	-0.097082695	0.086335948	1.0103172
a_year[1]	0.3447453799	0.24481847	0.013408352	0.781868725	1.0025634
a_year[2]	0.0105818367	0.19971928	-0.254104964	0.343015666	1.0021416
a_year[3]	0.0867594047	0.20137140	-0.179915101	0.438432473	1.0015427
a_year[4]	-0.2705610665	0.19911205	-0.539956209	0.060496149	1.0015578
a_year[5]	0.2668523932	0.20953005	-0.004062827	0.620827535	1.0020852
a_track[1]	0.0771035618	0.24103581	-0.309157396	0.458591486	0.9998387
a_track[2]	0.0505887694	0.24896356	-0.356054806	0.446207739	1.0002484

a_track[3]	0.1570650323	0.24408794	-0.227885175	0.549147036	1.0007047
a_track[4]	0.1282839628	0.24521024	-0.267806221	0.513210121	1.0032709
a_track[5]	0.1710958333	0.26778476	-0.249021714	0.597453779	1.0008352
a_track[6]	-0.3984764435	0.26761472	-0.821828788	0.020540419	1.0002823
a_track[7]	-0.0294659788	0.23545700	-0.403628242	0.342535781	1.0004741
a_track[8]	0.0065311518	0.22508257	-0.347830000	0.369835044	1.0008885
a_track[9]	0.0210336094	0.23309039	-0.354357982	0.395878435	1.0006910
a_track[10]	-0.3000281065	0.23597436	-0.685677997	0.076712412	1.0032751
a_track[11]	0.1839649358	0.24318873	-0.186487811	0.577219985	1.0008721
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a_track[16]	0.1163637261	0.24974986	-0.275102495	0.514214978	1.0003776
a_track[17]	-0.1305199369	0.23856314	-0.504151493	0.244212051	0.9996986
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a_track[19]	-0.1097434764	0.22705900	-0.471989888	0.258176653	1.0013157
a_track[20]	-0.2499971370	0.20800166	-0.588190927	0.075069318	1.0012768
a_track[21]	-0.0077251858	0.23146626	-0.379128195	0.358011160	1.0006114
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a_track[25]	-0.2607264050	0.25037110	-0.661972590	0.126942202	1.0014801
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a_track[27]	-0.0314509430	0.27269374	-0.458158239	0.396408386	1.0011979
a_track[28]	-0.1266699614	0.27147033	-0.561528958	0.300472619	1.0015518
a_track[29]	0.1125096113	0.27388728	-0.319632699	0.549554268	1.0046216
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a_track[37]	-0.1255489320	0.21670693	-0.465597921	0.217900148	1.0005674
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a_track[217]	-0.0308719303	0.26523976	-0.451730025	0.389117001	1.0016606

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a_track[401]	0.0654311317	0.28512293	-0.380091135	0.528661453	1.0015752
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a_track[406]	-0.0224241563	0.28882733	-0.493302089	0.441928886	1.0011236
a_track[407]	0.0873467434	0.27279792	-0.327944864	0.535037421	0.9997222
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a_track[544] 6834.8494
a_track[545] 6526.8986
a_track[546] 4358.0092
a_track[547] 6846.4501

a_track[548] 7410.3764
a_track[549] 5732.6965
a_track[550] 6454.0404
a_track[551] 6600.0320
a_track[552] 7419.7806
a_track[553] 7488.5019
a_track[554] 6884.4747
a_track[555] 6579.3169
a_track[556] 6858.1270
a_track[557] 6539.7246
a_track[558] 7221.1562
a_track[559] 6334.8688
a_track[560] 6236.3677
a_track[561] 6680.1469
a_track[562] 7132.3451
a_track[563] 7173.1399
a_track[564] 5062.0090
a_track[565] 6784.0494
a_track[566] 6961.5486
a_track[567] 6446.6434
a_track[568] 8549.7828
a_track[569] 7309.5759
a_track[570] 6164.8459
a_track[571] 7008.7063
a_track[572] 5762.5100
a_track[573] 2984.8731
a_track[574] 6608.1176
a_track[575] 8053.0839
a_track[576] 5748.5423
a_track[577] 5907.0539
a_track[578] 7337.7281
a_track[579] 5309.3740
a_track[580] 7964.2436
a_track[581] 5653.3292
a_track[582] 7195.4051
a_track[583] 5925.1731
a_track[584] 8090.8415
a_track[585] 7949.9715
a_track[586] 6518.8643
a_track[587] 6181.5086
a_track[588] 5468.5747
a_track[589] 6342.9521
a_track[590] 6482.5279

```

a_track[591] 5121.6090
a_track[592] 7183.5066
b_tide      3711.2321
b_moon      3632.4439
sigma       2614.2167
sigma_month  291.3128
sigma_year   1750.7355
sigma_track  262.9728

```

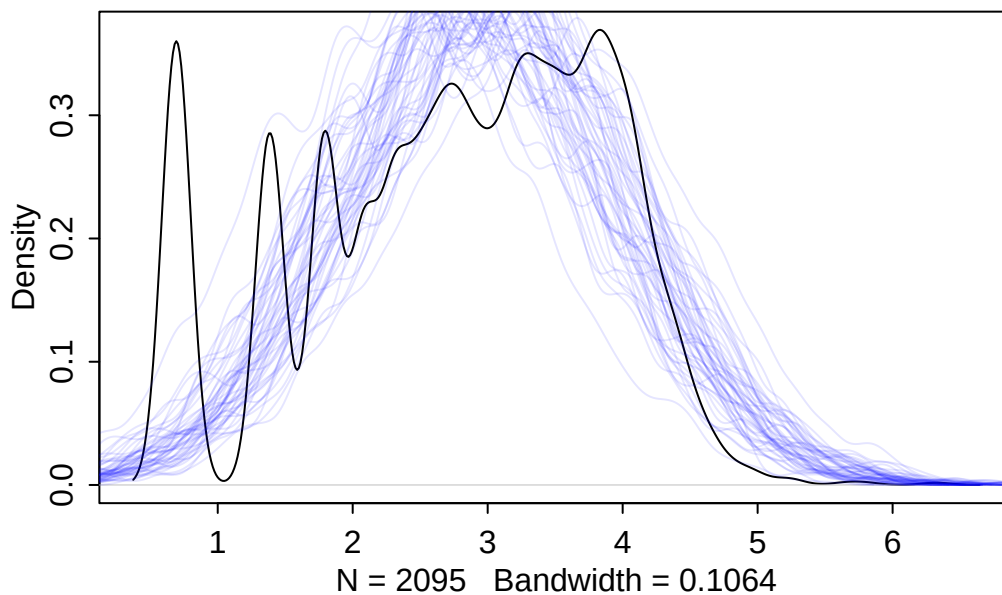
```
precis(m_bout, pars = c("b_tide", "b_moon"))
```

	mean	sd	5.5%	94.5%	rhat	ess_bulk
b_tide	0.07198959	0.03181048	0.02095148	0.12236898	1.001621	3711.232
b_moon	-0.02976223	0.02856272	-0.07480081	0.01679662	1.000273	3632.444

```
y_sim <- sim(m_bout, n = 1000)
```

```
dens(d_bout$Y)
```

```
for(i in 1:50) dens(y_sim[i,], add = TRUE, col = col.alpha("blue", 0.1))
```



```
post <- extract.samples(m_bout)

mean(exp(post$b_tide))
```

```
[1] 1.075188
```

```
mean(exp(post$b_moon))
```

```
[1] 0.9710724
```

```
## Probability of long-stay vs transit

# response variable (bout level)
threshold <- quantile(bout_data$duration_sec, 0.75, na.rm = TRUE)

bout_data <- bout_data %>%
  mutate(
    long_stay = ifelse(duration_sec >= threshold, 1, 0)
  )

# build data
d_longstay <- list(
  N = nrow(bout_data),
  Y = bout_data$long_stay,

  tide = bout_data$tide_s,
  moon = bout_data$moon_s,
  month = bout_data$month_id,
  year = bout_data$year_id,
  track = bout_data$track_id,

  n_month = length(unique(bout_data$month_id)),
  n_year = length(unique(bout_data$year_id)),
  n_track = length(unique(bout_data$track_id))
)

d_longstay <- list(
  N = nrow(bout_data),
  Y = bout_data$long_stay,
```

```

tide = bout_data$tide_s,
moon = bout_data$moon_s,
month = bout_data$month_id,
year = bout_data$year_id,
track = bout_data$track_id,

n_month = length(unique(bout_data$month_id)),
n_year = length(unique(bout_data$year_id)),
n_track = length(unique(bout_data$track_id))
)

# Bayesian logistic model
library(rethinking)

m_longstay <- ulam(
  alist(
    Y ~ dbern(p),

    logit(p) <- a +
      a_year[year] +
      a_track[track] +
      b_tide * tide +
      b_moon * moon,

    a ~ dnorm(0, 1),

    a_year[year] ~ dnorm(0, sigma_year),
    a_track[track] ~ dnorm(0, sigma_track),

    b_tide ~ dnorm(0, 1),
    b_moon ~ dnorm(0, 1),

    sigma_year ~ dexp(1),
    sigma_track ~ dexp(1)
  ),
  data = d_longstay,
  chains = 4,
  cores = 4,
  iter = 4000,
  control = list(adapt_delta = 0.95)
)

```

Running MCMC with 4 parallel chains, with 1 thread(s) per chain...

```
Chain 1 Iteration:    1 / 4000 [ 0%] (Warmup)
Chain 2 Iteration:    1 / 4000 [ 0%] (Warmup)
Chain 3 Iteration:    1 / 4000 [ 0%] (Warmup)
Chain 4 Iteration:    1 / 4000 [ 0%] (Warmup)
Chain 4 Iteration:  100 / 4000 [ 2%] (Warmup)
Chain 3 Iteration:  100 / 4000 [ 2%] (Warmup)
Chain 4 Iteration:  200 / 4000 [ 5%] (Warmup)
Chain 1 Iteration:  100 / 4000 [ 2%] (Warmup)
Chain 2 Iteration:  100 / 4000 [ 2%] (Warmup)
Chain 4 Iteration:  300 / 4000 [ 7%] (Warmup)
Chain 3 Iteration:  200 / 4000 [ 5%] (Warmup)
Chain 4 Iteration:  400 / 4000 [10%] (Warmup)
Chain 1 Iteration:  200 / 4000 [ 5%] (Warmup)
Chain 2 Iteration:  200 / 4000 [ 5%] (Warmup)
Chain 4 Iteration:  500 / 4000 [12%] (Warmup)
Chain 1 Iteration:  300 / 4000 [ 7%] (Warmup)
Chain 2 Iteration:  300 / 4000 [ 7%] (Warmup)
Chain 1 Iteration:  400 / 4000 [10%] (Warmup)
Chain 4 Iteration:  600 / 4000 [15%] (Warmup)
Chain 3 Iteration:  300 / 4000 [ 7%] (Warmup)
Chain 1 Iteration:  500 / 4000 [12%] (Warmup)
Chain 4 Iteration:  700 / 4000 [17%] (Warmup)
Chain 1 Iteration:  600 / 4000 [15%] (Warmup)
Chain 2 Iteration:  400 / 4000 [10%] (Warmup)
Chain 1 Iteration:  700 / 4000 [17%] (Warmup)
Chain 3 Iteration:  400 / 4000 [10%] (Warmup)
Chain 4 Iteration:  800 / 4000 [20%] (Warmup)
Chain 2 Iteration:  500 / 4000 [12%] (Warmup)
Chain 1 Iteration:  800 / 4000 [20%] (Warmup)
Chain 2 Iteration:  600 / 4000 [15%] (Warmup)
Chain 3 Iteration:  500 / 4000 [12%] (Warmup)
Chain 1 Iteration:  900 / 4000 [22%] (Warmup)
Chain 3 Iteration:  600 / 4000 [15%] (Warmup)
Chain 2 Iteration:  700 / 4000 [17%] (Warmup)
Chain 1 Iteration: 1000 / 4000 [25%] (Warmup)
Chain 4 Iteration:  900 / 4000 [22%] (Warmup)
Chain 2 Iteration:  800 / 4000 [20%] (Warmup)
Chain 3 Iteration:  700 / 4000 [17%] (Warmup)
Chain 1 Iteration: 1100 / 4000 [27%] (Warmup)
Chain 3 Iteration:  800 / 4000 [20%] (Warmup)
Chain 1 Iteration: 1200 / 4000 [30%] (Warmup)
```

Chain 2 Iteration: 900 / 4000 [22%] (Warmup)
 Chain 4 Iteration: 1000 / 4000 [25%] (Warmup)
 Chain 2 Iteration: 1000 / 4000 [25%] (Warmup)
 Chain 2 Iteration: 1100 / 4000 [27%] (Warmup)
 Chain 3 Iteration: 900 / 4000 [22%] (Warmup)
 Chain 2 Iteration: 1200 / 4000 [30%] (Warmup)
 Chain 1 Iteration: 1300 / 4000 [32%] (Warmup)
 Chain 2 Iteration: 1300 / 4000 [32%] (Warmup)
 Chain 3 Iteration: 1000 / 4000 [25%] (Warmup)
 Chain 2 Iteration: 1400 / 4000 [35%] (Warmup)
 Chain 4 Iteration: 1100 / 4000 [27%] (Warmup)
 Chain 1 Iteration: 1400 / 4000 [35%] (Warmup)
 Chain 2 Iteration: 1500 / 4000 [37%] (Warmup)
 Chain 3 Iteration: 1100 / 4000 [27%] (Warmup)
 Chain 2 Iteration: 1600 / 4000 [40%] (Warmup)
 Chain 3 Iteration: 1200 / 4000 [30%] (Warmup)
 Chain 2 Iteration: 1700 / 4000 [42%] (Warmup)
 Chain 1 Iteration: 1500 / 4000 [37%] (Warmup)
 Chain 3 Iteration: 1300 / 4000 [32%] (Warmup)
 Chain 4 Iteration: 1200 / 4000 [30%] (Warmup)
 Chain 2 Iteration: 1800 / 4000 [45%] (Warmup)
 Chain 2 Iteration: 1900 / 4000 [47%] (Warmup)
 Chain 3 Iteration: 1400 / 4000 [35%] (Warmup)
 Chain 1 Iteration: 1600 / 4000 [40%] (Warmup)
 Chain 3 Iteration: 1500 / 4000 [37%] (Warmup)
 Chain 3 Iteration: 1600 / 4000 [40%] (Warmup)
 Chain 1 Iteration: 1700 / 4000 [42%] (Warmup)
 Chain 4 Iteration: 1300 / 4000 [32%] (Warmup)
 Chain 3 Iteration: 1700 / 4000 [42%] (Warmup)
 Chain 2 Iteration: 2000 / 4000 [50%] (Warmup)
 Chain 2 Iteration: 2001 / 4000 [50%] (Sampling)
 Chain 3 Iteration: 1800 / 4000 [45%] (Warmup)
 Chain 1 Iteration: 1800 / 4000 [45%] (Warmup)
 Chain 3 Iteration: 1900 / 4000 [47%] (Warmup)
 Chain 1 Iteration: 1900 / 4000 [47%] (Warmup)
 Chain 4 Iteration: 1400 / 4000 [35%] (Warmup)
 Chain 1 Iteration: 2000 / 4000 [50%] (Warmup)
 Chain 1 Iteration: 2001 / 4000 [50%] (Sampling)
 Chain 2 Iteration: 2100 / 4000 [52%] (Sampling)
 Chain 4 Iteration: 1500 / 4000 [37%] (Warmup)
 Chain 3 Iteration: 2000 / 4000 [50%] (Warmup)
 Chain 3 Iteration: 2001 / 4000 [50%] (Sampling)
 Chain 1 Iteration: 2100 / 4000 [52%] (Sampling)

```

Chain 4 Iteration: 1600 / 4000 [ 40%] (Warmup)
Chain 1 Iteration: 2200 / 4000 [ 55%] (Sampling)
Chain 1 Iteration: 2300 / 4000 [ 57%] (Sampling)
Chain 4 Iteration: 1700 / 4000 [ 42%] (Warmup)
Chain 1 Iteration: 2400 / 4000 [ 60%] (Sampling)
Chain 1 Iteration: 2500 / 4000 [ 62%] (Sampling)
Chain 1 Iteration: 2600 / 4000 [ 65%] (Sampling)
Chain 3 Iteration: 2100 / 4000 [ 52%] (Sampling)
Chain 1 Iteration: 2700 / 4000 [ 67%] (Sampling)
Chain 4 Iteration: 1800 / 4000 [ 45%] (Warmup)
Chain 1 Iteration: 2800 / 4000 [ 70%] (Sampling)
Chain 1 Iteration: 2900 / 4000 [ 72%] (Sampling)
Chain 1 Iteration: 3000 / 4000 [ 75%] (Sampling)
Chain 1 Iteration: 3100 / 4000 [ 77%] (Sampling)
Chain 1 Iteration: 3200 / 4000 [ 80%] (Sampling)
Chain 4 Iteration: 1900 / 4000 [ 47%] (Warmup)
Chain 1 Iteration: 3300 / 4000 [ 82%] (Sampling)
Chain 1 Iteration: 3400 / 4000 [ 85%] (Sampling)
Chain 1 Iteration: 3500 / 4000 [ 87%] (Sampling)
Chain 2 Iteration: 2200 / 4000 [ 55%] (Sampling)
Chain 1 Iteration: 3600 / 4000 [ 90%] (Sampling)
Chain 3 Iteration: 2200 / 4000 [ 55%] (Sampling)
Chain 4 Iteration: 2000 / 4000 [ 50%] (Warmup)
Chain 4 Iteration: 2001 / 4000 [ 50%] (Sampling)
Chain 1 Iteration: 3700 / 4000 [ 92%] (Sampling)
Chain 1 Iteration: 3800 / 4000 [ 95%] (Sampling)
Chain 1 Iteration: 3900 / 4000 [ 97%] (Sampling)
Chain 4 Iteration: 2100 / 4000 [ 52%] (Sampling)
Chain 1 Iteration: 4000 / 4000 [100%] (Sampling)
Chain 1 finished in 67.1 seconds.
Chain 4 Iteration: 2200 / 4000 [ 55%] (Sampling)
Chain 3 Iteration: 2300 / 4000 [ 57%] (Sampling)
Chain 4 Iteration: 2300 / 4000 [ 57%] (Sampling)
Chain 3 Iteration: 2400 / 4000 [ 60%] (Sampling)
Chain 4 Iteration: 2400 / 4000 [ 60%] (Sampling)
Chain 2 Iteration: 2300 / 4000 [ 57%] (Sampling)
Chain 3 Iteration: 2500 / 4000 [ 62%] (Sampling)
Chain 4 Iteration: 2500 / 4000 [ 62%] (Sampling)
Chain 3 Iteration: 2600 / 4000 [ 65%] (Sampling)
Chain 3 Iteration: 2700 / 4000 [ 67%] (Sampling)
Chain 4 Iteration: 2600 / 4000 [ 65%] (Sampling)
Chain 2 Iteration: 2400 / 4000 [ 60%] (Sampling)
Chain 3 Iteration: 2800 / 4000 [ 70%] (Sampling)

```

```

Chain 4 Iteration: 2700 / 4000 [ 67%] (Sampling)
Chain 3 Iteration: 2900 / 4000 [ 72%] (Sampling)
Chain 3 Iteration: 3000 / 4000 [ 75%] (Sampling)
Chain 2 Iteration: 2500 / 4000 [ 62%] (Sampling)
Chain 4 Iteration: 2800 / 4000 [ 70%] (Sampling)
Chain 3 Iteration: 3100 / 4000 [ 77%] (Sampling)
Chain 3 Iteration: 3200 / 4000 [ 80%] (Sampling)
Chain 4 Iteration: 2900 / 4000 [ 72%] (Sampling)
Chain 3 Iteration: 3300 / 4000 [ 82%] (Sampling)
Chain 2 Iteration: 2600 / 4000 [ 65%] (Sampling)
Chain 4 Iteration: 3000 / 4000 [ 75%] (Sampling)
Chain 3 Iteration: 3400 / 4000 [ 85%] (Sampling)
Chain 3 Iteration: 3500 / 4000 [ 87%] (Sampling)
Chain 4 Iteration: 3100 / 4000 [ 77%] (Sampling)
Chain 3 Iteration: 3600 / 4000 [ 90%] (Sampling)
Chain 2 Iteration: 2700 / 4000 [ 67%] (Sampling)
Chain 3 Iteration: 3700 / 4000 [ 92%] (Sampling)
Chain 2 Iteration: 2800 / 4000 [ 70%] (Sampling)
Chain 3 Iteration: 3800 / 4000 [ 95%] (Sampling)
Chain 4 Iteration: 3200 / 4000 [ 80%] (Sampling)
Chain 2 Iteration: 2900 / 4000 [ 72%] (Sampling)
Chain 3 Iteration: 3900 / 4000 [ 97%] (Sampling)
Chain 2 Iteration: 3000 / 4000 [ 75%] (Sampling)
Chain 4 Iteration: 3300 / 4000 [ 82%] (Sampling)
Chain 3 Iteration: 4000 / 4000 [100%] (Sampling)
Chain 3 finished in 105.9 seconds.
Chain 2 Iteration: 3100 / 4000 [ 77%] (Sampling)
Chain 4 Iteration: 3400 / 4000 [ 85%] (Sampling)
Chain 4 Iteration: 3500 / 4000 [ 87%] (Sampling)
Chain 4 Iteration: 3600 / 4000 [ 90%] (Sampling)
Chain 4 Iteration: 3700 / 4000 [ 92%] (Sampling)
Chain 4 Iteration: 3800 / 4000 [ 95%] (Sampling)
Chain 4 Iteration: 3900 / 4000 [ 97%] (Sampling)
Chain 4 Iteration: 4000 / 4000 [100%] (Sampling)
Chain 4 finished in 111.5 seconds.
Chain 2 Iteration: 3200 / 4000 [ 80%] (Sampling)
Chain 2 Iteration: 3300 / 4000 [ 82%] (Sampling)
Chain 2 Iteration: 3400 / 4000 [ 85%] (Sampling)
Chain 2 Iteration: 3500 / 4000 [ 87%] (Sampling)
Chain 2 Iteration: 3600 / 4000 [ 90%] (Sampling)
Chain 2 Iteration: 3700 / 4000 [ 92%] (Sampling)
Chain 2 Iteration: 3800 / 4000 [ 95%] (Sampling)
Chain 2 Iteration: 3900 / 4000 [ 97%] (Sampling)

```

Chain 2 Iteration: 4000 / 4000 [100%] (Sampling)
Chain 2 finished in 191.1 seconds.

All 4 chains finished successfully.
Mean chain execution time: 118.9 seconds.
Total execution time: 191.2 seconds.

```
precis(m_longstay, pars = c("b_tide", "b_moon"))
```

	mean	sd	5.5%	94.5%	rhat	ess_bulk
b_tide	0.07611505	0.06216283	-0.02467956	0.17277481	1.003183	1428.539
b_moon	-0.04543503	0.05630745	-0.13675193	0.04422193	1.007208	515.889

```
# convert to odds ratios  
post <- extract.samples(m_longstay)  
  
mean(exp(post$b_tide))
```

```
[1] 1.081171
```

```
mean(exp(post$b_moon))
```

```
[1] 0.9570958
```

```
##add grouping variables to bout_data  
  
bout_data <- bout_data %>%  
  mutate(  
    month_num = as.integer(month),  
    nov_mar = ifelse(month_num %in% c(11, 12, 1, 2, 3), 1, 0),  
    nov_2023_vs_2025 = case_when(  
      month_num == 11 & year == 2023 ~ "Nov2023",  
      month_num == 11 & year == 2025 ~ "Nov2025",  
      TRUE ~ NA_character_  
    )  
  )  
  
bout_data <- bout_data %>%  
  mutate(year = as.integer(as.character(year)))  
  
names(bout_data)
```

```

[1] "file_name"          "cell_id"           "bout_id"
[4] "n_points"           "duration_sec"       "tide_height_m"
[7] "moon_illumination" "month"              "year"
[10] "log_duration"        "month_id"           "year_id"
[13] "track_id"            "tide_s"             "moon_s"
[16] "long_stay"           "month_num"          "nov_mar"
[19] "nov_2023_vs_2025"

```

Analysis: Comparison November 2023 vs November 2025

```

# descriptive summaries

nov_compare <- bout_data %>%
  filter(!is.na(nov_2023_vs_2025)) %>%
  group_by(nov_2023_vs_2025) %>%
  summarise(
    n_bouts = n(),
    n_cells = n_distinct(cell_id),
    mean_duration = mean(duration_sec, na.rm = TRUE),
    median_duration = median(duration_sec, na.rm = TRUE),
    total_duration = sum(duration_sec, na.rm = TRUE)
  )

# areas visited per track

track_nov_compare <- bout_data %>%
  filter(!is.na(nov_2023_vs_2025)) %>%
  group_by(track_id, nov_2023_vs_2025) %>%
  summarise(
    n_cells_visited = n_distinct(cell_id),
    n_bouts = n(),
    mean_duration = mean(duration_sec, na.rm = TRUE),
    total_duration = sum(duration_sec, na.rm = TRUE),
    .groups = "drop"
  )

## Inferential model

nov_bout <- bout_data %>%
  filter(!is.na(nov_2023_vs_2025)) %>%
  mutate(
    period_id = ifelse(nov_2023_vs_2025 == "Nov2025", 1, 0),

```

```

    tide_s = as.numeric(scale(tide_height_m)),
    moon_s = as.numeric(scale(moon_illumination)),
    track_id = as.integer(factor(track_id))
  )

d_nov_bout <- list(
  N = nrow(nov_bout),
  Y = log(nov_bout$duration_sec),
  period = nov_bout$period_id,
  tide = nov_bout$tide_s,
  moon = nov_bout$moon_s,
  track = nov_bout$track_id,
  n_track = length(unique(nov_bout$track_id))
)

m_nov_bout <- ulam(
  alist(
    Y ~ dnorm(mu, sigma),
    mu <- a + b_period * period + b_tide * tide + b_moon * moon + a_track[track],

    a ~ dnorm(0, 1),
    b_period ~ dnorm(0, 1),
    b_tide ~ dnorm(0, 1),
    b_moon ~ dnorm(0, 1),

    a_track[track] ~ dnorm(0, sigma_track),

    sigma ~ dexp(1),
    sigma_track ~ dexp(1)
  ),
  data = d_nov_bout,
  chains = 4, cores = 4, iter = 2000
)

```

Running MCMC with 4 parallel chains, with 1 thread(s) per chain...

```

Chain 1 Iteration:    1 / 2000 [  0%]  (Warmup)
Chain 1 Iteration:   100 / 2000 [  5%]  (Warmup)
Chain 1 Iteration:   200 / 2000 [ 10%]  (Warmup)

Chain 2 Iteration:    1 / 2000 [  0%]  (Warmup)

```

Chain 2 Iteration: 100 / 2000 [5%] (Warmup)
 Chain 3 Iteration: 1 / 2000 [0%] (Warmup)
 Chain 3 Iteration: 100 / 2000 [5%] (Warmup)
 Chain 3 Iteration: 200 / 2000 [10%] (Warmup)
 Chain 4 Iteration: 1 / 2000 [0%] (Warmup)
 Chain 4 Iteration: 100 / 2000 [5%] (Warmup)
 Chain 1 Iteration: 300 / 2000 [15%] (Warmup)
 Chain 1 Iteration: 400 / 2000 [20%] (Warmup)
 Chain 2 Iteration: 200 / 2000 [10%] (Warmup)
 Chain 2 Iteration: 300 / 2000 [15%] (Warmup)
 Chain 2 Iteration: 400 / 2000 [20%] (Warmup)
 Chain 2 Iteration: 500 / 2000 [25%] (Warmup)
 Chain 3 Iteration: 300 / 2000 [15%] (Warmup)
 Chain 3 Iteration: 400 / 2000 [20%] (Warmup)
 Chain 3 Iteration: 500 / 2000 [25%] (Warmup)
 Chain 4 Iteration: 200 / 2000 [10%] (Warmup)
 Chain 4 Iteration: 300 / 2000 [15%] (Warmup)
 Chain 4 Iteration: 400 / 2000 [20%] (Warmup)
 Chain 4 Iteration: 500 / 2000 [25%] (Warmup)
 Chain 1 Iteration: 500 / 2000 [25%] (Warmup)
 Chain 1 Iteration: 600 / 2000 [30%] (Warmup)
 Chain 1 Iteration: 700 / 2000 [35%] (Warmup)
 Chain 2 Iteration: 600 / 2000 [30%] (Warmup)
 Chain 2 Iteration: 700 / 2000 [35%] (Warmup)
 Chain 2 Iteration: 800 / 2000 [40%] (Warmup)
 Chain 2 Iteration: 900 / 2000 [45%] (Warmup)
 Chain 3 Iteration: 600 / 2000 [30%] (Warmup)
 Chain 3 Iteration: 700 / 2000 [35%] (Warmup)
 Chain 3 Iteration: 800 / 2000 [40%] (Warmup)
 Chain 4 Iteration: 600 / 2000 [30%] (Warmup)
 Chain 4 Iteration: 700 / 2000 [35%] (Warmup)
 Chain 1 Iteration: 800 / 2000 [40%] (Warmup)
 Chain 1 Iteration: 900 / 2000 [45%] (Warmup)
 Chain 1 Iteration: 1000 / 2000 [50%] (Warmup)
 Chain 1 Iteration: 1001 / 2000 [50%] (Sampling)
 Chain 2 Iteration: 1000 / 2000 [50%] (Warmup)
 Chain 2 Iteration: 1001 / 2000 [50%] (Sampling)
 Chain 2 Iteration: 1100 / 2000 [55%] (Sampling)
 Chain 3 Iteration: 900 / 2000 [45%] (Warmup)
 Chain 3 Iteration: 1000 / 2000 [50%] (Warmup)
 Chain 3 Iteration: 1001 / 2000 [50%] (Sampling)
 Chain 4 Iteration: 800 / 2000 [40%] (Warmup)
 Chain 4 Iteration: 900 / 2000 [45%] (Warmup)

```

Chain 1 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 2 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 3 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 4 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 4 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 1 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 2 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 3 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 4 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 1 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 1 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 2 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 3 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 4 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 1 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 2 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 3 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 3 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 4 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 1 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 3 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 4 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 4 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 1 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 1 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 2 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 3 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 4 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 1 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 2 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 2 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 3 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 3 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 1 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 2 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 3 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 4 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 1 finished in 1.4 seconds.
Chain 3 finished in 1.4 seconds.
Chain 2 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 4 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 2 finished in 1.5 seconds.
Chain 4 Iteration: 1900 / 2000 [ 95%] (Sampling)

```

Chain 4 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 4 finished in 1.8 seconds.

All 4 chains finished successfully.
Mean chain execution time: 1.5 seconds.
Total execution time: 1.9 seconds.

```
# bout frequency

track_nov_counts <- bout_data %>%
  filter(!is.na(nov_2023_vs_2025)) %>%
  group_by(track_id, nov_2023_vs_2025) %>%
  summarise(
    n_bouts = n(),
    mean_tide = mean(tide_height_m, na.rm = TRUE),
    mean_moon = mean(moon_illumination, na.rm = TRUE),
    .groups = "drop"
  )

nrow(bout_data)
```

```
[1] 2095
```

```
table(is.na(bout_data$nov_2023_vs_2025), useNA = "ifany")
```

```
FALSE  TRUE
  147   1948
```

```
table(bout_data$nov_2023_vs_2025, useNA = "ifany")
```

```
Nov2023 Nov2025    <NA>
      30      117   1948
```

```
track_nov_compare
```

```
# A tibble: 67 x 6
  track_id nov_2023_vs_2025 n_cells_visited n_bouts mean_duration
    <int>   <chr>                <int>   <int>         <dbl>
1         1 Nov2023                 6     6           33
2         2 Nov2023                 5     6           33
3         3 Nov2023                 6     6           33
4         4 Nov2023                 5     5          39.6
5         5 Nov2023                 2     3          49.3
6         6 Nov2023                 4     4           3.5
7       432 Nov2025                 1     1           14
8       433 Nov2025                 2     2            9
9       434 Nov2025                 1     1           20
10      435 Nov2025                 4     4          16.5
# i 57 more rows
# i 1 more variable: total_duration <dbl>
```

```
track_nov_counts
```

```
# A tibble: 67 x 5
  track_id nov_2023_vs_2025 n_bouts mean_tide mean_moon
    <int>   <chr>                <int>   <dbl>   <dbl>
1         1 Nov2023                 6     2.11  0.990
2         2 Nov2023                 6     2.16  0.990
3         3 Nov2023                 6     2.16  0.990
4         4 Nov2023                 5     2.16  0.990
5         5 Nov2023                 3     2.21  0.990
6         6 Nov2023                 4     2.21  0.990
7       432 Nov2025                 1     1.13  0.980
8       433 Nov2025                 2     1.13  0.980
9       434 Nov2025                 1     1.13  0.980
10      435 Nov2025                 4     1.13  0.980
# i 57 more rows
```

```
track_nov_summary <- track_nov_compare %>%
  group_by(nov_2023_vs_2025) %>%
  summarise(
    mean_cells_visited = mean(n_cells_visited, na.rm = TRUE),
    mean_bouts = mean(n_bouts, na.rm = TRUE),
    mean_duration = mean(mean_duration, na.rm = TRUE),
    mean_total_duration = mean(total_duration, na.rm = TRUE),
    n_tracks = n(),
    .groups = "drop"
```

```
)

track_nov_summary

# A tibble: 2 x 6
  nov_2023_vs_2025 mean_cells_visited mean_bouts mean_duration
  <chr>             <dbl>         <dbl>         <dbl>
1 Nov2023           4.67           5           31.9
2 Nov2025           1.75           1.92          18.0
# i 2 more variables: mean_total_duration <dbl>, n_tracks <int>
```

```
## count/tide/moon dataset
track_nov_counts_summary <- track_nov_counts %>%
  group_by(nov_2023_vs_2025) %>%
  summarise(
    mean_bouts = mean(n_bouts, na.rm = TRUE),
    mean_tide = mean(mean_tide, na.rm = TRUE),
    mean_moon = mean(mean_moon, na.rm = TRUE),
    n_tracks = n(),
    .groups = "drop"
  )

track_nov_counts_summary
```

```
# A tibble: 2 x 5
  nov_2023_vs_2025 mean_bouts mean_tide mean_moon n_tracks
  <chr>             <dbl>     <dbl>     <dbl>     <int>
1 Nov2023           5         2.17     0.990         6
2 Nov2025           1.92     0.968     0.752        61
```

```
precis(m_nov_bout, pars = c("b_period", "b_tide", "b_moon"))
```

	mean	sd	5.5%	94.5%	rhat	ess_bulk
b_period	1.019999	0.6947889	-0.08382123	2.1369237	1.004535	941.3168
b_tide	0.902850	0.3771717	0.30652187	1.5089572	1.005453	928.3116
b_moon	-0.432721	0.1778416	-0.71839405	-0.1545763	1.004904	1099.6336

Analysis: November to March period vs rest of the year

```

bout_data <- bout_data %>%
  mutate(
    month_num = as.integer(month),
    season = ifelse(month_num %in% c(11, 12, 1, 2, 3), 1, 0)
  )

season_bout <- bout_data %>%
  mutate(
    tide_s = as.numeric(scale(tide_height_m)),
    moon_s = as.numeric(scale(moon_illumination)),
    track_id = as.integer(factor(track_id)),
    year_id = as.integer(factor(year))
  )

d_season_bout <- list(
  N = nrow(season_bout),
  Y = log(season_bout$duration_sec),
  season = season_bout$season,
  tide = season_bout$tide_s,
  moon = season_bout$moon_s,
  track = season_bout$track_id,
  year = season_bout$year_id,
  n_track = length(unique(season_bout$track_id)),
  n_year = length(unique(season_bout$year_id))
)

m_season_bout <- ulam(
  alist(
    Y ~ dnorm(mu, sigma),
    mu <- a +
      b_season * season +
      b_tide * tide +
      b_moon * moon +
      a_track[track] +
      a_year[year],

    a ~ dnorm(0, 1),
    b_season ~ dnorm(0, 1),
    b_tide ~ dnorm(0, 1),
    b_moon ~ dnorm(0, 1),

    a_track[track] ~ dnorm(0, sigma_track),

```

```

a_year[year] ~ dnorm(0, sigma_year),

sigma ~ dexp(1),
sigma_track ~ dexp(1),
sigma_year ~ dexp(1)
),
data = d_season_bout,
chains = 4,
cores = 4,
iter = 2000
)

```

Running MCMC with 4 parallel chains, with 1 thread(s) per chain...

```

Chain 1 Iteration:      1 / 2000 [  0%] (Warmup)

Chain 2 Iteration:      1 / 2000 [  0%] (Warmup)
Chain 3 Iteration:      1 / 2000 [  0%] (Warmup)
Chain 4 Iteration:      1 / 2000 [  0%] (Warmup)
Chain 1 Iteration:    100 / 2000 [  5%] (Warmup)
Chain 2 Iteration:    100 / 2000 [  5%] (Warmup)
Chain 4 Iteration:    100 / 2000 [  5%] (Warmup)
Chain 3 Iteration:    100 / 2000 [  5%] (Warmup)
Chain 3 Iteration:    200 / 2000 [ 10%] (Warmup)
Chain 1 Iteration:    200 / 2000 [ 10%] (Warmup)
Chain 2 Iteration:    200 / 2000 [ 10%] (Warmup)
Chain 4 Iteration:    200 / 2000 [ 10%] (Warmup)
Chain 3 Iteration:    300 / 2000 [ 15%] (Warmup)
Chain 1 Iteration:    300 / 2000 [ 15%] (Warmup)
Chain 2 Iteration:    300 / 2000 [ 15%] (Warmup)
Chain 3 Iteration:    400 / 2000 [ 20%] (Warmup)
Chain 4 Iteration:    300 / 2000 [ 15%] (Warmup)
Chain 1 Iteration:    400 / 2000 [ 20%] (Warmup)
Chain 2 Iteration:    400 / 2000 [ 20%] (Warmup)
Chain 4 Iteration:    400 / 2000 [ 20%] (Warmup)
Chain 3 Iteration:    500 / 2000 [ 25%] (Warmup)
Chain 2 Iteration:    500 / 2000 [ 25%] (Warmup)
Chain 1 Iteration:    500 / 2000 [ 25%] (Warmup)
Chain 3 Iteration:    600 / 2000 [ 30%] (Warmup)
Chain 4 Iteration:    500 / 2000 [ 25%] (Warmup)
Chain 2 Iteration:    600 / 2000 [ 30%] (Warmup)

```

Chain 4 Iteration: 600 / 2000 [30%] (Warmup)
 Chain 3 Iteration: 700 / 2000 [35%] (Warmup)
 Chain 2 Iteration: 700 / 2000 [35%] (Warmup)
 Chain 1 Iteration: 600 / 2000 [30%] (Warmup)
 Chain 2 Iteration: 800 / 2000 [40%] (Warmup)
 Chain 3 Iteration: 800 / 2000 [40%] (Warmup)
 Chain 4 Iteration: 700 / 2000 [35%] (Warmup)
 Chain 3 Iteration: 900 / 2000 [45%] (Warmup)
 Chain 4 Iteration: 800 / 2000 [40%] (Warmup)
 Chain 2 Iteration: 900 / 2000 [45%] (Warmup)
 Chain 1 Iteration: 700 / 2000 [35%] (Warmup)
 Chain 4 Iteration: 900 / 2000 [45%] (Warmup)
 Chain 3 Iteration: 1000 / 2000 [50%] (Warmup)
 Chain 2 Iteration: 1000 / 2000 [50%] (Warmup)
 Chain 3 Iteration: 1001 / 2000 [50%] (Sampling)
 Chain 2 Iteration: 1001 / 2000 [50%] (Sampling)
 Chain 1 Iteration: 800 / 2000 [40%] (Warmup)
 Chain 4 Iteration: 1000 / 2000 [50%] (Warmup)
 Chain 4 Iteration: 1001 / 2000 [50%] (Sampling)
 Chain 3 Iteration: 1100 / 2000 [55%] (Sampling)
 Chain 4 Iteration: 1100 / 2000 [55%] (Sampling)
 Chain 1 Iteration: 900 / 2000 [45%] (Warmup)
 Chain 4 Iteration: 1200 / 2000 [60%] (Sampling)
 Chain 2 Iteration: 1100 / 2000 [55%] (Sampling)
 Chain 3 Iteration: 1200 / 2000 [60%] (Sampling)
 Chain 4 Iteration: 1300 / 2000 [65%] (Sampling)
 Chain 1 Iteration: 1000 / 2000 [50%] (Warmup)
 Chain 1 Iteration: 1001 / 2000 [50%] (Sampling)
 Chain 3 Iteration: 1300 / 2000 [65%] (Sampling)
 Chain 2 Iteration: 1200 / 2000 [60%] (Sampling)
 Chain 4 Iteration: 1400 / 2000 [70%] (Sampling)
 Chain 1 Iteration: 1100 / 2000 [55%] (Sampling)
 Chain 3 Iteration: 1400 / 2000 [70%] (Sampling)
 Chain 4 Iteration: 1500 / 2000 [75%] (Sampling)
 Chain 1 Iteration: 1200 / 2000 [60%] (Sampling)
 Chain 2 Iteration: 1300 / 2000 [65%] (Sampling)
 Chain 4 Iteration: 1600 / 2000 [80%] (Sampling)
 Chain 3 Iteration: 1500 / 2000 [75%] (Sampling)
 Chain 1 Iteration: 1300 / 2000 [65%] (Sampling)
 Chain 4 Iteration: 1700 / 2000 [85%] (Sampling)
 Chain 2 Iteration: 1400 / 2000 [70%] (Sampling)
 Chain 3 Iteration: 1600 / 2000 [80%] (Sampling)
 Chain 1 Iteration: 1400 / 2000 [70%] (Sampling)

```

Chain 4 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 4 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 1 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 3 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 2 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 4 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 4 finished in 27.8 seconds.
Chain 1 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 3 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 2 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 1 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 3 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 1 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 2 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 3 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 3 finished in 31.4 seconds.
Chain 1 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 1 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 1 finished in 32.7 seconds.
Chain 2 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 2 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 2 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 2 finished in 35.7 seconds.

```

All 4 chains finished successfully.
 Mean chain execution time: 31.9 seconds.
 Total execution time: 35.8 seconds.

```
precis(m_season_bout, pars = c("b_season", "b_tide", "b_moon"))
```

	mean	sd	5.5%	94.5%	rhat	ess_bulk
b_season	-0.04783723	0.06380293	-0.14985652	0.05391531	1.0005062	4203.439
b_tide	0.07358312	0.03086472	0.02464421	0.12313864	1.0001727	4291.622
b_moon	-0.02911302	0.02807705	-0.07398676	0.01539199	0.9997855	4317.636

```

season_summary <- bout_data %>%
  mutate(season_label = ifelse(season == 1, "Nov-Mar", "Rest")) %>%
  group_by(season_label) %>%
  summarise(
    n_bouts = n(),
    mean_duration = mean(duration_sec, na.rm = TRUE),

```

```

    median_duration = median(duration_sec, na.rm = TRUE),
    total_duration = sum(duration_sec, na.rm = TRUE),
    .groups = "drop"
)

```

```
season_summary
```

```
# A tibble: 2 x 5
```

	season_label	n_bouts	mean_duration	median_duration	total_duration
	<chr>	<int>	<dbl>	<dbl>	<dbl>
1	Nov-Mar	965	27.2	20	26220
2	Rest	1130	25.3	16	28614

```
## areas visited
```

```

track_space_use <- bout_data %>%
  mutate(
    season = ifelse(nov_mar == 1, "Nov-Mar", "Rest")
  ) %>%
  group_by(track_id, year, season) %>%
  summarise(
    n_cells_visited = n_distinct(cell_id),
    n_bouts = n(),
    mean_duration = mean(duration_sec, na.rm = TRUE),
    mean_tide = mean(tide_height_m, na.rm = TRUE),
    mean_moon = mean(moon_illumination, na.rm = TRUE),
    .groups = "drop"
  )

```

```
## Summarize by season
```

```

track_space_summary <- track_space_use %>%
  group_by(season) %>%
  summarise(
    mean_cells = mean(n_cells_visited, na.rm = TRUE),
    mean_bouts = mean(n_bouts, na.rm = TRUE),
    mean_duration = mean(mean_duration, na.rm = TRUE),
    n_tracks = n(),
    .groups = "drop"
  )

```

```
track_space_summary
```

```
# A tibble: 2 x 5
  season mean_cells mean_bouts mean_duration n_tracks
  <chr>      <dbl>      <dbl>      <dbl>      <int>
1 Nov-Mar    3.75      4.47      26.4       216
2 Rest       2.43      3.01      24.7       376
```

```
ggplot(track_space_use, aes(x = season, y = n_cells_visited)) +
  geom_boxplot() +
  labs(
    x = "Season",
    y = "Number of cells visited"
  ) +
  theme_minimal()
```

